

1 **Title: Genomic analyses reveal distinct evolutionary pattern of ducks and identify**
2 **genetic changes underlying domestication and speciation**

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Abstract

Domestic ducks were considered to be tamed from the mallard or the descendant of the mallard and the spot-billed duck. Domestic ducks showed remarkable phenotypic variation in morphology, physiology and behavior. However, molecular genetics for ducks' origin and their phenotypic variation are still poorly studied.

Here we presented the mallard and the spot-billed genomes and performed whole-genome sequencing on eight domestic duck breeds and eight wild duck species.

Surprisingly, Analyses on these data supports a model that domestic ducks diverged from their closely wild lineage (mallard duck and spot-billed duck) at the last glacial period (LGP, 100~300 kilo years ago (Kyr)). Wild lineage further speciated into mallard duck and spot-billed duck about 70 Kyr, whereas domestic lineage decreased their population size through the LGP. A scan of wild duck genomes by comparing with domestic duck genomes identified a number of locus that may have been affected by positive selection in ancestral wild ducks after their divergence from domestic lineages.

Function analyses suggested that those genes usually affecting organs development and energy metabolism that may involving long-distance flight ability. Further selective sweep analyses identified two genes associated with egg production and three genes related to feeding modulation being under selection in domestic ducks. These analyses unravel a distinct evolutionary pattern of ducks and two wild duck *de novo* genomes also provide a novel resource for speciation study.

Significance

Recently, many researches have found that traditional domestication hypothesis is incompatible with genetic data. Moreover, why those domestic animals been domesticated and does the remarkable phenotypic variations between domestic and its closest wild relatives just mainly caused by domestic's genetic adaptations to human environment? For example, does domestic duck loss flight and migration ability during domestic process? Through reveal the evolutionary history of ducks, we demonstrate that both the traditional hypotheses on duck domestication are incompatible with our data. Instead, there is a very long divergence time between wild and domestic ducks, longer than the divergence time between mallard and spot-billed duck. The comparison between domestication and speciation process are also informative for understanding the genetic basis of biodiversity

Introduction

As two major concerns of evolutionary study, both domestication and speciation (or population divergence) benefited from the advance of sequencing technology(1-9). The genome assembly and characterizing of genetic variation in many domestic and wild animals have facilitated the revealing of their complex demographic history, which provides profound knowledges for both domestication and speciation studies (6-8, 10). We previously drew the draft of a Beijing duck genome and identified genomic variation within this assembly(11). However, as the most economical important waterfowl and one of the principal natural hosts of influenza A viruses, genetic variations of duck were not well characterized. And debates about domestic ducks being

domesticated from the mallard duck or the descendant of mallard duck and spot-billed duck are still continuing.

Until now there are more than 140 wild duck species across the world and most of them can hybridize each other (12, 13). Phylogeographic structure of dabbling ducks using 5'-end sequencing of the mtDNA control region and ODC-6 of nuclear DNA demonstrated that the mallard duck diverged from its closely population (spot-billed duck) recently, implying they are good resource for speciation studies (14). Domestic ducks are considered to originated from East Asian and Europe and show broad phenotypic difference from these wild duck species, such as loss of flight and migration ability, reducing of brain size, increasing of body size and fertility(15-19). Previous studies based on similar outlooks habits and distributions provide two hypotheses about domestic ducks' origin. One hypothesis suggested that domestic ducks were originated from the mallards, while the other argued that they were tamed from descents of the mallard (*Anas platyrhynchos*) and spot-billed (*Anas zonorhynchos*)(20). Until now, evolutionary patterns of wild and domestic ducks in genome-wide scale are poorly understand.

Results

Here, we report high-quality drafts of the mallard duck and spot-billed duck respectively, and compare them to the genome of Beijing ducks and other birds (SI Appendix; Fig. S1; Fig. S3-S11; Table S1-S9). We also generate a total of 1,219 Gb whole-genome resequencing data of 35 wild ducks from eight species (gadwall (n=1), baikal teal (n=1), one pintail (n=1), falcated teal (n=1), common teal (n=1), wigeons

(n=3), spot-billed duck (n=13) and mallard duck (n=14)), 94 domesticated ducks from six Chinese breeds, 20 ducks from two commercial breeds (Cherry and Campebell ducks), together with 20 individuals from two outgroups (ten Muscovy ducks and ten Taihu geese)(Fig. S2; Table S10) (mean coverage for mallard, spot-billed and Beijing duck is 12x).

Variants were called using GATK for each population and different combination separately, applying several quality filters. 44,852,612 SNPs were identified in the 10 duck populations (eight domestic duck breeds and two wild duck populations) 20,676,377 and 42,012,846 SNPs were detected in domestic ducks and two wild duck populations (the mallard duck and spot-billed duck), respectively, with 18,093,115 SNPs shared (Fig. S12). Moreover, mallard and spot-billed shared comparative number SNPs with domestic ducks (SI Appendix).

Domestic duck originated in a lineage distinct from the mallard duck and spot-billed duck

We initially explored the genetic relationships of domestic and wild ducks by performed a principal component analysis (PCA) using genome wide SNPs from 27 Shaoxing, 27 Beijing ducks representing domestic ducks and 35 wild ducks. Expectedly, the domestic ducks (Shaoxing and Beijing ducks) clustered to mallard and spot-billed ducks, while separated clearly from other wild species (Fig. 1a). Such genetic relationships were also identified by further PCA analysis and tree analysis based on the IBS (identical by state) distance of those 89 individuals (Fig. 1b; SI Appendix; Fig. S13, S14). It supported that mallard duck and spot-billed duck had a more closely genetic relationship with

domestic populations than other wild species.

We then asked whether domestic ducks were originated from mallard duck or the descents of the mallard and spot-billed ducks (20). We performed a 3-population test for admixture by taking the mallard duck and spot-billed duck as ancestral populations, Beijing and Shaoxing as target populations through a Block Jackknife analysis that corrected for linkage disequilibrium among SNPs (21, 22). Surprisingly, this effort obtained significantly positive scores for these four groups ($f_3 = 0.007279$, Z score = 23.834; Table S11) and did not support that domestic ducks were descents of the mallard duck and spot-billed ducks.

We further performed genetic structure analysis using the ADMIXTURE to estimate ancestry among individuals of the four populations (Beijing, Shaoxing, the mallard duck and spot-billed duck) with K (the number of ancestry components) ranging from two to seven(23). This analysis indicated that domestic ducks had a homogeneous genetic background whereas the wild ducks showed a heterogeneous genetic background which is consistent with PCA (Fig. 1a,d). However, the wild ancestor duck started to divide into mallard and spot-billed when K were increased to four that is following the separation between the wild ducks and domestic ducks with when $K = 2$ which have lowest cross-validation error(Fig. 1d; SI Appendix; Fig. S15). These observations neither support domestic ducks were offspring of mallard and spot-billed ducks which means that traditional domestication hypothesis may not fit to genetic data whereas the F_{ST} based phylogenetic tree indicate a population history compatible with the genetic structure analysis (Fig. 1 c-e).

To eliminate the effect of population drift within each domestic population which may also lead the separation between domestic and wild ducks previous than the two wild ducks' split in genetic structure analysis, and further confirm the tree topology of domestic duck and the two wild ducks, we randomly selected and polarized 20,333 unlinked SNPs from these 10 duck populations with two outgroups (Muscovy ducks) for a diffusion approach demographic simulation ($\partial a \partial i$)(Methods)(24). We estimated the divergent time of the domestic ducks (quantize all eight domestic breeds into one population) and two wild duck populations (mallard and spot-billed) using two simple but informative models, which one took and the other did not take migration into account (Fig. 2a; SI Appendix). One hundred bootstrap simulations of the two models support that both the divergent times between two wild duck populations and domestic duck were almost equal (100~300 Kyr) and longer than the corresponding one between mallard and spot-billed ducks (~70 Kyr, Fig. 2b-c; SI Appendix; Table S12, Fig. S16-17). These results further supported that the ancestor of domestic duck split from the ancestor of mallard and spot-billed ducks before the two wild ducks' divergence.

Moreover, we performed a haploid pairwise sequential Markovian coalescent (PSMC) analysis to reconstruct the demographic history and estimated the effective population size (N_e) change along time among the mallard, spot-billed, Beijing and Shaoxing duck. This analysis suggested that ancestors of these four populations hold a comparative level in effective population size before 110 Kyr ago (beginning of LGP). However, ancestors of the two wild ducks and domestic ducks seemed to be under differently demographic histories after LGP. During this period, the effective populations of

Beijing and Shaoxing duck ancestors decreased dramatically and continually. It is contrast to the case in the mallard and spot-billed duck ancestor, where their effective populations didn't decrease or even experienced once or twice times of dramatic increasing (Fig. 2d; SI Appendix; Fig. S18). PSMC can also be informally used to infer divergent times when N_e trajectories that are overlapping diverge moving forward in time towards the present (25, 26). Interestingly, the divergent time was roughly coincident with the one inferred from $\partial a \partial i$. Such consistency further encouraged us to infer that the ancestor of domestic duck diverged from the ancestor of mallard duck and spot-billed duck at about the beginning of the LGP. Similar to N_e estimated by the PSMC analysis, we found that populations size of domestic population decreased whereas these of both the two wild duck species increased after split between wild and domestic population when using $\partial a \partial i$ to simulated a more complex model (i.e. populations' size change exponentially after split) of divergence between populations (SI Appendix).

The early ancestors of mallard duck and spot-billed duck might benefit from adaptive selections

It is interesting that different tendency of population size change occurred after divergence between wild lineage and domestic lineage during LGP. One possibility might be that nature selection granted the mallard duck and spot-billed duck ancestors, but not domestic duck ancestors, new advantage traits to adjust environment change after their split. Consequently, environment changes lead the domestic ancestor population size decreased, whereas wild ancestor population size even increased (Fig.

2d; SI Appendix). We detected positive selections in the early ancestors of these two wild populations after their divergence with domestic lineage, and identified a total of 424 putative selective sweep regions, with a total size of 10.86 Mb, using thresholds of 4 standard deviations away from the mean (Fig. 3a-b; Fig. S19; Methods)(27, 28). Among them, 168 selective sweeps harbored 202 protein coding genes, while the remained ones don't contain annotated protein coding genes (SI Appendix; Table S13). Recent studies indicated that positive selections in intergenic and intragenic regions were associated with phenotypic variation (4, 7, 27). We then extended 180 kb in both upstream and downstream the regions containing no annotated genes. This effort identified 162 closest genes in 156 of these extended regions (Table S13). After that, we performed gene ontology (GO) analysis of 306 genes (12 of them detected in domestic selection were removed, Methods) in or closest (extended 180 kb in both upstream and downstream) to selective sweeps regions (SI Appendix) and highlighted 343 enriched categories related to 236 biological process (Table S14, S15). Interestingly, the most remarked categories were related to a series of organs and system, such as morphogenesis and development of lung, heart, metanephros, aortic and cardiac muscle, and functions of blood circulatory system, respiratory system, peripheral nervous system and ventricular system development, rhythmic excitation (Table 2). Since long distance flight need adaptive changes in body shape, energy metabolism and spatial navigation, we proposed that selection on those genes associated with above functions might enhance the ability to fly long distance in the ancestor of the mallard duck and spot-billed duck (SI Appendix).

We further investigated functions of genes in putative selective sweep regions by surveying published literature and listed several genes might be related to the flight activity. For example, *KDR* associated with lymph vessel development were required for lung inflation at birth and enhance the function of respiratory system (29). Selection of two genes including *DNAJA1* stimulates ATP hydrolysis(30) and *MRPS27* help in protein synthesis within the mitochondrion(31) might contribute to supply high energy for sustained flight. We also noted that *PARK2* were under selection when use a lower threshold ($S=-3$). *PARK2* promotes the autophagic degradation of dysfunctional depolarized mitochondria which might suggest an adaptation to facilitate degradation of damaged mitochondria caused by products of elevated metabolic rate. Previous studies found some genes play important roles in energetic metabolism as a ferrous iron transporter (i.e *SLC39A14*) or positive regulation of fever generation (i.e. *PTGS2*)(32, 33). Positive selection on these genes in the early mallard and spot-billed duck ancestor might enhance their ability to transport oxygen in blood and adapt to cold and hypoxic environments of high altitude during flying. Interestingly, we found some genes involving retina and forebrain regionalization may imply functions related to magnetic navigation ability.

In addition, we examined functional activity of four mutations locating in intron, UTR or downstream of *EFNA5*, *ISL1* and *MRPS27*, respectively. Three of the four sites with wild type showed significantly higher gene expression than domestic genotype in luciferase activity assay (Fig. 3d; Table S22; Fig. S29).

The genomic islands of mallard and spot-billed duck divergence

We estimated that the mallard ducks diverged from the spot-billed ducks about 70 Kys
 using two split models (Fig. 2a-b). This divergent time was consistent with that one
 counted using mtDNA sequences, and even less than the corresponding one (about 340
 Kys~2 Myrs) of two flycatcher species, which were studied extensively in speciation
 divergence(5, 10, 14). Fixed difference (d_f) for 40kb bins, an indication of species
 divergence, were used to detect ‘divergence peak’ regions which show highly elevated
 divergence up to 50 times than genomic median, presented a genomic landscape that is
 sharp contrast to the case of flycatchers, where using the same criterion. Fewer d_f s
 (4,386) with larger proportion (80%) contained in “divergence peak” is distributed in
 smaller genomic regions (0.6% of duck genome) than flycatchers (SI Appendix). This
 difference might due to that the two wild ducks is in an earlier stage of speciation than
 flycatchers(10). Moreover, almost all of these “divergent peaks” were located in
 scaffolds homologous to the chicken “Z” chromosome. This may indicate an analogous
 situation to flycatchers’ speciation (Discussion Section; Fig. S22).

Other genomic parameters (F_{ST} (fixation index), d_{xy} (sequence divergence), π
 (nucleotide diversity) and Tajima’s D) for 40kb bins shown a similar pattern to many
 speciation studies. Observations revealed highly heterogeneous genomic landscapes of
 differentiation when using elevated F_{ST} (4-fold standard deviations away from the mean,
 “differentiation islands”) as an indication of genomic divergence (5, 9, 34-36)(Fig. 4,
 Fig. S23; Table S16), and significant difference between Z chromosomes and
 autosomes, nucleotide diversity and SNP density of differentiation islands were reduced
 (Fig. 5c-f; SI Appendix; Table S17). However, d_{xy} were not elevated and even reduced

in most of differentiation islands, which is consistent with the observation in flycatchers whereas different with the case of Darwin's finches(9, 34). Moreover, Tajima's D of high differentiation were not reduced in both mallard and spot-billed, that is different from observations on other species(9, 34)(Discussion Section; Fig. 5f).

Identification of selective sweep regions in domestic duck

To detect genomic regions under domestic selection, we searched the duck genome for regions with reduced heterozygosity (Hp) and increased fixation index (F_{ST}) using methods similar to the chicken and dog genome (Fig. 3c;SI Appendix)(3, 4). We Z-transformed the Hp and F_{ST} distributions of autosomes and sex-chromosomes separately, and focused on putatively selective sweep regions falling at least four standard deviations away from the mean ($Z(Hp) < -4$ and $Z(F_{ST}) > 4$)(Fig. 3c;Methods;Fig. S24-S25). However, we found only small proportion selective sweep regions were predicted using both the two signals (Fig. S26). This may attributed to long time different evolutionary history since their split (100~300 Kyr). We therefore focused on these 137 regions with extremely low levels of Hp (average length = 63.6230 kb, average Hp = 0.2100 for autosomes, 0.2470 for sex-chromosomes)(SI Appendix). We found that 131 genes and 289 miRNAs contained in 80 of these CDRs, while 47 genes closest to one of 33 CDRs when we extended 180 kb in both upstream and downstream of each these 57 CDRs containing none annotated protein coding genes (Table S18, S20). Further gene ontology analysis indicated that 162 genes in or closest to these 137 CDRs were enriched to 145 biological process categories (p-value <0.05) representing developmental signaling and nervous system, lipid, carbohydrate

and protein metabolic process, cell division and immune response (Table S21). Among them, two genes (*NLGN1* and *PER2*) involve into circadian sleep/wake cycle or is associated with eggs production(37), three genes (*MRAP2*, *MC4R* and *RMII*) are associated with the modulation of feeding behavior or reducing of food intake in response to dietary excess, and four genes (*HSD17B4*, *HEXB*, *B4GALT1* and *CYP19A1*) are related to sexual characteristics, male courtship behavior or androgen metabolic process (38, 39)(Table S18-S21).

Among these candidate domestication genes, we examined functional activity of two mutations locating in introns of *GRIK1* and *LYST*, respectively. Domestic genotype of *LYST* mutation showed significantly higher gene expression acidity than wild genotype in luciferase activity assay (Fig. 3d; Table S21)

Reanalysis provide further evidence on evolutionary relationship of domestic duck, mallard wild duck and spot-billed wild duck

As the genomic average genetic distance between mallard and spot-billed (mean F_{ST} : 0.0637) is significantly lower than it in mallard/spot-billed vs domestic pairs (mean F_{ST} : 0.2832, 0.2498)(Fig. 4b). To further investigate whether this higher genomic "similarity" between mallard and spot-billed is caused by hybridization or phylogeny as we inferred above. We stratified the genomic divergence values of each windows by ΔDAF (or absolute ΔDAF) (derived allele frequency difference between the two populations) bins. According to recent research on introgression issue in chimpanzee, the chance of derive alleles being introduced through gene flow increases with the frequency in the donor population. Gene flow should introduce into receptor populations at low frequencies

(40). So, we may expect lower divergence value in these higher Δ DAF (absolute Δ DAF) bins as it represented more probability of introgression. We did find decreased genetic distance (F_{ST} , d_{xy}) between mallard/spot-billed and domestic ducks in higher Δ DAF bins (Fig. S30b). However, both F_{ST} and d_{xy} between mallard and spot-billed is increased with the Δ DAF (Fig. S30a). This observation suggested that this higher genomic “similarity” between the two wild duck species maybe not caused by introgression. Furthermore, we found that the mean Δ DAF value didn’t show significant difference between high differentiation regions and genomic background in the comparison of mallard and spot-billed genomes, which is consistent with recent research that gene flow is not the major factor in the formation of the genomic islands (Table S23) (34).

To further deciphering the introgression effect and It’s influence on evolutionary relationship inference of domestic and the two wild ducks, we performed ABBA-BABA test (assume that mallard, spot-billed, domestic duck is P1, P2, P3 respectively) using Muscovy duck as outgroup. It also suggested that the genetic relationship between mallard and spot-billed ducks are closer than it between the two wild species and domestic(Fig. S31a). We also found evidence of gene flow between domestic and wild ducks which is consistent with above analysis. (Fig. 1d, D-statistics: -0.2972; jackknife std: 0.4189). In order to characterized the gene flow effect on the phylogenetic tree reconstruction, we used the same idea, Δ DAF, to test is there evidence that these derived allele sharing is caused by introgression (Fig. S31a). We stratified the number of BBAA (ABBA, BABA) sites by the Δ DAF bins between mallard and spot-billed ducks

populations (spot-billed and domestic, mallard and domestic, respectively). The most majority proportion of BBAA sites were distributed in lower ΔDAF bins ($-0.4 \sim 0.4$), imply the major reason for excess derived allele sharing due to phylogeny rather than introgression (Fig. S31b left). As the majority number of SNPs distributed at lower ΔDAF bins. We further measured the proportion distribution of the BBAA sites among all SNPs (i.e. polymorphic in the compared two populations. i.e. Polymorphic sites in spot-billed and domestic, mallard and domestic, mallard and spot-billed for ABBA, BABA, BBAA, respectively). If the introgression exists between mallard and spot-billed (mallard and domestic, spot-billed and domestic), we may expect the proportion of BBAA sites would increase with the ΔDAF (or absolute ΔDAF in negative). We did find that the proportion of BABA, ABBA sites increased with ΔDAF , which indicate that the derived allele sharing between wild (mallard or spot-billed) and domestic populations is partly due to the introgression. Whereas the distribution of BBAA proportion across ΔDAF bins showed a different pattern that the BBAA rate has only little difference across ΔDAF bins, which indicate that absent evidence of introgression between mallard and spot-billed (Fig. S31b).

Discussion

How domestic animals originate and lead to striking morphological and behavioral changes are hot topics of evolutionary biology. Recently, the rapid development of high-throughput sequencing technologies has significantly boosted knowledges for origination and domestic process in dog, pig and rabbit, as well as chicken (6, 7, 25, 41, 42). Here we demonstrate the domestication process of duck and surprisingly found

that domestic ducks were neither originated from the mallard ducks nor tamed from descents of mallard and spot-billed ducks. Instead, compelling evidence indicate that ducks were domesticated from an ancient wild duck lineage, which was close to the ancestor of mallard and spot-billed ducks. And we also found evidence shows that, compare to domestic duck, the ancestor of wild duck might acquire enhanced flight ability through early selection. It may contribute to their long-distance migration ability and the preservation of *Ne* through the LGP climate changes which may leading to the decrease of domestic lineage. In the context that accumulated studies have revealed that extant wild animal may not be the wild progenitors of its domestic forms (dog, horse, pig)(6, 43-45). We propose that our conclusion may help to explain the extinction (or hardly to find) of those progenitors and highlight the importance of the genetic architecture of wild for the understanding of remarkable phenotype changes in domestication (18, 25, 43, 46, 47). Meanwhile, we found that duck domestication was accompanied by selection at genes affecting neuronal activities (e.g. *SH3GL2* (42)), lekking behavior (e.g. *HEXB*, *B4GALT1* and *HSD17B4*). Moreover, Genes associated with lipid, carbohydrate, protein metabolic process and food intake control were under selection (e.g. *RMII*, *MRAP2* and *MC4R* (8)). In summary, an adaption to food-rich and artificial feeding environment with development of agriculture may contribute to duck domestication.

The rapid speciation in mallard and spot-billed, which species may occur in only 70 kyr, provide excellent resource to speciation studies(34, 48, 49). As recent studies of this area have focused on high differentiation islands, our observation also shown the

348 obvious extraordinary features of high differentiation regions. Think about the dynamic
349 changes of these islands across speciation process may help to understand the drive
350 process of speciation and the evolves of genomic differentiation landscape. In our
351 observations, all “divergent peaks” were located in scaffolds homologous to ‘Z’ chicken
352 chromosome (Fig. S22). Studies on flycatchers suggest that both species-specific male
353 plumage traits and species recognition are located on the Z chromosome(50, 51).
354 Similar with flycatchers, male plumage traits of mallard and spot-billed duck are quite
355 different. This may indicate that “divergent peaks” on Z chromosome is associated to
356 the difference in male-plumage traits between two wild duck species and their
357 speciation. Although it’s still unclear what trigger the speciation process, we propose
358 that those limited few genes contained in these high “differentiation islands” and
359 “divergent peaks” may provide critical clue due to they may in a very early speciation
360 stage. Linked selection is suggested to dominantly drive the evolution of the genomic
361 differentiation rather than gene flow(9). Not elevated dxy of “differentiation islands” in
362 observations again, support this suggestion (Fig. S23). Furthermore, Tajima’s D is not
363 reduced in differentiation islands could also be best explained by new strong positive
364 selection (Fig. 4). Tajima’s D is used to detect selection of new mutations and correlated
365 to difference between π and the expectation of π which is related to number of SNPs
366 (52). Due to short divergence time, the mallard and spot-billed ducks did not deposit
367 much new mutations to increase (θ_w) after selection (new mutations increase diversity
368 slowly and in a low frequency, mainly increase population mutation rate (θ_w) rather
369 than nucleotide diversity (π)), thus results in allele frequency spectra didn’t skew

towards rare alleles. Whereas collared and pied flycatcher (about 340 Kyr~2 Myr) and Darwin's finches reduce Tajima's D in their genomic divergent islands due to a long time after initial selection may associated with the driving force of speciation (10, 34). In the end, we argue that our article has presented distinct genetic resource between wild and domestic duck. Genetic resource of wild duck may have been significantly shaped through the global climate change during LGP, which making wild duck evolved new property to adaption to environment change, also consist of the stability of the ecology system. We also note evidence in genetic structure showed that mallard duck and spot-billed duck (especially spot-billed duck) have interspecific assignment to domestic ducks (Fig. 1d; Fig. S15. Especially when K=3,4). This may pose a significant threat to genetic diversity of wild stocks.

Methods

Sequence assembly and annotation

Sequencing was carried out with Illumina technology using genomic DNA of a wild female mallard and a wild female spot-billed collected by Aoji Duck Farm (Ningbo, Zhejiang) with hunting permission granted by the local government. Sequences from paired-end reads of 17 libraries for each individual with insert size ranging from 250 bp to 20,000 bp were assembled with SOAPdenovo and SSPACE software (Table S1). The super-scaffolds were positioned along the chromosomes based on the duck cytogenetic map and the comparative genomic map between duck and chicken (11, 53). A reference gene set for the mallard genome and the spot-billed genome were retrieved through merging the homologous set, the transcript set (predicted with 21 duck

transcriptomes) and the *de novo* set(11).

***daði* simulation and PSMC analysis**

For *daði* simulation analysis, SNPs were sampled randomly each per 100 kb from the SNPs identified in wild populations (mallard and spot-billed) and domestic populations (Cherry Valley, Campbell and 6 Chinese duck breeds). SNPs were further filtered according to the coverage of resequencing data at their position with thresholds of 20-fold for pooled data and total number of genotyped alleles more than 50, 36, 26, 28 for Beijing, Shaoxing, mallard duck and spot-billed duck populations, respectively. We then inferred ancestral alleles of the above sampling filtered SNP using Muscovy ducks as outgroups using the following thresholds: (1) more than 30-fold coverage two outgroups; (2) the allele was fixed and same in outgroups. After that, we quantized the frequency of each SNP of 8 domestic duck populations into one population and simulated the divergence among spot-billed, mallard and “domestic” using split without migration, split with migration and IM models 50 bootstraps each (Fig. 3a; SI Appendix). *Nref* was calculated with the following format:

$$Nref = \frac{\text{Watterson's estimator } \Theta}{4 * \text{mutation rate} * L};$$
 L was length of the SNPs from and diluted by rate of SNP sampling.

For the PSMC analysis(54, 55), SNPs were filtered according to the coverage of resequencing data at their position with thresholds of 10- to 100-fold for Beijing, spot-billed and mallard, or 5- to 80-fold for Shaoxing. Effective population sizes were assessed by PSMC model analysis. We assumed that mutation rate per year to be 9.97e-10 and that the generation time to be one year for both *daði* simulation and PSMC

analysis(56).

Positive selection in early ancestor of mallard and spot-billed ducks

To detect selection that occurred in wild duck lineage (mallard and spot-billed ducks) after their divergence with domestic lineage, we selected SNPs according to the inferred ancestral alleles and using strict thresholds (28). We then calculated the derived allele frequency of 18,083,967 selected SNPs in both wild lineage and domestic lineage, and defined selective sweep regions of the early mallard duck and spot-billed duck ancestors using the thresholds of -0.2819 (4-fold standard deviations away from mean) for autosomes and -0.3089 for sex-chromosome (1.5-fold standard deviations away from mean), respectively.

As expected, the derived allele frequencies in both lineages are highly correlated (Fig. S18). This correlation was used to calculate the expected derived allele frequency in the domestic lineage as a function of the derived allele frequency in wild lineage. We then calculated the observed derived allele frequency in domestic lineage, and the log-transformed value of the ratio for the observed/expected derived allele frequency in sliding 20-kb windows with a step size of 10kb. After Z-transform the genome-wide distributions to arrive at a measure S. As mentioned previous, this method naturally disregards regions of recent selection or strong purifying selection in wild lineage. However, recent selective sweeps regions in domestic lineage could also reduce S value, thus we exclude regions that show lower zHp (CDRs) in domestic lineage.

Availability of supporting data and materials

Data availability

The genome assembly has been deposited in GenBank PRJNA392350. The duck, Taihu goose and Muscovy duck genome resequencing data under the bioproject PRJNA315043.

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Author contributions

Y.H.H and N.L. designed the project. Z.L.L, H.F.L., L.C., H.B.C., W.T.S. and E.G.R collected and purified DNA samples. Q.W.L., Y.Z, G.Y.F., H.F.L., Y.S.S., W.B.C., X.L. and X.X. performed genome assembly, gene annotation and gene family evolution of the spot-billed and mallard genome. R.L performed phylogenetic evolution, selective sweep and speciation analysis. E.G.R. and X.X.W. carried out luciferase activity analysis. Y.H.H., R.L. and Q.W.L. wrote the manuscript. Y.H.H. and R.L. revised the manuscript.

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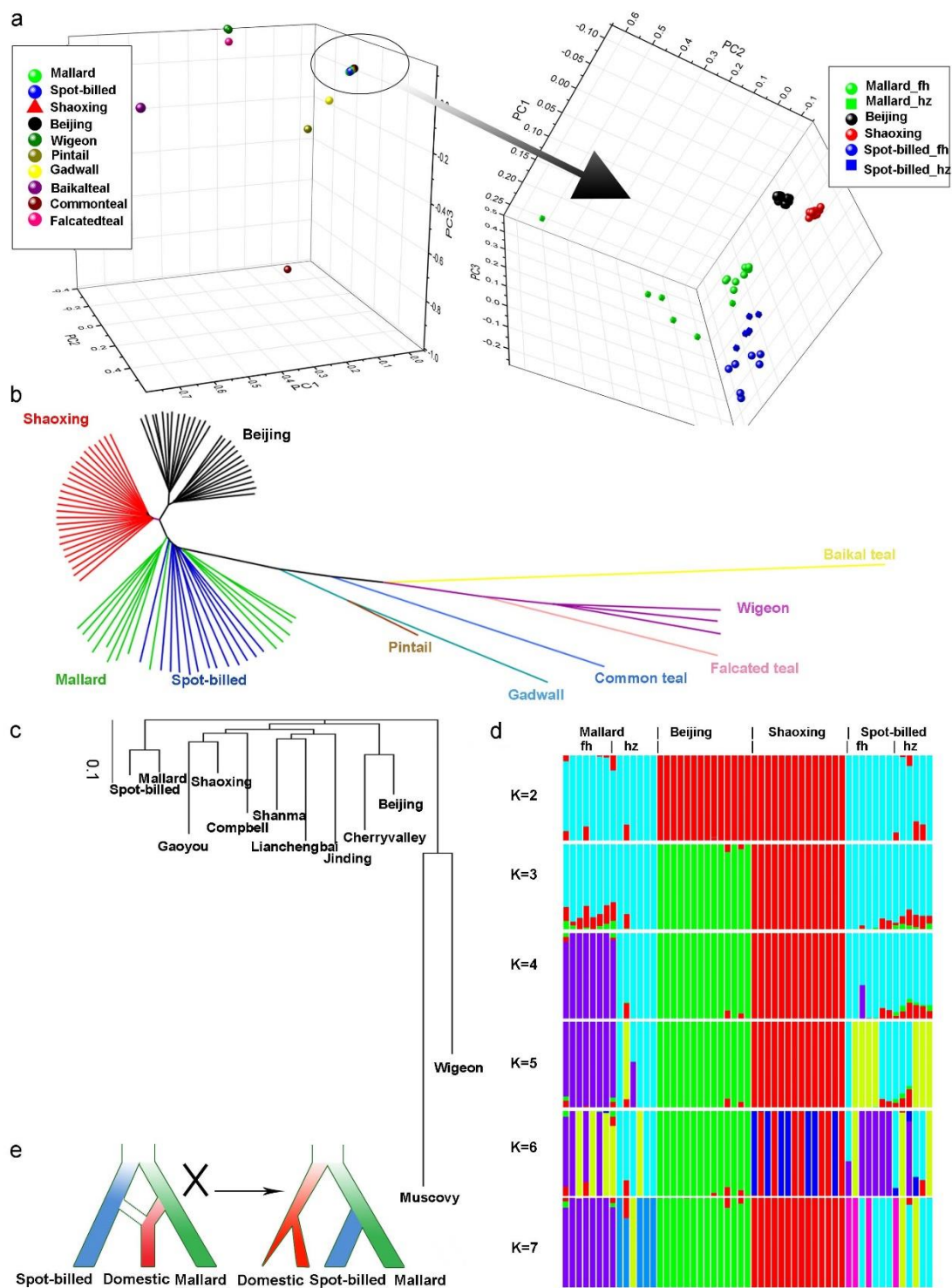
Reference

1. Seehausen O, *et al.* (2014) Genomics and the origin of species. *Nat Rev Genet* 15(3):176-192.
2. Andersson L (2001) Genetic dissection of phenotypic diversity in farm animals. *Nat Rev Genet* 2(2):130-138.
3. Axelsson E, *et al.* (2013) The genomic signature of dog domestication reveals adaptation to a starch-rich diet. *Nature* 495(7441):360-364.
4. Rubin CJ, *et al.* (2010) Whole-genome resequencing reveals loci under selection during chicken domestication. *Nature* 464(7288):587-591.
5. Ellegren H, *et al.* (2012) The genomic landscape of species divergence in *Ficedula* flycatchers. *Nature* 491(7426):756-760.
6. Frantz LA, *et al.* (2015) Evidence of long-term gene flow and selection during domestication from analyses of Eurasian wild and domestic pig genomes. *Nat Genet* 47(10):1141-1148.
7. Carneiro M, *et al.* (2014) Rabbit genome analysis reveals a polygenic basis for phenotypic change during domestication. *Science* 345(6200):1074-1079.
8. Rubin CJ, *et al.* (2012) Strong signatures of selection in the domestic pig genome. *Proc Natl Acad Sci U S A* 109(48):19529-19536.
9. Burri R, *et al.* (2015) Linked selection and recombination rate variation drive the evolution of the genomic landscape of differentiation across the speciation continuum of *Ficedula* flycatchers. *Genome Res* 25(11):1656-1665.
10. Nadachowska-Brzyska K, *et al.* (2013) Demographic divergence history of pied flycatcher and collared flycatcher inferred from whole-genome re-sequencing data. *PLoS Genet* 9(11):e1003942.
11. Huang Y, *et al.* (2013) The duck genome and transcriptome provide insight into an avian influenza virus reservoir species. *Nature genetics* 45(7):776-783.
12. Trump (2017) <http://afowlshome.com/>.
13. Lavretsky P, McCracken KG, & Peters JL (2014) Phylogenetics of a recent radiation in the mallards and allies (Aves: Anas): inferences from a genomic transect and the multispecies coalescent. *Mol Phylogenet Evol* 70:402-411.
14. Zhuravlev YN & Kulikova IV (2014) Waterfowl Population Structure: Phylogeographic Inference. *Achievements in the Life Sciences* 8(2):123-127.
15. Madge S & Burn H (1988) *Waterfowl: an identification guide to the ducks, geese, and swans of the world* (Houghton Mifflin).
16. Richards J (2017) www.waterfowl.org.uk. (<http://www.waterfowl.org.uk/index.html>).
17. Albarella U (2005) *Alternate fortunes? The role of domestic ducks and geese from Roman to Medieval times in Britain. Documenta Archaeobiologiae III. Feathers, Grit and Symbolism* (ed. by G.Grupe & J.Peters), 249-58.
18. Larson G & Fuller DQ (2014) The Evolution of Animal Domestication. *Annu Rev Ecol Evol S* 45:115-136.

- 499 19. Stern DL (2000) Evolutionary developmental biology and the problem of variation. *Evolution*
500 54(4):1079-1091.
- 501 20. Steve M (1992) Waterfowl: an identification guide to the ducks, geese and swans of the world.
502 *Houghton mifflin in harcourt*:3.
- 503 21. Reich D, Thangaraj K, Patterson N, Price AL, & Singh L (2009) Reconstructing Indian population
504 history. *Nature* 461(7263):489-494.
- 505 22. Rodgers JL (1999) The Bootstrap, the Jackknife, and the Randomization Test: A Sampling
506 Taxonomy. *Multivariate Behav Res* 34(4):441-456.
- 507 23. Alexander DH, Novembre J, & Lange K (2009) Fast model-based estimation of ancestry in
508 unrelated individuals. *Genome Res* 19(9):1655-1664.
- 509 24. Gutenkunst RN, Hernandez RD, Williamson SH, & Bustamante CD (2009) Inferring the joint
510 demographic history of multiple populations from multidimensional SNP frequency data. *PLoS*
511 *Genet* 5(10):e1000695.
- 512 25. Freedman AH, *et al.* (2014) Genome sequencing highlights the dynamic early history of dogs.
513 *PLoS Genet* 10(1):e1004016.
- 514 26. Wang GD, *et al.* (2016) Out of southern East Asia: the natural history of domestic dogs across
515 the world. *Cell Res* 26(1):21-33.
- 516 27. Groenen MA, *et al.* (2012) Analyses of pig genomes provide insight into porcine demography
517 and evolution. *Nature* 491(7424):393-398.
- 518 28. Green RE, *et al.* (2010) A draft sequence of the Neandertal genome. *Science* 328(5979):710-
519 722.
- 520 29. Jakus Z, *et al.* (2014) Lymphatic function is required prenatally for lung inflation at birth. *J Exp*
521 *Med* 211(5):815-826.
- 522 30. Baaklini I, *et al.* (2012) The DNAJA2 substrate release mechanism is essential for chaperone-
523 mediated folding. *J Biol Chem* 287(50):41939-41954.
- 524 31. Davies SM, *et al.* (2012) MRPS27 is a pentatricopeptide repeat domain protein required for the
525 translation of mitochondrially encoded proteins. *FEBS letters* 586(20):3555-3561.
- 526 32. McCarthy RC & Kosman DJ (2015) Iron transport across the blood–brain barrier: development,
527 neurovascular regulation and cerebral amyloid angiopathy. *Cellular and molecular life sciences*
528 72(4):709-727.
- 529 33. Liuzzi JP, Aydemir F, Nam H, Knutson MD, & Cousins RJ (2006) Zip14 (Slc39a14) mediates non-
530 transferrin-bound iron uptake into cells. *Proc Natl Acad Sci U S A* 103(37):13612-13617.
- 531 34. Han F, *et al.* (2017) Gene flow, ancient polymorphism, and ecological adaptation shape the
532 genomic landscape of divergence among Darwin's finches. *Genome Res* 27(6):1004-1015.
- 533 35. Martin SH, *et al.* (2013) Genome-wide evidence for speciation with gene flow in *Heliconius*
534 butterflies. *Genome Res* 23(11):1817-1828.
- 535 36. Lavretsky P, *et al.* (2015) Speciation genomics and a role for the Z chromosome in the early
536 stages of divergence between Mexican ducks and mallards. *Mol Ecol* 24(21):5364-5378.
- 537 37. Nakao N, *et al.* (2007) Circadian clock gene regulation of steroidogenic acute regulatory protein
538 gene expression in preovulatory ovarian follicles. *Endocrinology* 148(7):3031-3038.
- 539 38. Does RM & Garcia Y (2015) Views on the co-evolution of the melanocortin-2 receptor, MRAPs,
540 and the hypothalamus/pituitary/adrenal-interrenal axis. *Mol Cell Endocrinol* 408:12-22.
- 541 39. Sebag JA, Zhang C, Hinkle PM, Bradshaw AM, & Cone RD (2013) Developmental control of the
542 melanocortin-4 receptor by MRAP2 proteins in zebrafish. *Science* 341(6143):278-281.

- 543 40. de Manuel M, *et al.* (2016) Chimpanzee genomic diversity reveals ancient admixture with
544 bonobos. *Science* 354(6311):477-481.
- 545 41. Xiang H, *et al.* (2014) Early Holocene chicken domestication in northern China. *Proc Natl Acad*
546 *Sci U S A* 111(49):17564-17569.
- 547 42. Wang GD, *et al.* (2013) The genomics of selection in dogs and the parallel evolution between
548 dogs and humans. *Nat Commun* 4:1860.
- 549 43. Warmuth V, *et al.* (2012) Reconstructing the origin and spread of horse domestication in the
550 Eurasian steppe. *Proc Natl Acad Sci U S A* 109(21):8202-8206.
- 551 44. Botigue LR, *et al.* (2017) Ancient European dog genomes reveal continuity since the Early
552 Neolithic. *Nat Commun* 8:16082.
- 553 45. Frantz LA, *et al.* (2016) Genomic and archaeological evidence suggest a dual origin of domestic
554 dogs. *Science* 352(6290):1228-1231.
- 555 46. Ni Leathlobhair M, *et al.* (2018) The evolutionary history of dogs in the Americas. *Science*
556 361(6397):81-85.
- 557 47. Almathen F, *et al.* (2016) Ancient and modern DNA reveal dynamics of domestication and
558 cross-continental dispersal of the dromedary. *Proc Natl Acad Sci U S A* 113(24):6707-6712.
- 559 48. Harr B (2006) Genomic islands of differentiation between house mouse subspecies. *Genome*
560 *Research* 16(6):730-737.
- 561 49. Cruickshank TE & Hahn MW (2014) Reanalysis suggests that genomic islands of speciation are
562 due to reduced diversity, not reduced gene flow. *Molecular Ecology* 23(13):3133-3157.
- 563 50. Saetre GP, *et al.* (2003) Sex chromosome evolution and speciation in Ficedula flycatchers. *Proc*
564 *Biol Sci* 270(1510):53-59.
- 565 51. Saether SA, *et al.* (2007) Sex chromosome-linked species recognition and evolution of
566 reproductive isolation in flycatchers. *Science* 318(5847):95-97.
- 567 52. Misawa K & Tajima F (1997) Estimation of the amount of DNA polymorphism when the neutral
568 mutation rate varies among sites. *Genetics* 147(4):1959-1964.
- 569 53. Huang Y, *et al.* (2006) A genetic and cytogenetic map for the duck (*Anas platyrhynchos*).
570 *Genetics* 173(1):287-296.
- 571 54. Lohmueller KE, Bustamante CD, & Clark AG (2010) The effect of recent admixture on inference
572 of ancient human population history. *Genetics* 185(2):611-622.
- 573 55. Li H & Durbin R (2011) Inference of human population history from individual whole-genome
574 sequences. *Nature* 475(7357):493-496.
- 575 56. Nadachowska-Brzyska K, Li C, Smeds L, Zhang G, & Ellegren H (2015) Temporal Dynamics of
576 Avian Populations during Pleistocene Revealed by Whole-Genome Sequences. *Curr Biol*
577 25(10):1375-1380.

578



580
581 **Figure 1: Inference of phylogenetic topology in wild and domestic ducks.** Beijing
582 and Shaoxing ducks representing domestic ducks were used the Principal Component

583 Analysis (PCA), phylogenetic analysis based on identity-by-state (IBS) and genetic
584 structure analyses. “_fh” and “_hz” mean spot-billed and mallard collected from
585 Fenghua and Hangzhou respectively. (a) PCA plot constructed for Principal
586 Component 1 (PC1), PC2 and PC3 using 10,525 high-quality filtered SNPs from 27
587 Beijing and 27 Shaoxing ducks and 35 wild ducks (one bailkal teal, one gadwall, one
588 pintail one falcated teal, common teal, three wigeon, 13 spot-billeds and 14 mallards)
589 shows that domestic ducks are close to spot-billed and mallard wild ducks. (b)
590 Neighbor-joining tree based on IBS distance of 80,505,576 SNPs. (c) Neighbor-joining
591 tree based on genome wide pairwise fixation index (F_{ST}) of 44,852,612 SNP. (d)
592 Genetic structure analyses of 13 spot-billed and 14 mallard wild ducks, together with
593 14 Beijing and 14 Shaoxing domestic ducks, using admixture ranging K from two to
594 seven. (e) The inferred demographic models for wild and domestic ducks inferred based
595 on genetic analysis of **a-d** (right). The traditional demographic model, hypothesized
596 that domestic ducks were originated from mallard or the descendant of a mixture of
597 spot-billed and mallard (left) are incompatible with observations in neighbor-joining
598 trees (**b-c**) and genetic data(**d**).

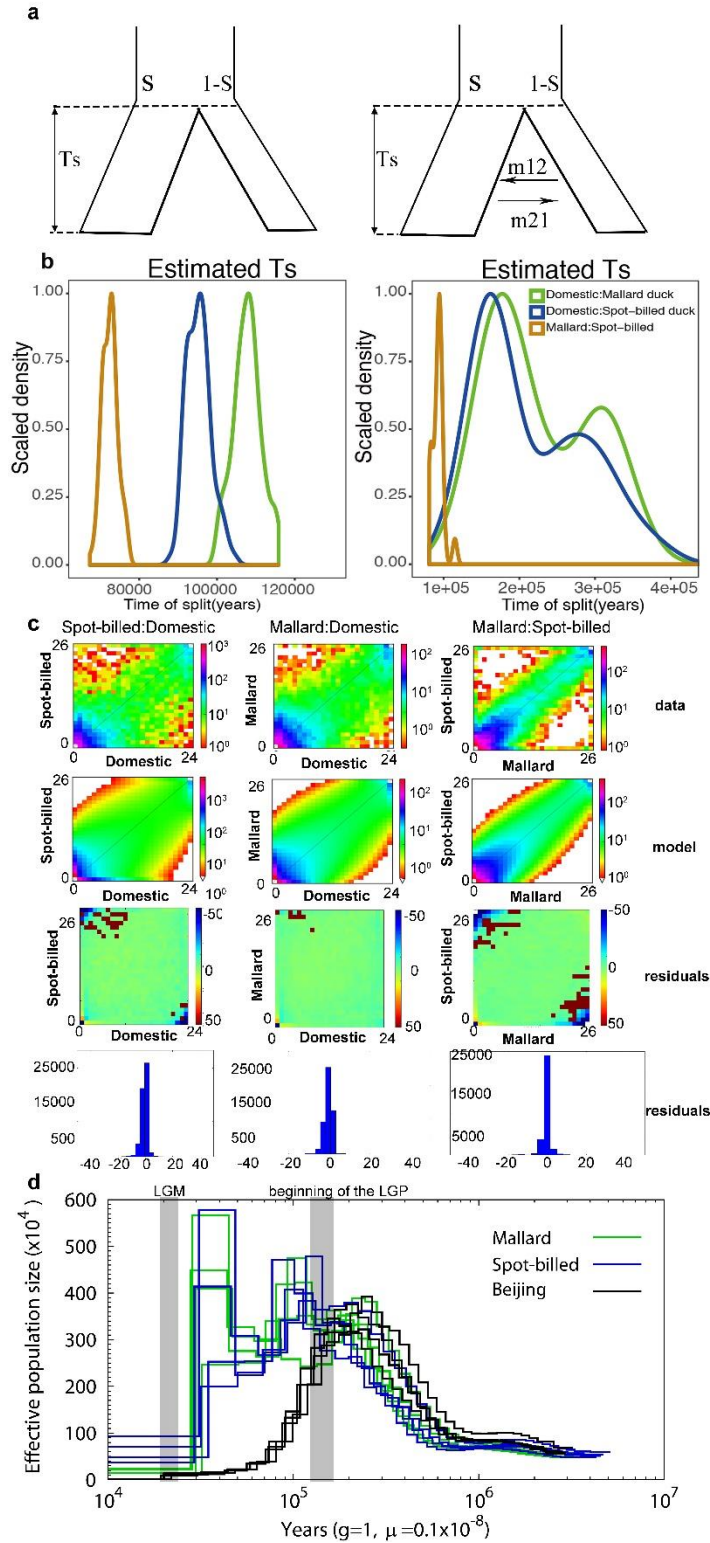


Figure 2: Inference of joint demographic history of ducks. Wild ducks included spot-billed and mallard ducks. Domestic ducks included Beijing, Cherryvalley, Compbel, Gaoyou, Jingding, Lianchengbai, Shaoxing and Shanma. (a) Illustration of

603 the two models (with/without migration) used in the history inference through the $\partial a \partial i$
604 analysis. Parameters corresponding to the code that were provided in supporting
605 information. **(b)** Distribution of estimate T_s (divergence time) of each model in 100
606 simulations between spot-billed and mallard (orange), mallard and domestic lineage
607 (green), spot-billed and domestic lineage (blue), respectively. **(c)** Joint Site Frequency
608 Spectrum (SFS) for observed data, simulation data and residuals. **(d)** Demographic
609 history inference of mallard duck, spot-billed duck and Beijing duck through the PSMC
610 analysis.

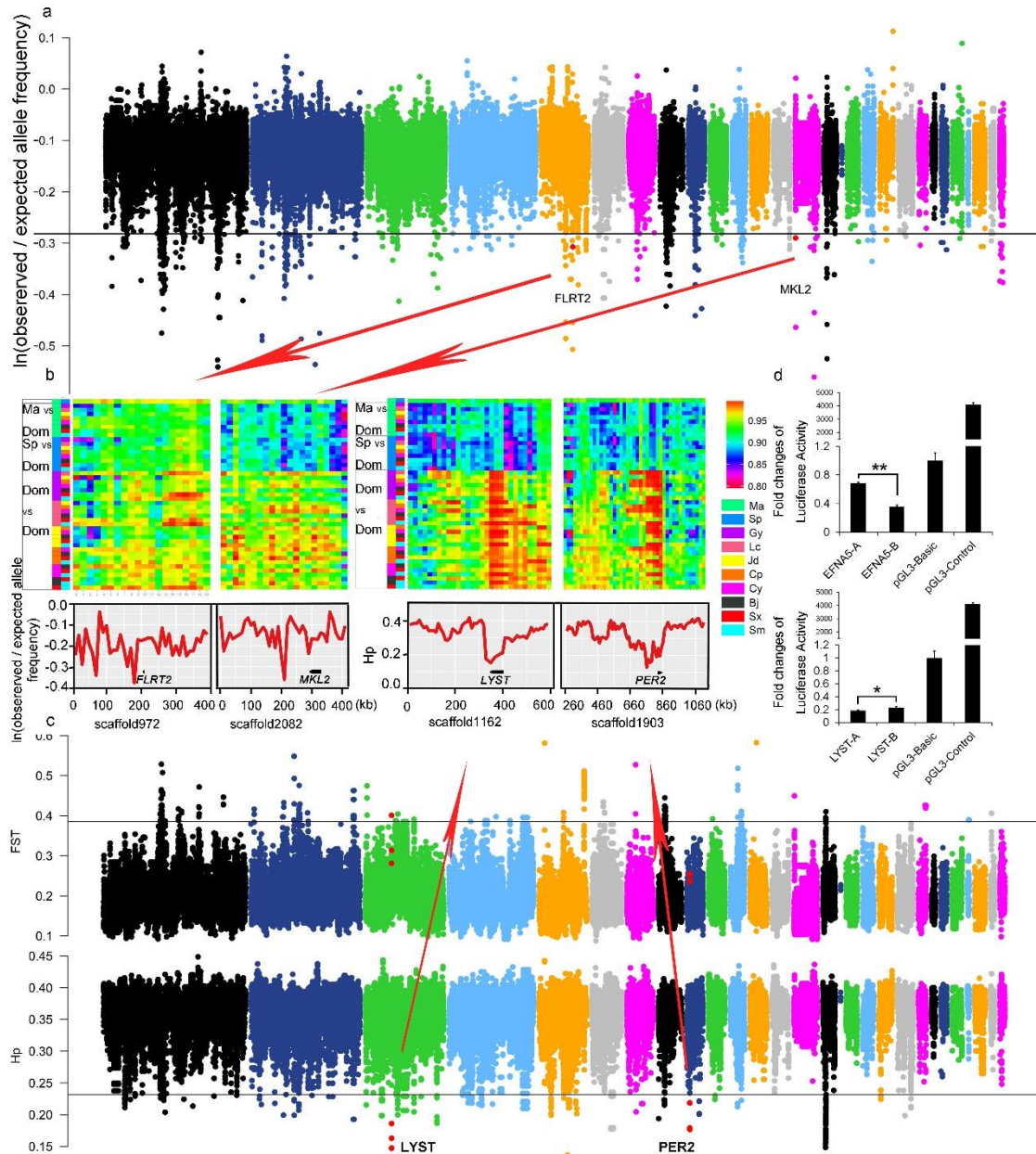


Figure 3: Summary of selective sweep analyses of the early wild duck lineage and domestic duck. (a) Distribution of $\ln(\text{observed}/\text{expected derived allele frequency})$ that is calculated in 20 kb window with a sliding size of 10 kb along scaffolds that is arranged according to their alignment to the chicken autosomes. The horizontal black line indicates the threshold at $\ln(\text{observed}/\text{expected derived allele frequency}) = -0.3136$ (mean of $\ln$ values - 4 standard deviations). Outlier regions that were detail visualized

618 by identity scores (IS) and re-computing values in more smaller window size in **(b)**
 619 were highlight with red dots. **(b)** Details of four putative selective sweep regions around
 620 the *FLRT2*, *MKL2*, *LYST* and *PER2* genes. Upside of each panel shown degree of
 621 haplotype sharing in pairwise comparisons among populations. Color boxes (left)
 622 indicate the comparison performed on that row. “Ma”, “Sp”, “Dom”, represent mallard,
 623 spot-billed and domestic duck respectively. Y-axis (downside) of *FLRT2*, *MKL2*, which
 624 is associated with retina/heart morphogenesis that were under selection in wild ancestor,
 625 are $\ln(\text{observed/expected derived allele frequency})$, whereas y-axis of *LYST*, *PER2* that
 626 were under selection in domestic duck are h_p . **(c)** Distribution of F_{ST} between the wild
 627 and the domestic duck, and h_p in the domestic duck that is calculated in 40 kb with a
 628 sliding size 20 kb along scaffolds that is arranged according their alignment to the
 629 chicken autosomes. The horizontal black line indicates the threshold at $F_{ST} = 0.3858$ or
 630 $h_p = 0.2319$ (mean of $\ln$ values - 4 standard deviations). Red dots indicate the location
 631 of outlier genes (4 standard deviations away from h_p means) that were detail visualized
 632 in **(b)**. **(d)** Comparison in luciferase activity of major alleles of *MRPS27* and *EFNA5*
 633 in wild (spot-billed and mallard) and domestic (Beijing) duck in DF1 (chicken
 634 embryonic fibroblast) cells. “A” and “B” represent wild duck and domestic duck,
 635 respectively. “*” $P \leq 0.05$.

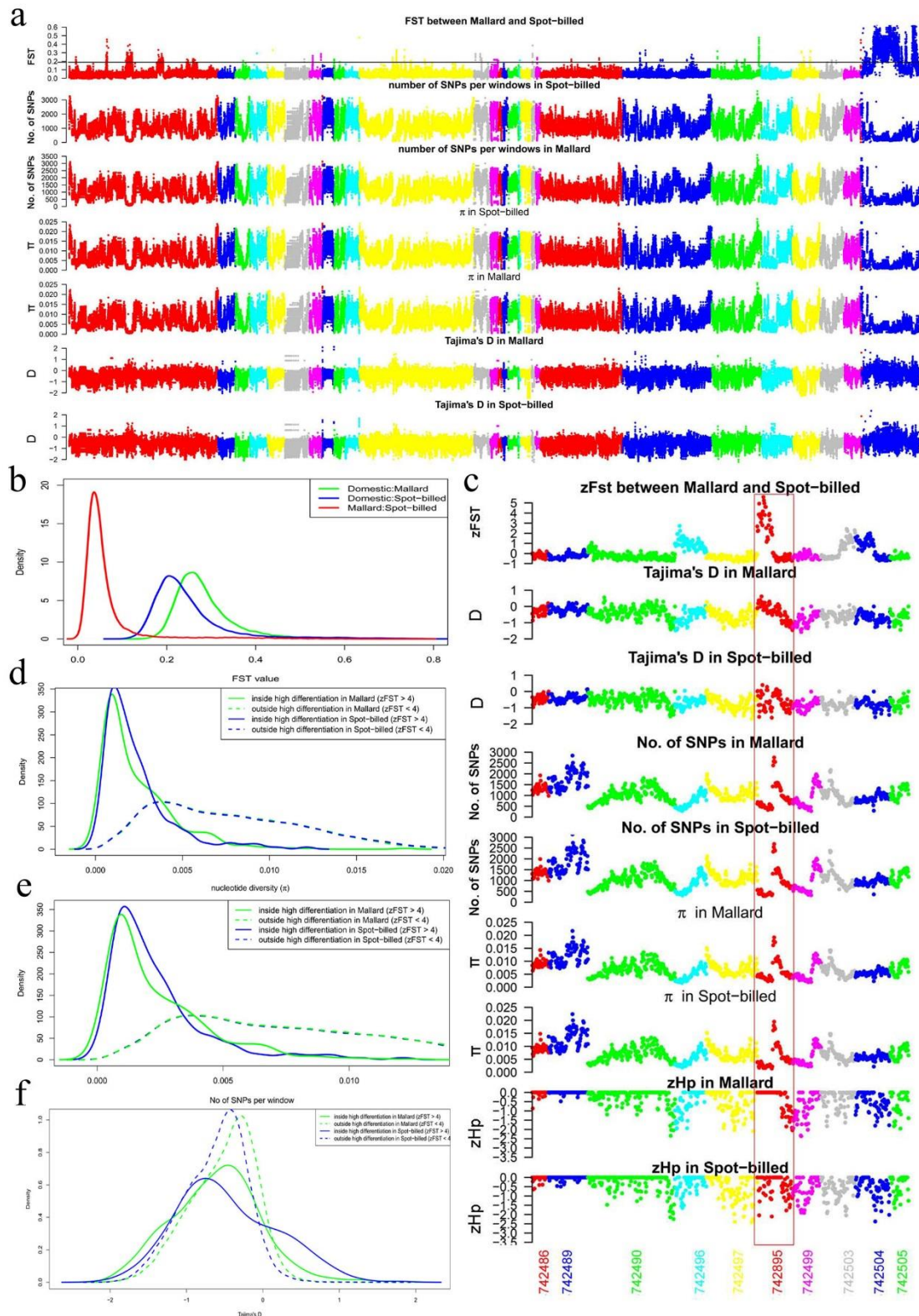


Figure 4: The genomic landscape of species divergence between spot-billed and mallard ducks. (a) Distribution of population genomic parameters of the Beijing duck assembly, in 40 kb windows sliding 20 kb steps. The blue scaffolds in the right were

640 referred as duck Z scaffolds. The black horizontal line indicates the threshold at $F_{ST} =$
641 0.1888 (mean of $F_{ST} + 4$ standard deviations, referred it as $Z(F_{ST}) = 4$). The plots show
642 the F_{ST} between spot-billed and mallard, number of SNPs per window, nucleotide
643 diversity (π), Tajima's D for spot-billed and mallard respectively. **(b)** Distribution of
644 pairwise F_{ST} for 40 kb windows sliding 20 kb steps. The red line shows the plot in the
645 first row of **(a)**. The blue and green line show the pairwise F_{ST} between spot-
646 billed/mallard and domestic ducks (combination of 8 domestic breeds). **(c)** Distribution
647 of population genomic parameters along ten scaffolds. **(d-f)** Distribution of nucleotide
648 diversity (π), number of SNPs per window and Tajima's D inside/outside high
649 differentiation regions in Beijing duck assembly.

Table 1. Comparison in features of the mallard, spot-billed and Beijing duck genomes

Genomic features	Mallard	Spot-billed	Beijing duck
Assembled genome size (Gb)	1.27	1.31	1.11
Contig N50 (bp)	38,271	33,061	26,114
Scaffold N50 (bp)	2,489,142	2,054,876	1,233,631
GC content (%)	41.90	41.90	40.90
Repeat rate (%)	10.71	10.55	7.31
Heterozygosity rate ($\times 10^{-3}$)	3.60	3.54	2.40
Predicted protein-coding genes	21,056	21,123	20,629

Table 2. Top 32 enriched biological process terms among genes under positive selection in the early spot-billed and mallard ancestors

Gene ontology term	P-value	Number	Size
positive regulation of tyrosine phosphorylation of Stat3 protein	6.29E-05	4	12
regulation of activin receptor signaling pathway	0.000382	2	2
positive regulation of cytosolic calcium ion concentration involved in phospholipase C-activating G-protein coupled signaling pathway	0.000387	3	8
detection of virus	0.001131	2	3
D-aspartate import	0.001131	2	3
metanephros morphogenesis	0.002232	2	4
L-glutamate import	0.002232	2	4
negative regulation of extrinsic apoptotic signaling pathway in absence of ligand	0.003441	3	16
positive regulation of granulocyte macrophage colony-stimulating factor production	0.003672	2	5
respiratory system development	0.003672	2	5
determination of left/right symmetry	0.003701	5	53
spinal cord motor neuron differentiation	0.004119	3	17
lung morphogenesis	0.004119	3	17
ionotropic glutamate receptor signaling pathway	0.004872	3	18
ventricular cardiac muscle tissue morphogenesis	0.004872	3	18
dorsal/ventral pattern formation	0.005176	4	36
negative regulation of peptidyl-threonine phosphorylation	0.005437	2	6
serotonin metabolic process	0.005437	2	6
lung epithelium development	0.005437	2	6
cytokine-mediated signaling pathway	0.006291	4	38
nucleotide-excision repair	0.006613	3	20
neutrophil chemotaxis	0.006613	3	20
negative regulation of epithelial cell proliferation	0.006904	4	39
positive regulation of heart rate	0.007514	2	7
inositol phosphate-mediated signaling	0.007514	2	7
bud elongation involved in lung branching	0.007514	2	7
limb morphogenesis	0.009839	3	23
long-chain fatty acid metabolic process	0.00989	2	8
dicarboxylic acid transport	0.00989	2	8
interleukin-6-mediated signaling pathway	0.00989	2	8
thrombin receptor signaling pathway	0.00989	2	8

response to wounding	0.012416	3	25
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654 Full lists of enriched terms are presented in Table S15. Terms that considered to be associated with
655 enhancement of long range flight ability were in bold. Size represents total number genes of this term.
656

Supporting information for: Genomic analyses reveal distinct evolutionary pattern of ducks and identify genetic changes underlying domestication and speciation (SI Appendix)

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1. Genome sequencing and assembly

We generated 262.45 Gb (~209-fold) pair-end reads of a female mallard duck and 275.20 Gb pair-end reads (~220-fold) of a female spot-billed duck, respectively (Table S1, Fig. S2-S3). Using methods similar to those applied to the Beijing duck genome (1), we generated a high-quality mallard assembly consisting of 61,591 scaffolds and covering 1.27 Gb, and estimated this assembly having a heterozygosity rate 3.61×10^{-3} higher than the corresponding in Beijing duck genome (2.40×10^{-3}). The contig N50 and scaffold N50 values of the mallard genome assembly were 38.27 Kb and 2.49 Mb, respectively (Table 1; Table S2). We also generated a high-quality spot-billed genome assembly containing 67,685 scaffolds and covering 1.31 Gb. Similarly, we found that spot-billed assembly had a high heterozygosity rate (3.54×10^{-3}). The contig N50 and scaffold N50 values of the spot-billed assembly were 33.06 Kb and 2.05 Mb, respectively (Table 1; Table S2). We then estimated coverage of the mallard duck and spot-billed duck assemblies with alignment to seven finished BACs (completed independently using Sanger sequencing technology) and 319,996 duck ESTs assembled in our previous duck genome project using BLAT (1, 2). This analysis suggested that 7 BACs covering 640 Kb on chromosomes 1, 3 and 4 were aligned over more than 92% of their lengths (Fig. S4-S5, Table S3), and greater than 98% of 319,996 ESTs were aligned to both the mallard duck and spot-billed duck assemblies (Table S4).

We annotated transposable elements (TE) of the mallard duck and spot-billed duck assemblies using a combined pipeline. This effort found that the mallard duck and spot-billed duck assemblies contained ~136 Mb and ~139 Mb TE sequences, respectively, accounting for about 10% of their assemblies, respectively (Table S6). Among those TE elements, the LINE (long interspersed elements) type CR1 (chicken repeat 1) transposons are of the most abundance, a situation similar as ones in Beijing duck and chicken. Phylogenetic analysis of CR1 in mallard duck, spot-billed duck, Beijing duck and chicken revealed two duck-specific CR1 clades (clade 1 and 3), in which large variation of copy numbers were shown between the two wild ducks and Beijing duck. This implies that, as a main class of TE, CR1 evolved rapidly among different avian species and distinct duck lineages. (Fig. 1a; Fig. S6).

We predicted 21,056 and 21,123 protein-coding genes, which constitutes approximately 2.21% of the mallard genome and 2.13% of the spot-billed genome, using the BGI pipelines (Table S7). These numbers are slightly larger than the corresponding in the domestic duck (Beijing duck) genome (20,629 protein-coding genes) (Table 1). Of the 21,056 mallard genes, 12,817 were mapped to categories established by the Gene Ontology (GO) project, 15,878 had orthologs in the Kyoto Encyclopedia of Genes and Genomes (KEGG) database, and 14,894 were supported by the duck ESTs produced in our previous duck genome project (Fig. S7). These numbers are similar to these of the spot-billed, where 12,815 genes were mapped to GO project, 15,803 had orthologs in KEGG database and 14,827 were supported by the duck ESTs (Fig. S8).

We then constructed gene families using the above mallard duck and spot-billed

reference gene sets, our previous Beijing duck gene set, together with a combined outgroup gene sets from chicken, turkey, flycatcher, ostrich, mouse and human (Fig. S9, Table S8). Interestingly, two wild ducks were similar and their ancestors were clustered to the domestic (Beijing) duck (Fig. 1b). We further examined large-scale differences in gene complements between one domestic duck and two wild ducks. We found that 1,305 gene families were specific to wild duck and this number was larger than the corresponding of Beijing duck (497 gene families) (Fig. S10). Among 19,405 gene families detected in the above three duck genomes, 25 were significantly expanded in wild duck. GO enrichment analysis showed that those expanded genes were significantly over represented in biological processes like chromatin organization, cell-cell adhesion, ATP synthesis and immune response. This implies enhanced genome regulation, energy metabolic activity and response to stimulation along with the speciation of the two wild ducks, which might be beneficial to their adaption to new environments (Table S9; Fig. 1c-d).

1.1 Genome sequencing

We extracted genomic DNA with standard molecular biology techniques from blood and random fragmented DNA for library construction. For short insert size libraries, 5 µg of DNA were fragmented to 250-800bp, end-repaired, A-tailed and ligated to Illumina paired-end adapters (Illumina). The ligated fragments were size selected at 250, 350 and 800 bp on agarose gel and amplified by PCR to yield the corresponding short insert size libraries. For mate-pair library construction, 20-40µg DNA were sheared to the desired insert size using nebulization for 2 kb or HydroShear (Covaris) for 5 kb, 10 kb and 20 kb. Then the DNA fragments were end-repaired using biotinylated nucleotide analogues and circularized by intramolecular ligation. Circular DNA molecules were sheared with Adaptive Focused Acoustic (Covaris) to an average size of 500 bp. Biotinylated fragments were purified with Dynabeads® M-280 Streptavidin beads (Invitrogen), end-repaired, A-tailed and ligated to Illumina paired-end adapters, PCR amplification and agarose gel electrophoresis for size-selection. Then sequence all these libraries with Illumina HiSeq2000 platform.

1.2 Raw data filtering and genome size estimation

To reduce the effect of sequencing error or low-quality reads for genome assembly, we took a series of checking and filtering steps on raw reads. We filtered: 1) Reads having an 'N' over 10% of their length; 2) Reads with more than 40 bp low quality base (quality score ≤ 7); 3) Reads with more than 10 bp adapter sequences (allowing ≤ 3 bp mismatches); 4) Short insert size paired-end reads that were overlapped (≥ 10 bp); 5) Redundant duplicated reads generated by PCR amplification in the library construction (Read1 and Read2 of two pairs of paired-end reads were identical, respectively). In total, 275.20 and 262.45 Gb of data were kept for D2B and D2L *de novo* genome assembly respectively.

Genome size can be estimated using K-mer analysis method, the principle is briefly depicted as follows: A K-mer is an artificial sequence division of K bp iteratively from sequencing reads. A read with L bp in length composes (L-K+1) K-mers. The frequency

of each K-mer can then be calculated from the input data. The resulted K-mer frequencies should generally follow a Poisson distribution, except for a high proportion of low frequencies due to sequencing errors. The genome size G can be estimated as $G = K_num / K_expect$, where K_num is the total number of K-mer, and K_expect is the expected value of the poisson distribution. We performed K-mer analysis ($K=17$) using about 21-27X filtered data, and estimated the mallard and spot-billed genome size to be about 1.15Gb and 1.14Gb, respectively (Figure S1).

1.3 Genome assembly and super-scaffold construction

At first, we assembled the two genomes using SOAP *de novo*(3) for contig and scaffold construction. Detailed assembly steps and the data used in each step: 1) Constructed *de Bruijn* graph with reads from the short insert size libraries using K-mer size 29 bp, which means the parameter $-K$ is set as 29. 2) Simplified the *de Bruijn* graph for contigs construction by removal tips, merging bubbles, removal the low coverage of the connection and resolve the small repeats. 3) Constructed scaffolds with short pair-end and mate-pair libraries step by step. 4) Filled in gaps inside the constructed scaffolds with gap closure step.

1.4 Quality evaluation of the genome assembly

We made use of known EST and BAC sequences to evaluate the accuracy of genome assembly. We mapped 7 BACs(2) and 319,996 ESTs of Beijing duck onto the two assemblies using BLAT (4) with default parameters. We found that, for those 7 BACs totally spanning 640 Kb, more than 92% of the spanning were aligned well to the assemblies (Figure S3 and S4). In addition, around 94% of the ESTs were mapped to individual scaffold of the assemblies (Table S6).

Excessive heterozygosity at regional homologous chromosomes usually lead to failure of collapsing the two haplotype sequences, and thus results in redundant sequences in the assembly. To determine whether there is any case of redundant sequences in our assemblies, we performed a self-to-self alignment of the assemblies. Then found the total length of the scaffolds with the coverage rate is more than 50% is 11.3Mb and 10.3Mb in female mallard and a female spot-billed genome, respectively. That analysis indicated there was almost no redundant sequence in the two assemblies and we masked these scaffolds with coverage rate more than 50% for further deep analysis.

2. Repeat prediction, gene structure and gene functional annotation

2.1 Transposable elements annotation

We annotated transposable elements (TE) using the following steps:

- 1) We first scanned for repeat elements using LTR_FINDER(5) with the default parameters. The resulted TE sequences were used as TE database for RepeatMasker(6) search in step 2.
- 2) We then used RepeatMasker to identify repeat elements against the Repbase database(7) as well as the TE sequences founded in step1.
- 3) We further searched the assembly against the protein repetitive database provided in

RepeatMasker using RepeatProteinMask.
4) At last, we combined all the three set of predictions (two from step2 and one from step3) to generated an integrated repeat annotation result for each genome assembly.
5) Tandem repeats were identified by Tandem repeat finder(8) using the defaults of “Match=2, Mismatch=7, Delta=7, PM=80, PI=10, Minscore=50, and MaxPeriod=12”.

2.2 Variation of CR1 (chicken repeat 1) transposon among ducks

From the result of repetitive elements annotation in section 2.1, we found that CR1 (chick repeat 1), a subtype of LINE retrotransposon, accounts for both the majority of both TE content and variation of TE content among mallard, spot-billed duck, Beijing duck and chicken assembly. In fact, we found there were more than 90Mb (7.4%) of CR1 subtype repeat in two wild ducks, but only approximately 60Mb (5.5%) in Beijing duck. Therefore, we further studied the variation of CR1 among those four species in a perspective of phylogeny. We downloaded non-LTR retrovirus reverse transcriptase (RT) of chicken retrotransposon CR1, complete consensus sequence from NCBI. We extracted the 252 amino acids (aa) long RT_nLTR_like protein sequences from the consensus, and then mapped it to the four genome assemblies using tblastn. We identified 2579, 2294, 617, 1737 RT_nLTR_like domains in the four genome assemblies (evalue < 1e-5, identity >50%, alignment length >200bp), respectively. We then performed multiple sequence alignment of all the identified RT_nLTR_like domains using MUSCLE(9) (version 3.8.31). Based on the alignment, a phylogenetic tree was subsequently reconstructed using FastTree.

2.3 Gene structural annotation

First, we applied the homolog based gene prediction method using protein datasets of *Anas platyrhynchos*, *Gallus gallus*, *Struthio camelus*, *Meleagris gallopavo*, and *Ficedula albico*. All the protein sequences were firstly mapped to the assembly using BLASTN with the parameters as “-e 1e-5 -F F -m 8”. Then the aligned protein sequences were realigned to the assembly using GeneWise(10) for accurate spliced alignments.

Second, we used Beijing Duck transcriptome reads to polish our target gene annotation. We mapped the transcriptome reads to the two wild duck genome assemblies using TopHat(11) with its default settings and then constructed transcripts using Cufflinks. The process is the same as one preformed in the Beijing Duck genome project(12). Finally, we combined all the gene models (five homologous gene models sets and the transcript gene models set) by GLEAN(13) to generate an consensus gene set.

2.4 Gene functional annotation

We aligned protein sequences of the final gene set to various protein databases with known functional knowledge, including InterPro(14) (iprscan_4.7) (profilescan, blastprodom, hmmsmart, hmmpanther, hmmpfam, fprintscan, patternScan), Gene Ontology(15), Swiss-Prot(16), TrEMBL(16) and KEGG(17) (release76), to identify conserved domains, putative biological functions and molecular pathways they may involve. In total, there are more than 87% of the annotated protein-coding gene models

had at least one hit with known function (Table S5).

3. Evolutionary analysis

Mallard and spot-billed ducks, together with five other avian species (*Gallus gallus*, *Struthio camelus*, *Meleagris gallopavo*, *Ficedula albico* and *Anas platyrhynchos*, which is the Beijing duck) and two mammals (*Homo sapiens*, *Mus musculus*), were used in evolutionary analysis.

3.1 Gene family clustering

Protein-coding genes for *Gallus gallus*, *Struthio camelus*, *Meleagris gallopavo*, and *Ficedula albico* were downloaded from Ensembl release 80. The gene sets for *Anas platyrhynchos* was obtained from the BGI inner database. For gene loci with alternative splicing isoforms, only the transcript with the longest translation product was retained. We carried out an all-to-all alignment of all the collected protein sequences using BLASTP (E-value < 1E-7), and conjoined fragmental alignments using Solar(18). Then a simplified version of Treefam(19) methodology was used to cluster individual sequence into families based on the conjoined alignment (Table S11).

3.2 Phylogenetic tree reconstruction and divergent time estimation

After gene family clustering, single copy genes were selected to reconstruct phylogenetic tree of those nine species. Multiple sequence alignment of each gene family was performed by MUSCLE(9) (version 3.8.31). Phase1-base degenerate sites were extracted and concatenated to generate a super alignment matrix. We built phylogenetic tree using MrBayes(20) which takes advantage of both codon-based and amino acid-based algorithms and adjusts them to the topology of the species tree, to form a more accurate consensus tree according to phase1-base degenerate site. Divergence time of species were estimated using molecular clock model implemented by PAML mcmctree(21).

3.3 Segmental duplications

Segmental duplications (SDs) are duplicated blocks of genomic DNA typically ranging in size from 1–200 kb(22). To explore any difference in the content of SDs among two wild duck and Beijing duck genome, we detected SDs in their genome assembly using a whole-genome-alignment based method. We performed self-to-self whole genome alignment of these three assemblies using BLASTZ with the parameters: T=2 C=2 H=2000 Y=3400 L=6000 K=2200. Repetitive elements were masked in advance. We then filtered out self-alignment results (>0.85 similarity and >1000 aligned bp) and obtained more credible align block pairs. Finally, we realigned and filtered these block pairs (>0.9 similarity and >1000 aligned bp), producing a final version of SD regions in each genome. In summary, we identified 59.64 Mb (coving 4,367 genes) and 66.05 Mb (coving 4,864 genes) segmental duplications in mallard and spot-billed genome respectively, which are much larger than Beijing duck's 8.39 Mb (coving 1,133 genes). We further did pair-wise alignment among SDs detected in all these three assemblies to

identify SDs specific to each of them.

4. Resequencing and population genomic analyses

4.1 Genome sequencing, Polymorphism identification, accuracy verification and its combine between populations

Genomic DNA libraries were prepared according to the manufacturer's instructions (Illumina). One DNA sample or pools of DNA samples from 10 individuals for each six duck breeds were used to generated pair-end reads with the Illumina technology (Table S1). Pair-end reads were mapped to the Beijing duck assembly (BGI_duck_1.0) using the bowtie2 with default parameters. After removing the reads which map to multiple place, we called SNPs using the Genome Analysis Toolkit (GATK) and filtered SNPs according to the coverage of resequencing data with thresholds of $\leq$ one third of the mean or $\geq$ three-fold of the mean and MQ value $\leq 28(23)$

We produce BAM file for each individual sample or population pool according its sequencing strategy (Table S11). VCF files for each population were using "UnifiedGenotyper" of GATK by taken all processed BAM files of individual samples as input, or taken pool BAM file of a population as input. And all BAM files were processed by "REMOVE_DUPLICATES", "use unique mapped reads" and "IndelRealigner". In the functional validation section, 10 SNP sits were selected to be used to experimental verification its functional activity. 9 of these 10 were true SNPs but one was INDEL instead. We think that these results could reflect the high accuracy of SNP calling in some extent. In the following selection analyses or $\partial a \partial i$ simulation, all SNPs are from regions that homologous to chicken autosomes. We combine the VCF files by using a way like outer join of SQL (Structured Query Language) (<https://github.com/vitogump/lifelit>). For the variations that present in one VCF but absent in the another one, we fill it as it fixed as ref allele when the coverage (information from BAM) of this site reached the threshold.

4.2 Domestic duck shared a comparative number of SNPs with the mallard duck and spot-billed duck

Detailed analyses suggested that domestic ducks shared large numbers of SNPs (18,093,115) with wild duck, and hold comparative numbers of common SNPs with the mallard (15,904,435) and spot-billed (16,062,465) ducks. Moreover, we found that the numbers of mallard specific SNPs (heterozygous in mallard and homozygous in spot-billed) are comparative to the numbers of spot-billed specific SNPs (homozygous in mallard and heterozygous in spot-billed) in each of eight tested domestic duck breeds (Fig. S12).

4.3 Estimation of nucleotide diversity and population mutation rate

We estimated nucleotide diversity (π) and Watterson's estimator Θ using the PoPoolation package, which corrects biases causing by pooling and sequencing errors, in 40 kb non-overlapping windows(24, 25). We required per position a minimum

coverage of 13,13,13,6 and 6 a maximum coverage of 150,150,500,100 and 100 for mallard, spot-billed, Beijing, Shaoxing, and other 8 domestic population each, and set min-count and min-covered-fraction to be 3 and 0.6 for all.

4.4 Analysis of linkage disequilibrium (LD), identical by state (IBS), principal component analysis (PCA), f3-statistics and population structure

LD level in Beijing, Shaoxing, mallard duck and spot-billed duck populations were measured through calculating the correlation coefficient (r^2) of alleles using the PLINK(26).

The parameters were set as follows: --ld-window-kb 500 --ld-window 99999 --ld-window-r2 0 --r2. To reduce effect of LD, SNPs were pruning by LD and subsequently used for the PCA, f3-statistics and population structure analyses (Fig. S11). We constructed phylogenetic trees by using neighbor-joining method with MEGA based on IBS distance matrix data of all individuals calculated by the PLINK using the above sampling SNPs(27). Principal component analysis (PCA) was performed using the GCTA64(28). The f3-statistics were calculated using the ADMIXTOOLS and genetic structure was inferred using the admixture program.

4.5 PCA, genetic structure and PSMC analysis

SNPs used in both PCA and genetic structure analysis were filtered using PLINK(26) (--indep-pairwise 500 50 0.4 --maf 0.05). First, we want to know the genetic relationship of all 8 wild duck species and these domestic populations as mentioned in main text Fig. 2a, which provided evidence that mallard duck and spot-billed duck having a far more closely genetic relationship with domestic duck than other wild species. To eliminate the unequal sample size effect, assembly difference effect and autosomes/sex difference effect, we supplemented PCA analysis by using same sample size for each population, with both mallard duck assembly and Beijing duck assembly (Figure S13). These results further supported our conclusion.

Second, in our genetic structure analysis, it is interesting that domestic lineage separated from wild duck first, and even subdivided into Beijing and Shaoxing previous than the mallard duck and spot-billed duck's subdivision. To further investigate this phenomenon and eliminate the coverage effect. We perform genetic structure analysis by using equal sample size and same coverage of each population with different populations combinations (Figure S14). These conclusions are consistent.

Third, in our PSMC analysis, we cutting the coverage of Beijing individuals into the same with Shaoxing individuals. They show the same curve with Shaoxing ducks (Figure S17). So, we propose that all domestic ducks should have the same PSMC curves.

Cutting sequence coverage was performed by samtools(29) (samtools view -bhs).

4.6 *ada*i simulation

There are two ways to compare the two phylogenetic trees (spot-billed, (mallard, domestic)) and (domestic, (mallard, spot-billed)). First, as some literates did, simulated the three populations split model and calculated the likelihoods to compare the fit of each model (30).

We think this is unsuitable to our date. Because this would take more parameters into model that would significantly increase the difficult to simulate and is very time consuming. Furthermore, the likelihoods are influenced by multiple parameters. We hardly give a property demographic model and constraint the parameters, since there is lack of reliable evidence (beside genetic data) about the real evolutionary history of ducks. Moreover, both PCA and structure analysis shows that wild duck populations are structured that is challenging with $\partial a \partial i$ because the dadi model treats populations each as distinct homogenous entities. So, the best way is simplifying the model and making it easy to be get convergent parameters.

We used an alternated way by simulating paired divergence and comparing the distribution of divergence time. If (Fig. 1e left) is the correct evolutionary relationship, then we expected longer divergence time between mallard duck and spot-billed than the ones between the two wild ducks and the domestic ducks. Otherwise if (Fig. 1e right) is the correct tree topology. We would expect the divergence time between two wild duck populations and domestic population were similar and longer than the one between mallard duck and spot-billed duck.

As we only focus on divergence time, we first used an extreme simple model with only two parameters (Fig. 2a left). This way would eliminate the influence between different parameters. Especially taking migration rate into simulation would have a significant impact in divergence time which could be purely due to mathematical property of $\partial a \partial i$ software.

The distribution of each estimated parameter and the confident intervals were presented in Figure S15 and Table S12, respectively. And the comparison of divergence time between the domestic and mallard/spot-billed duck, mallard duck and spot-billed duck were shown in Fig. 2b left.

We then add migration parameters into model (Fig. 2a right). The distribution of each estimated parameter was presented in Figure S16 that show bimodality distributions rather than normal, so we don't calculate the confident intervals. And the comparison of divergence time between the domestic and mallard/spot-billed duck, mallard duck and spot-billed duck under this model were shown in Figure 3b right. The maximum likelihood Ts between domestic and mallard/spot-billed duck, mallard duck and spot-billed duck are 333,504 years ago, 334,738 years ago and 74,915 years ago respectively. Combined the two models' result, we inferred that the ancestor of wild duck diverged from domestic duck about 100Kyr~300Kyr, and subsequently diverged into mallard duck and spot-billed duck about 70Kyr.

In conclude, dadi simulation on both models, considering or not considering the migration, are support the phylogenetic tree that is inferred from our previous population structure analysis. Although dadi software cannot distinguish short-divergence with low migration from long divergence with high-migration, this usually cause the estimate Ts not convergent or convergent into two peaks (smaller Ts accompanied by high migration or bigger Ts coupled with low migration rate). In our second simulations (with migration), estimate Ts does increased, when compared with first model, due to introduce migration. But the Ts still convergent into two peaks, higher estimated Ts with higher estimated migration rate whereas shorter Ts with lower

1115 migration rate. a reasonable explanation could be: although dadi have difficult to
 1116 determine longer or shorter Ts, thus having simulation on both situation and produce
 1117 two peaks of estimate Ts, the Ts between mallard and spot-billed is still significant
 1118 smaller than Ts between wild and domestic one in any case.

1119 In additional, we add parameters to allow population change exponentially (IM_2
 1120 model). We found that many of simulations with IM model can't get the convergent
 1121 estimates or get estimates with some parameters hit the boundary, especially for
 1122 divergence between mallard duck and spot-billed duck. This may be because as the
 1123 number of parameters (2 for "split_no_M" model, 4 for "split_M" and 7 for "IM_2"
 1124 model) increased, the correlation between parameters (which is a purely mathematical
 1125 property of) will significantly increase the difficulty to getting the correct estimates.
 1126 And the two wild populations are highly structured and exist many fixed difference
 1127 sites which is particular challenging with $\partial a \partial i$. Furthermore, the distribution of each
 1128 parameters shown two or more peaks. So, we hardly estimated results. But we found
 1129 that 29 of 39, 53 of 68 bootstrap fits yield increased domestic population size change
 1130 after they split with mallard duck/spot-billed duck. All bootstrap fits yield increased
 1131 wild duck population size change.

1132 (data not show)

1133 def split_nm(params,ns,pts):

```
1134     s,Ts=params
1135     xx=dadi.Numerics.default_grid(pts)
1136     phi=dadi.PhiManip.phi_1D(xx)
1137     phi=dadi.PhiManip.phi_1D_to_2D(xx,phi)
1138     phi=dadi.Integration.two_pops(phi,xx,Ts,nu1=s,nu2=(1-s))
1139     fs=dadi.Spectrum.from_phi(phi,ns,(xx,xx))
1140     return fs
```

1141 model code1: split without migration

1142

1143 def split_m(params,ns,pts):

```
1144     s,Ts,m12,m21=params
1145     xx=dadi.Numerics.default_grid(pts)
1146     phi=dadi.PhiManip.phi_1D(xx)
1147     phi=dadi.PhiManip.phi_1D_to_2D(xx,phi)
1148     phi=dadi.Integration.two_pops(phi,xx,Ts,nu1=s,nu2=(1-s),m12=m12,m21=m21)
1149     fs=dadi.Spectrum.from_phi(phi,ns,(xx,xx))
1150     return fs
```

1151 model code2: split with migration

1152

1153 def IM_2(params,ns,pts):

```
1154     s,nu1,nu2,TS,m12,m21=params
1155     xx=dadi.Numerics.default_grid(pts)
1156     phi=dadi.PhiManip.phi_1D(xx)
1157     phi=dadi.PhiManip.phi_1D_to_2D(xx,phi)
1158     nu1_func=lambda t: s *(nu1/s)**(t/TS)
```

```

1159     nu2_func=lambda t: (1-s) *(nu2/(1-s))**(t/TS)
1160     phi=dadi.Integration.two_pops(phi,xx,TS,nu1=nu1_func,nu2=nu2_func,m12=m1
1161 2,m21=m21)
1162     fs=dadi.Spectrum.from_phi(phi,ns,(xx,xx))
1163     return fs
1164 model code2: split with migration and population exponentially change.
1165 full program is provided in https://github.com/vitogump/lifelit.

```

1166 4.7 Selection analyses

1167 In the d_f calculation, we required all sampled individuals were sufficient covered for
1168 each d_f sites.

1169 Whole genome alignment to chicken were performed by LASTZ(31).

1170 4.7.1 Detail functional analysis on candidate selected genes in early ancestor of 1171 mallard and spot-billed ducks.

1172 We detected positive selections in the early ancestors of these two wild populations
1173 using methods invented in early modern humans and applied in European and Asian
1174 wild boards pig (3, 32). As the long time passed from the ancient selection occurrence
1175 in wild lineage, the accumulation of recombination, mutation and the other population
1176 effects may obscure the selection regions. We use a relative narrow window width to
1177 detect ancient selection. We filtered variation that passed the following criteria:

- 1178 1) No CpG sites
- 1179 2) SNPs were fixed in the tested Muscovy ducks with at least 20-fold coverage;
- 1180 3) At least 5 individuals were observed in each population or 10 reads were observed
1181 for pool populations
- 1182 4) At least 20 reads were observed when the site was fixed in a population
- 1183 5) Derived allele was detected in both wild and domestic lineage

1184 For windows with less than 10 variation sites were discarded.

1185 In addition to the functional analysis presented in main text. An overrepresented term
1186 of positive selection related to “dorsal/ventral pattern formation”, “determination of
1187 left/right symmetry” also imply selection on genes which making their body more
1188 suitable for flying. overrepresented on terms “response to wounding”, “egg activation”,
1189 “acute inflammatory response to antigenic stimulus”, “detection of virus” may indicate
1190 that selection on genes increased mallard and spot-billed ducks’ survivability in wild.

1191 In addition, we found that at least 32 genes (e.g. *IL11RA*, *PRKAA1*, *LPAR1*, and *NS1BP*)
1192 involved into immune response to infections and three of them (i.e. *ADAP*, *RIG-I* and
1193 *CX3CR1*) played critical roles against influenza A virus infections (Table S15; Fig.
1194 S21)(33, 34). Therefore, it seemed that positive selections on immune genes might help
1195 the early mallard and spot-billed duck ancestor keep evolutionary equilibrium with
1196 influenza A viruses.

1197 Some cryptochrome-containing cells within the retina are active at night when the birds
1198 perform magnetic orientation(35). In some migratory birds, a distinct part of the
1199 forebrain, where primarily processes input from the eye, is highly active at night(36).

1200 We also found three genes associated with retinal ganglion cell axon guidance (*EFNA5*,
1201 *ISL1* and *EPHA7* (S=-3)) and two genes (*CHRD* and *WNT7B*) related to forebrain

regionalization or development were under selection in the early mallard duck and spot-billed duck ancestor (Fig. 3d; Fig. S20)(37, 38). Gene family that function on short wavelength-sensitive opsin (Cryptochromes, which have been suggested to form the basis of light-dependent magnetic compass in birds, is excited by blue (short wavelength) light) expanded in wild ducks may also indicated an enhancement in (Table S9)(36).

4.7.2 Detail analysis on genomic islands of mallard and spot-billed duck divergence.

To elucidate genomic properties and speciation of these two wild duck species, we firstly calculated how many sites all mallard ducks homozygous for one allele and all spot-billed ducks homozygous for another (which we referred the density of these fixed difference sites as d_f) in 40kb sliding windows with 20kb step size along duck scaffolds (Methods). This effort found that a small number of dfs (4,386) and only 26 SNPs of these located within protein-coding regions distributed in small genomic regions (758 of totally 52,215 windows) between mallard and spot-billed. We refer the windows showing highly elevated divergence up to 50 times higher than the genomic mean as “divergent peak”. Divergent peaks contain most (3,522, 80%) of these dfs that distribute in 348 (0.6% of the whole duck genome) windows. This is sharp contrast to the case in *Ficedula flycatchers*, where using the same criterion, that 25% (59,936) dfs were distributed in the large genomic divergent islands covering 2.7% of the flycatcher genome. Such large difference in number of dfs and the pattern of genomic divergent islands in two these species might due to that the wild duck (the mallard and spot-billed duck) has a shorter divergence time and is in an earlier stage than the *Ficedula flycatcher*. As our sequence depth (~10X) is higher than the flycatchers’ research (~5x), which may leading to the reduce of the dfs. We also random sample the reads into about 5-fold coverage of duck genome and the distribution of dfs remain almost the same (data not shown).

In order to study the genomic divergence in duck autosomes, we then estimated the population differentiation, nucleotide diversity through calculating F_{ST} (fixation index), d_{xy} (sequence divergence), π (nucleotide diversity) and Tajima’s D values in sliding window that homologous to chicken autosomes. (Methods; Fig. 4a; Table S16). Scaffolds homologous to chicken Z chromosome showed higher level of F_{ST} (5.5-fold) and Tajima’s D (1.5-fold), but smaller number of SNP density (0.5-fold) and lower level of π (0.6-fold) than autosomes (Table S16, Fig. S23). These observations are consistent with change of F_{ST} , Tajima’s D and SNP density in flycatcher and wild ducks (Mexican and mallard duck)(39, 40). Analysis on autosomal differentiation regions shows that the distribution of F_{ST} between the mallard duck and spot-billed ranged from -0.0157 to 0.7785 with an average of 0.04930346 and distributed in a narrow peak, lower than F_{ST} between the wild ducks and domestic ducks ranged from 0.0880 to 0.8912 with an average of 0.2100 and distributed in a wide peak (Fig. 4b). Detailed analysis indicates that small part of genomic regions of the mallard duck and spot-billed duck have F_{ST} values being away from the mean F_{ST} at least 4 standard deviations (we referred these regions as high differentiation islands). These higher F_{ST} regions showed lower level of nucleotide diversity (π mallard: 0.0024 ± 0.0022 ; π spot-billed: 0.0023 ± 0.0020) and

SNP density (360 ± 305 and 334 ± 280 SNP per window in the mallard duck and spot-billed duck respectively) than genomic backgrounds (π mallard: 0.0076 ± 0.0042 ; π spot-billed: 0.0077 ± 0.0043 . 1134 ± 569 and 1161 ± 597 SNP per window in the mallard duck and spot-billed duck respectively) in both mallard duck and spot-billed duck (Fig. 4c-f; Fig. S23; Table S17). This is similar to the case in flycatchers, Darwin's finches and ducks, where high differentiation islands are characterized by reduced levels of nucleotide diversity and polymorphisms(32, 39-41). A noteworthy consequence of these coinciding features was that both Tajima's D of these high differentiation islands was not reduced in both mallard duck and spot-billed, whereas it is different with the case in flycatchers that Tajima's D of high differentiation islands was significantly reduced (Fig 4f; Table S17). Moreover, sequence divergence between mallard duck and spot-billed didn't exceed background levels in high differentiation islands that is consistent with the observation in flycatchers, whereas Darwin's finches' study show elevated dxy in high differentiation islands (Discussion Section; Fig. S23)(41).

4.7.2 Detail analysis on identification of candidate domestication regions

Limited by many factors, we used very conserved strategy and has been succeed applied in previous reports, may not the very advance in methods, to detect genomic regions under domestic selection. We first used the methods that combine the Z-transformed distributions of Hp and F_{ST} signals of autosomes and sex-chromosomes separately. And focused on putatively selective sweep regions falling at least four/two standard deviations away from the mean of the $Z(Hp)$ and $Z(F_{ST})$ of autosomes and sex-chromosomes separately. We applied this thresholds because Z including a reduction in effective populations size and recombination rate as previous discussed(42). Furthermore, we perform the statistic test on this outlier approaches using 4 times of zF_{ST} and zHp to define selective region, 4 times represent the extreme ends of the distribution of zF_{ST} and zHp , and P-value is less than 0.05, zF_{ST} ($P(zF_{ST} \geq 4) = 0.00548$), zHp ($P(zHp \leq -4) = 0.00322$). two signal combined identified 137 regions with extremely low levels of Hp (average length = 63.6230 kb, average Hp = 0.2100 for autosomes; Hp = 0.2470 for sex-chromosomes) and 145 regions with significantly elevated F_{ST} values (average length = 58.7670 kb, average $F_{ST} = 0.5390$ for autosomes, average $F_{ST} = 0.394$ for sex-chromosomes). However, only a small proportion putative selective sweeps regions werer predicted using both extremely low levels of Hp and significantly elevated F_{ST} values. Such observation might be partly attributed to that the domestic lineage and wild lineage (mallard duck and spot-billed ancestors) were under different evolutionary patterns after they had diverged about 100-200 Kyr. For example, the wild lineage seemed to benefit from positive selection and increased their effective population sizes once or twice times, while the domestic lineage dramatically declined their effective populations sizes during LGP (Fig. 2d,3a; Fig. S17). This complex demographic history of the domestic and wild lineage might affect the F_{ST} distribution. We counted the heterozygosity of these populations (Hp) in 40-kb windows (windows containing less than 10 SNPs were not included) in eight duck breeds respectively. We further calculated the Hp mean of these breeds for each window. Similarly, we calculated fixation index (F_{ST}) between each domestic duck breeds and the wild duck

population (combine spot-billed and mallard) using an estimator introduced in dog in 40-kb windows, and counted the F_{ST} of eight domestic duck breeds for each window(24). After that, we Z-transformed the H_p and F_{ST} to $Z(H_p)$ and $Z(F_{ST})$, and defined regions with $Z(H_p)$ or $Z(F_{ST})$ being four standards deviations for autosomes or two standard deviations for sex-chromosomes from the mean of the genome-wide ($Z(H_p)$) as putative selective sweep regions.

In H_p calculation, we didn't require min SNP numbers because the region with extreme little SNP could be Runs of homozygosity (ROH) that is also our interest.

For the regions under early stage selection in wild lineage or under recent selection in domestic lineage. We performed more detail analysis to visualize the genetic variations in Figure S23, S24.

The comparison between the details of selection regions in Figure S23, S24 reflect that the early stage selections mostly couldn't be visualized by current variations directly. But regions under recent selection in domestic lineage showed clear trace.

GO enrichment analysis were performed by using Fisher's Exact Test (see <https://github.com/vitogump/lifelit>).

4.7.3 Evolutionary relationship of domestic duck, mallard wild duck and spot-billed wild duck

Such inference was supported by both phylogenetic comparative analysis and population genetic analysis. Despite the incomplete lineage sorting (ILS) and gene flow may confound the analyses, the different SNP sets used in genetic structure analysis, f_3 -statistics, SNPs distributions analysis and tree construction that was sampled from genome-wide would avoid these confounding effects in phylogenic inference(43). Furthermore, Genetic structure show each individual of both the mallard duck and spot-billed duck almost consistent of purely one ancestry component. This is different from the case of admixture, where individuals of one population with components of other populations' ancestry (44-46), thus indicating no evidence of introgression between the mallard and spot-billed duck (Fig. 1d; Fig. S14). Subsequent $\partial a \partial i$ and PSMC simulations that rely on different data sets and theory reconstructed the perfect matched demographic history further support the robustness of our inference. In addition, the subsequent observations of F_{ST} on domestication and speciation analyses are consistent with or could be best explained by this phylogenic inference (Fig. 4b).

4.8 Estimation of genetic diversity and distance

Nucleotide diversity (π) was calculated using the vcfTools(47). Tajima's D was calculated using the Popgenome R package(48) Pairwise F_{ST} was calculated in 40-kb windows with a sliding size of 20-kb. Neighbor-joining tree based on pairwise F_{ST} values were constructed with phylip-3.696 "neighbor" function by taking the distance matrix of pairwise F_{ST} and integrating neighbor-joining trees of 27,374 windows.

4.9 select candidate casual mutation to verify its functional activity

For the sake of simplicity, we catalog the selection regions by its overlap with CDS, UTR, intron, intergenetic region. And rank SNPs by the ΔAF between wild lineage and domestic lineage. we selected the SNPs that located in UTR, intron, intergenetic

1333 region and closed to the genes that may play important functional in the duck
1334 evolution or domestication.

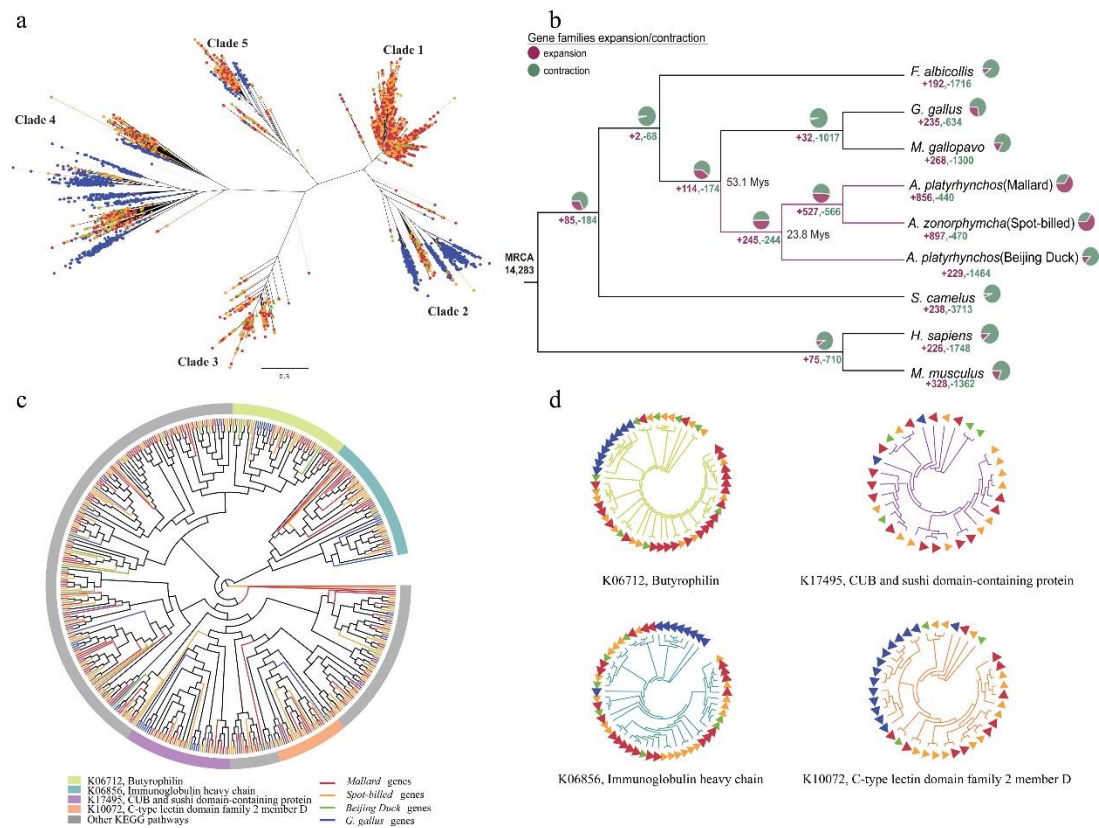


Fig. S1: Expansion and contraction of transportable element and gene in mallard, spot-billed and Beijing duck genome. Green, red, yellow and blue in a and c panel represent Beijing duck, mallard, spot-billed and chicken, respectively. **(a)** Phylogeny of CR1 retrotransposons in four bird genomes. **(b)** Numbers of gene family losses and gains across ten vertebrates. Pie chart and numbers nearby indicate the number of expanded and contracted gene families in each branch. **(c)** Phylogeny of gene families expanded in spot-billed and mallard. Top four families ranked by the number of gene gained are indicated in color arcs. KEGG pathway annotation of those four families are listed in the bottom. Other families are indicated in grey color. **(d)** Detailed phylogeny of four gene families expanded in spot-billed and mallard duck as indicated in (c).

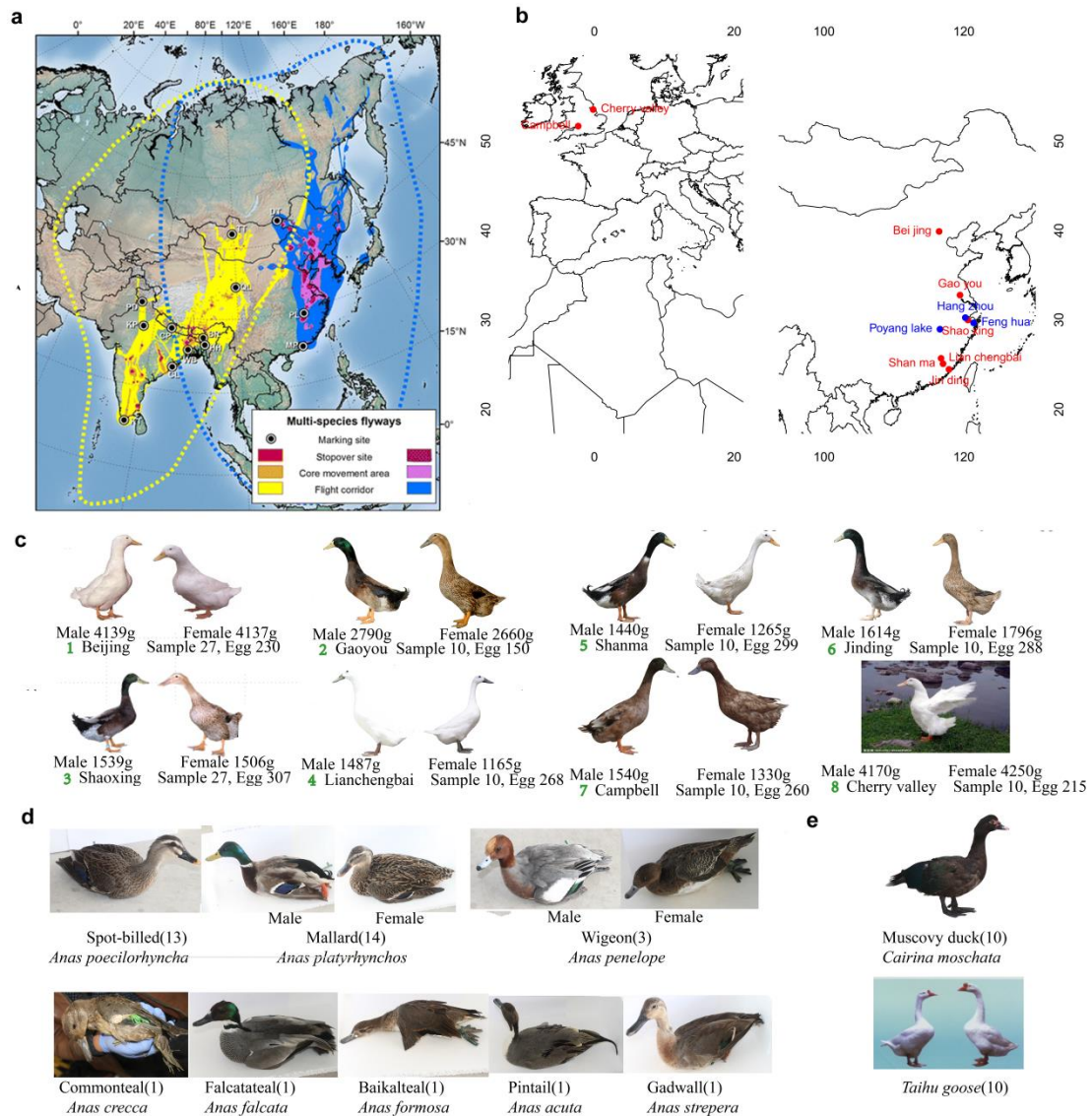


Fig. S2. Sample distribution and phenotypic variation in domestic and wild ducks.

Photos were taken from the sequenced samples except Taihu goose and Cherry valley being download from web. (a) Estimated migration routes of Anatidae in Central Asian Flyway (CAF) and East Asian-Australasian Flyway (EAAF)(49). Regions in the flyway of CAF displayed in yellow-red and for EAAF displayed in blue-purple. From darkest to lightest, colors represent 50%, 75% and 99% cumulative probability contours. (b) Geographical distribution of wild duck and domestic duck samples. Blue dots represent the place where wild duck samples collected from. Compare to a, our wild duck samples were collection from the main migration route of the Anatidae flyway. (c) Phenotype of the eight domestic duck breeds. Number of individuals, average weight (male/female), average eggs production per year were marked. (d) Photos and sample numbers of the eight wild duck species. (e) Photos and sample number of Muscovy duck and Taihu goose used as outgroup.

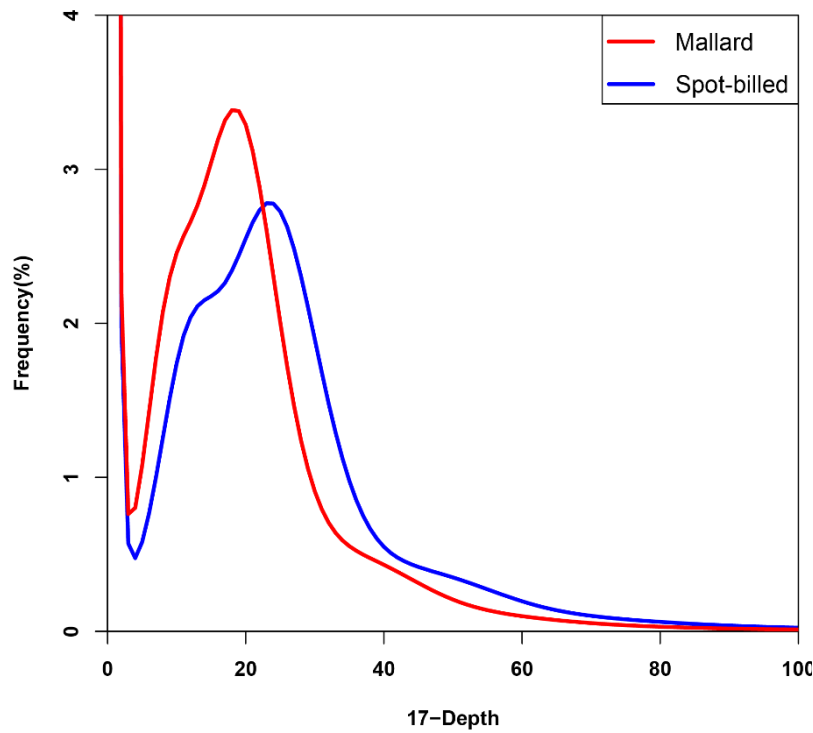


Fig. S3. Distribution of 17-mer frequency in the corrected pair-end reads of the mallard and the spot-billed genome.

Only the reads from short-size libraries (< 500 bp) were included in this analysis. We identified 22,312,838,068 kmers using 21-fold data of mallard and 28,137,616,208 kmers using 27-fold data of spot-billed. The D2B and D2L genome size can be estimated as $G = \frac{K_number}{K_depth}$ (the volume peak), which is estimated to be 1.14Gb and 1.15Gb, respectively.

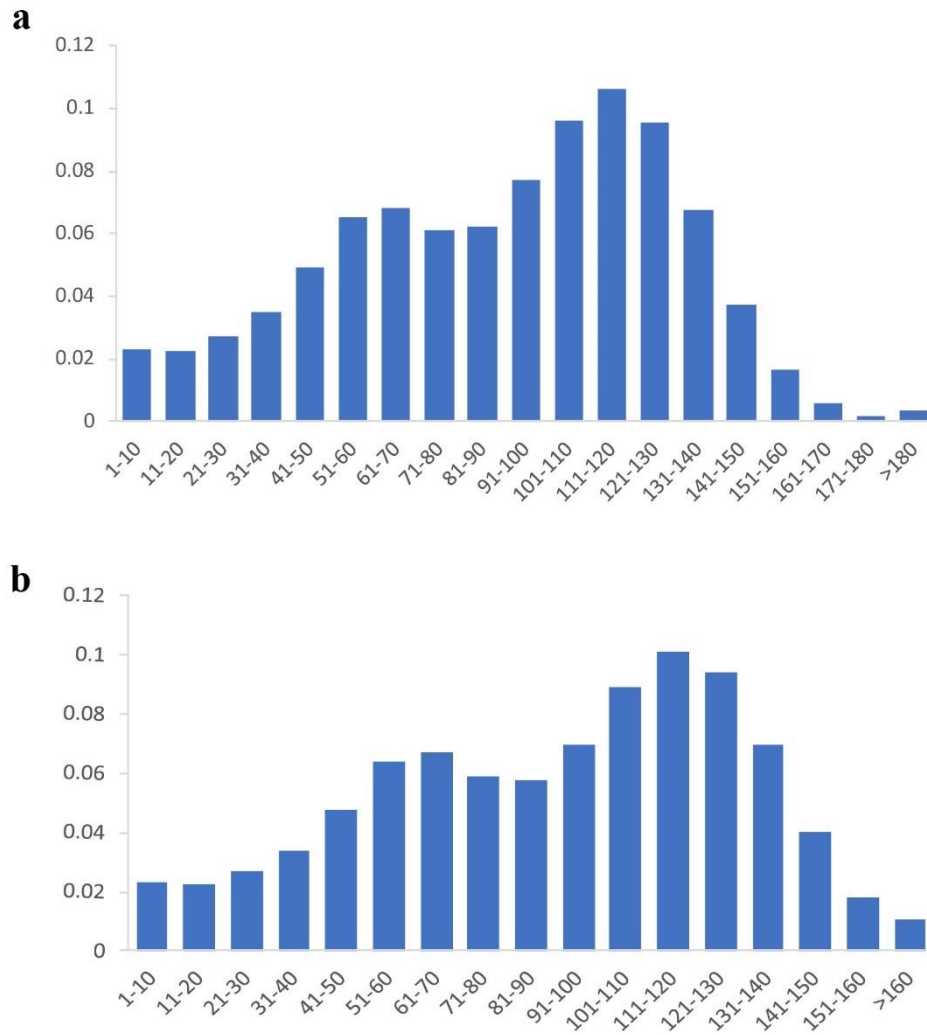


Fig. S4. Distribution of the sequencing depth of the mallard and the spot-billed assemblies.
 (a) the mallard assembly. (b) the spot-billed assembly.

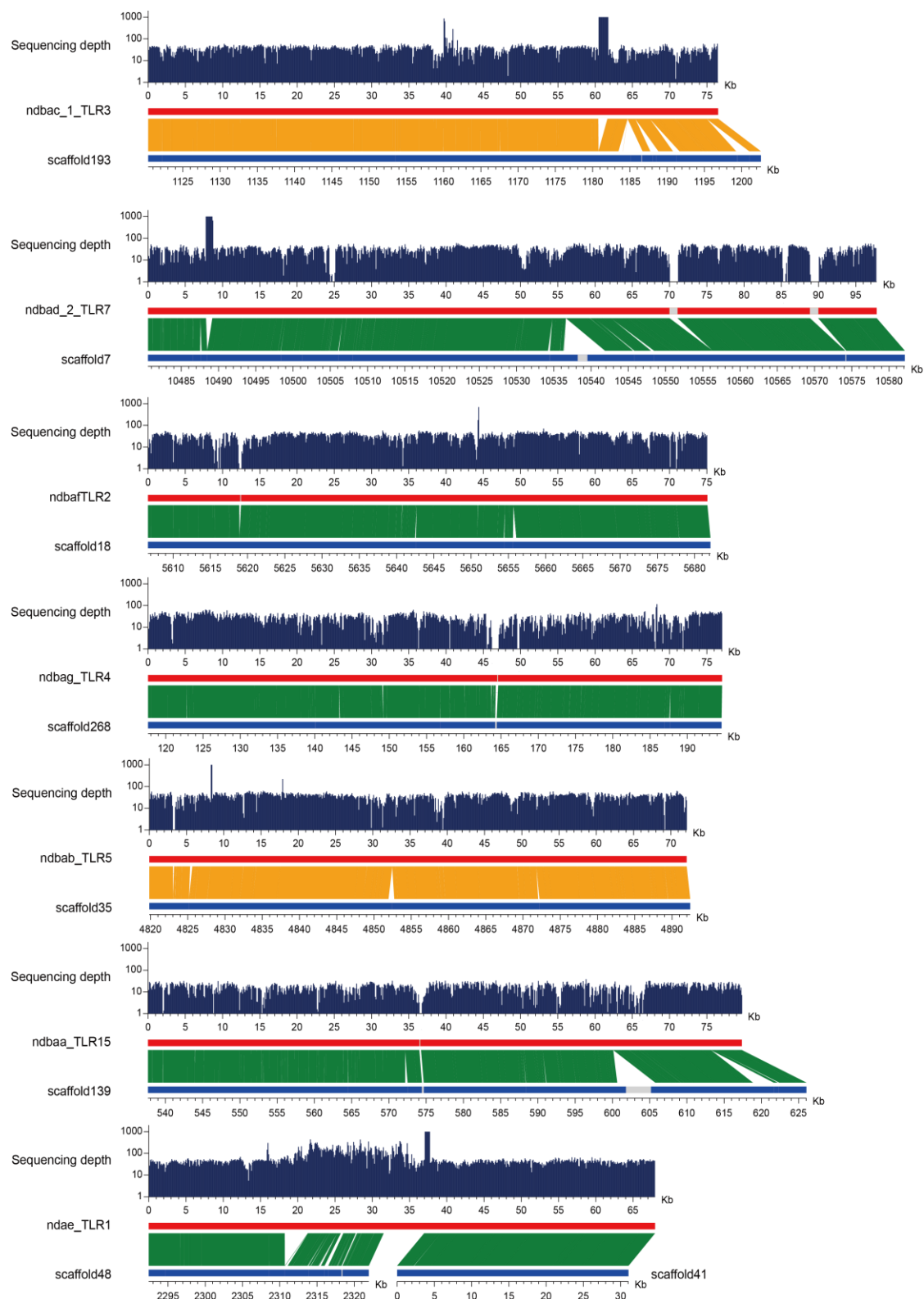


Fig. S5. Evaluation of the mallard assembly with sequences of seven BACs. Seven BACs were from chromosome 1, 3 or 4. The BAC sequences and assembly scaffolds are shown in red and dark blue, respectively. Orange represents positively and green was reversely aligned, the remaining unclosed gaps on the scaffolds are marked as white blocks.

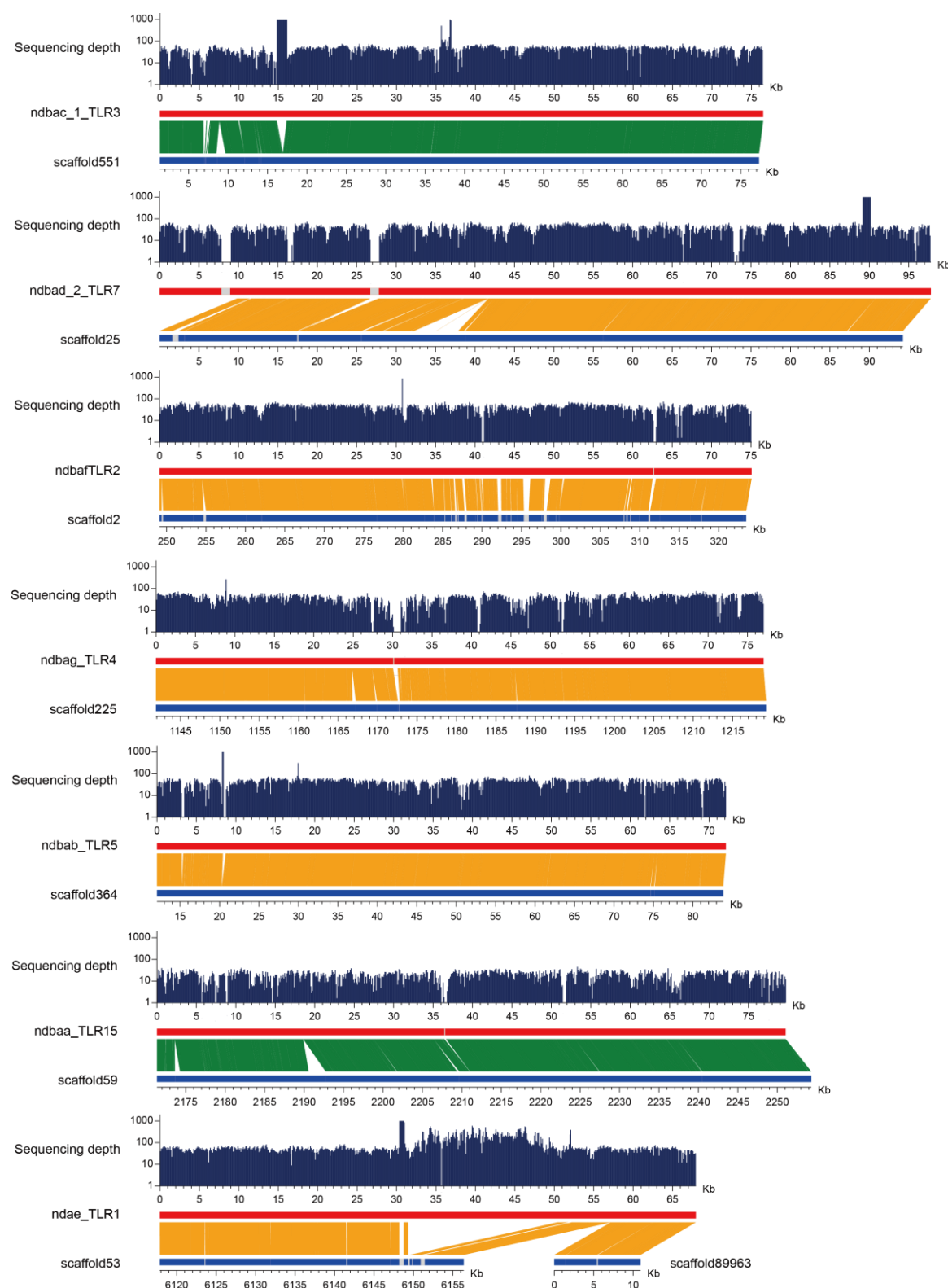


Fig. S6. Evaluation of the spot-billed assembly with sequences of seven BACs.
 Seven BACs were from chromosome 1, 3 or 4. The BAC sequences and assembly scaffolds are shown in red and dark blue, respectively. Orange represents positively and green was reversely aligned, the remaining unclosed gaps on the scaffolds are marked as white blocks.

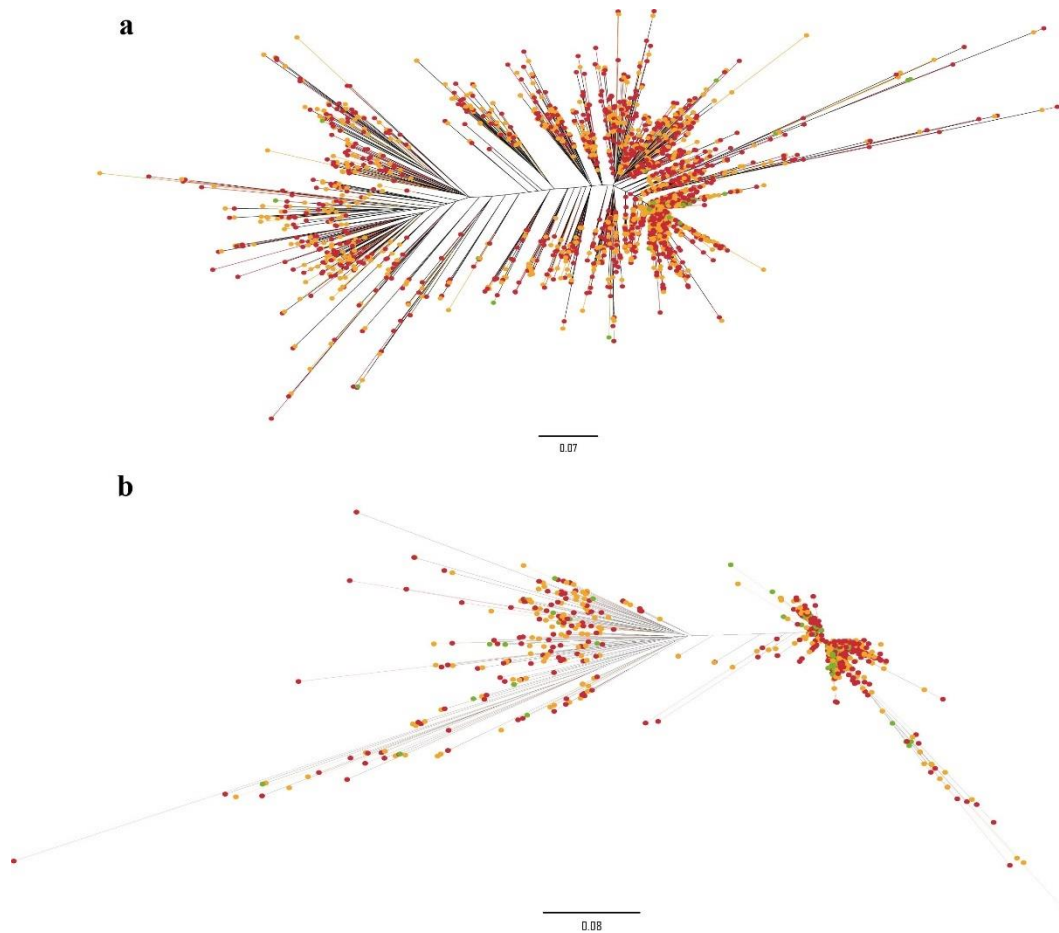


Fig. S7. Phylogeny of two CR1 retrotransposon clade in four bird genomes.
Green, red and yellow represent Beijing duck, mallard and spot-billed, respectively.

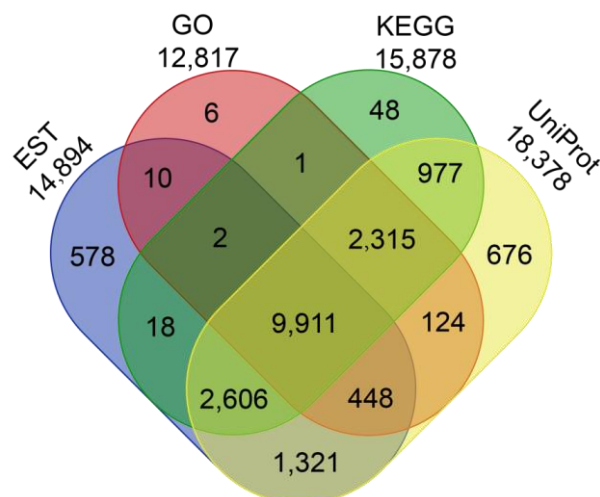


Fig. S8. Venn diagram showing the mallard reference genes annotated using three databases or supported by EST.

Total 21,056 mallard genes supported by 319,996 ESTs of Beijing duck and functioned annotation with three databases. And 19,041 genes were supported by one of them.

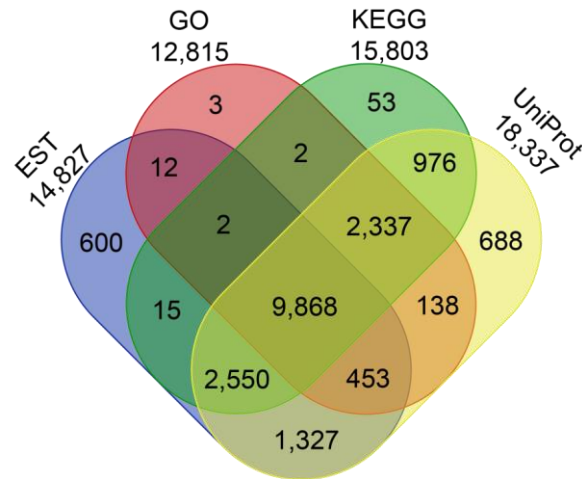


Fig. S9. Venn diagram showing the spot-billed reference genes annotated using three databases or supported by EST.

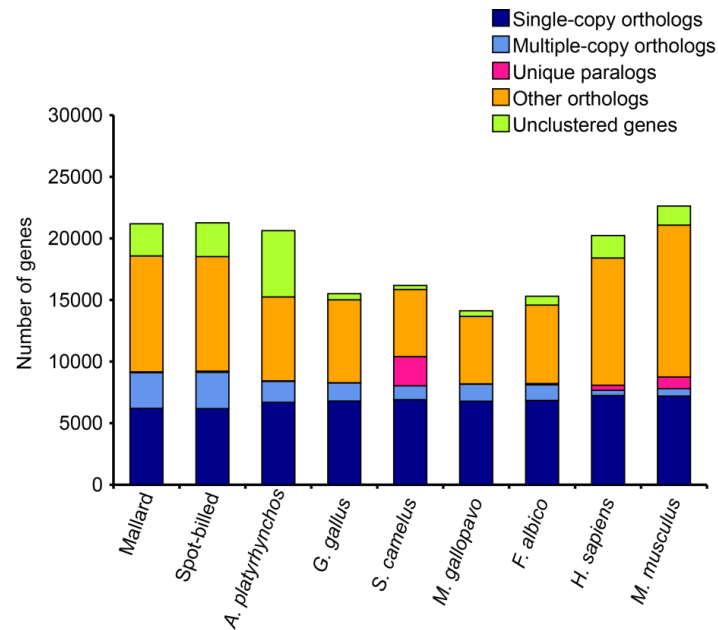
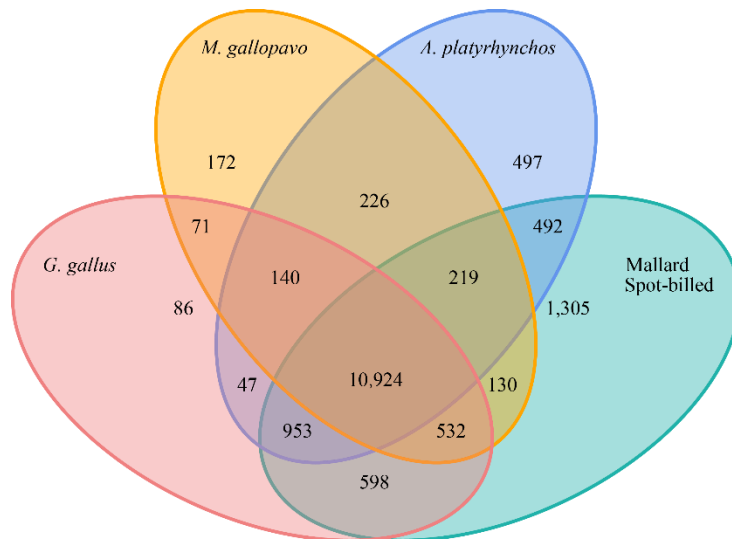


Fig. S10. Protein homologous comparison among the reference gene sets of nine species by TreeFam.



Number of gene families

Fig. S11. Venn diagram shows gene family clusters in five species by TreeFam.

The number of unique and shared gene families among five genomes.

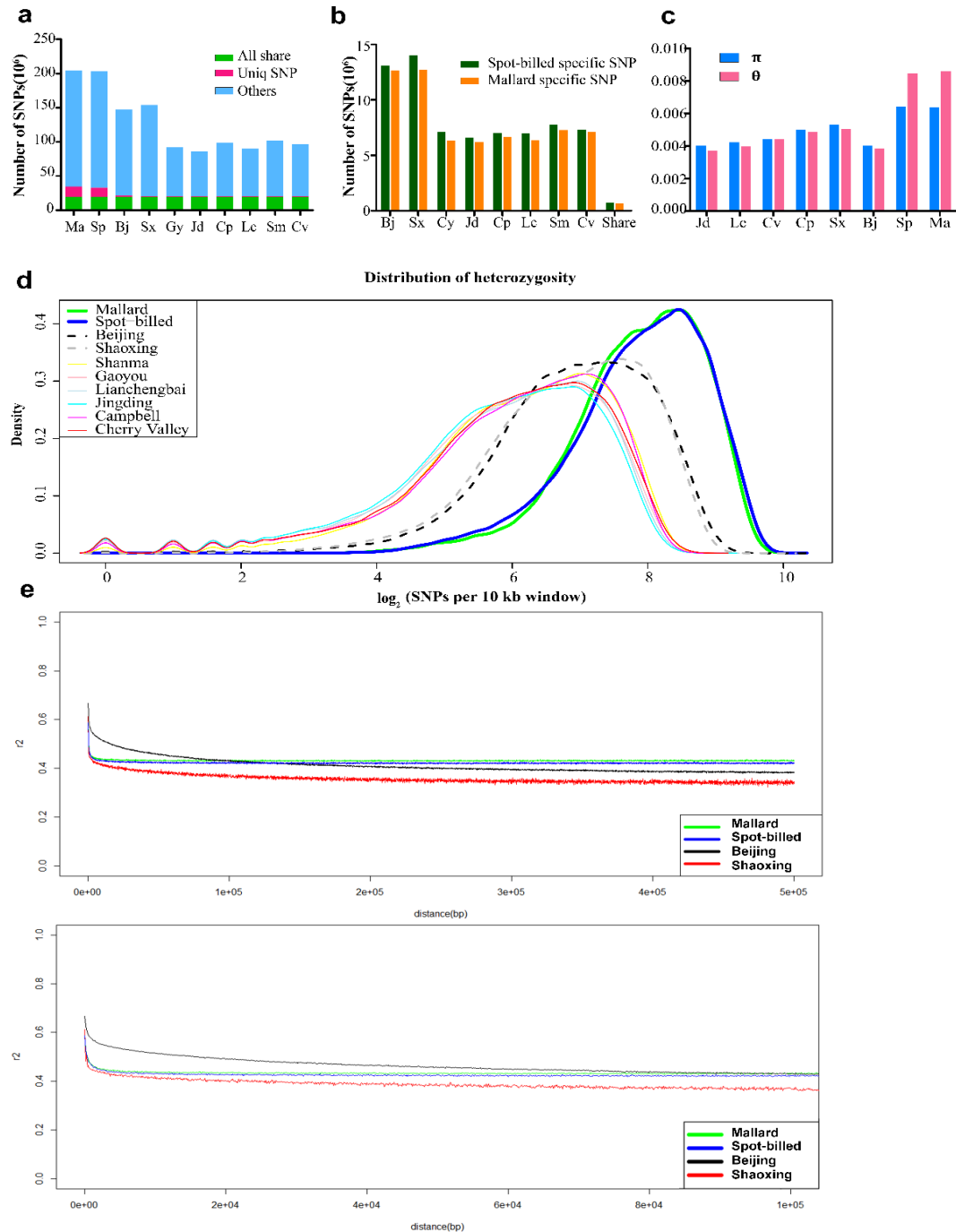


Fig. S12. Distribution of genetic variation and characteristics for two wild and eight domestic duck populations.

Mallard, Spot-billed, Beijing, Shaoxing, Gaoyou, Jinding, Campbell, Lianchengbai, Shanma and Cherry valley are represented by Ma, Sp, Bj, Sx, Gy, Jd, Cp, Lc, Sm, Cv. (a) Total number of SNPs. (b) Total number of mallard and spot-billed specific SNPs segregating in each domestic population. (c) Genetic diversity was measured by nucleotide diversity π and Watterson's estimator Θ . (d) Distribution of heterozygosity. (e) LD decay measured by r^2 estimator against distance (50 kb up and 10 kb down) between the SNPs in mallard, spot-billed, Beijing and Shaoxing populations.

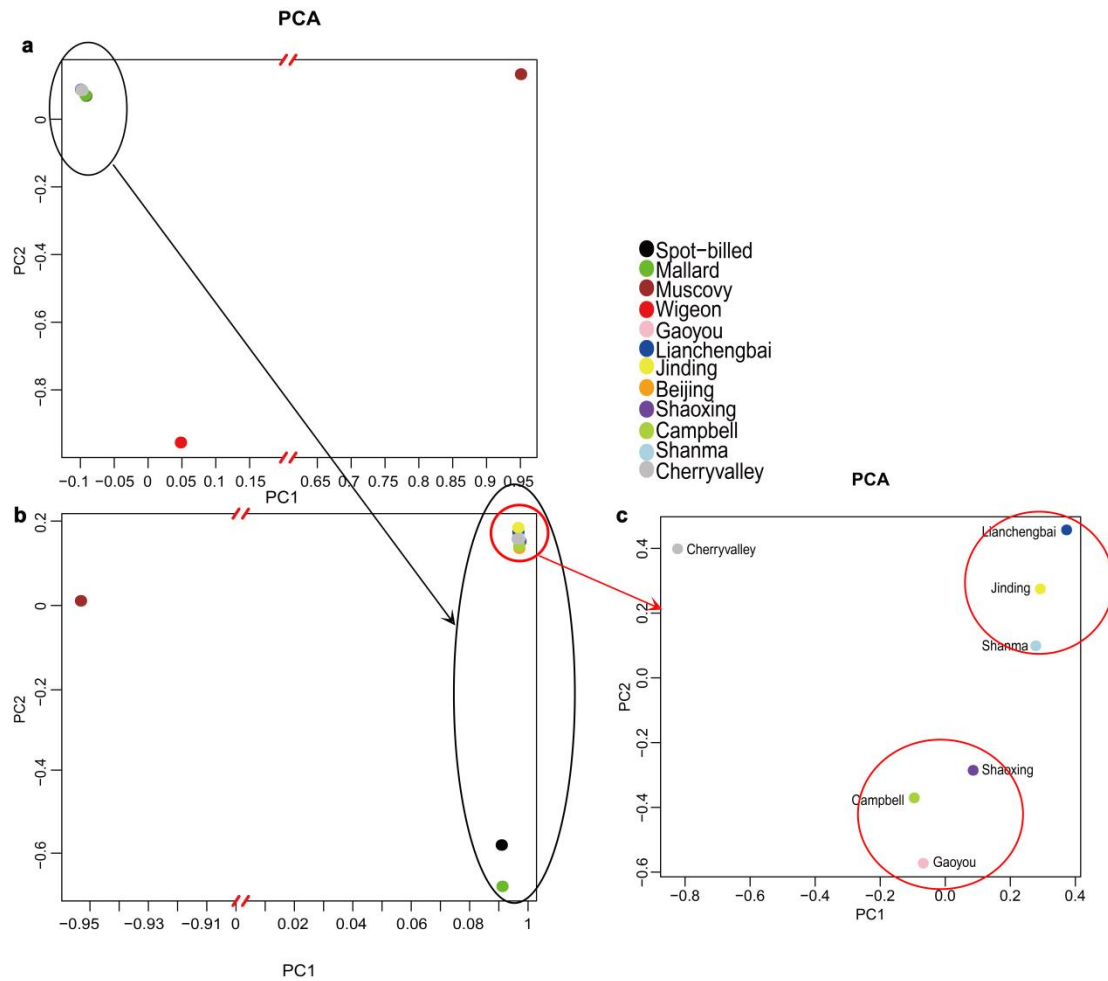


Fig. S13. Principal component analysis of wild and domestic ducks using population as unit.

(a) PCA analysis with genome wide SNPs set from 8 domestic duck breeds and three wild duck species, that having population data, together with Muscovy ducks. Two wild duck (mallard and spot-billed) and eight domestic duck populations were clustered in the top left, while the wigeon populations (consist of 3 individuals) and the muscovy duck population showing large distance with eight domestic duck populations. (b) PCA analysis on eight domestic duck populations and two wild duck (mallard and spot-billed) populations along with one muscovy duck population. It shows that 8 domestic duck populations cluster together and clearly separated from Mallard and Spot-billed in the second component. (c). PCA analysis on seven domestic duck populations. It shows that these populations clustered according to their geographical distributions.

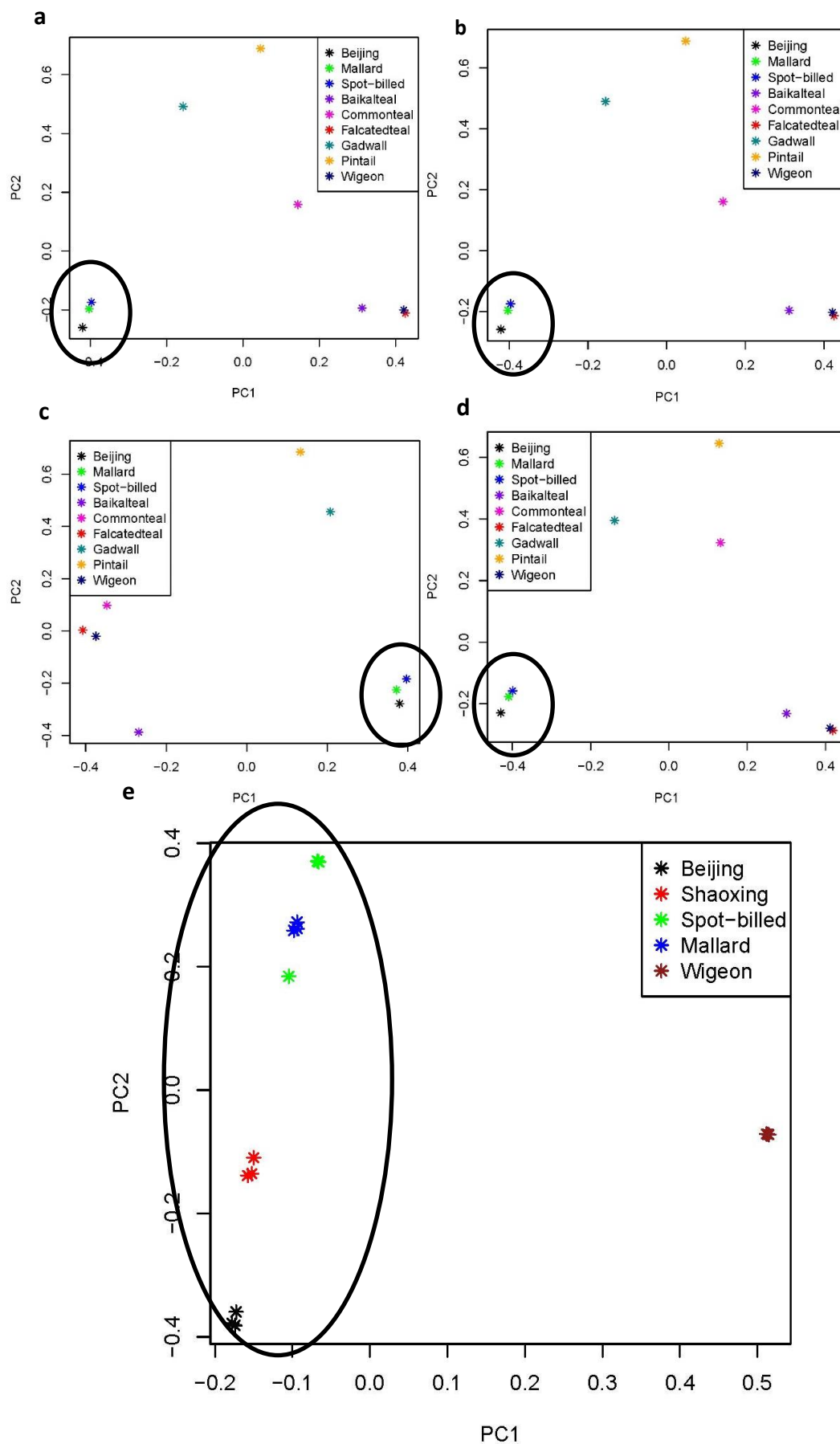


Fig. S14. Principal component analysis of wild and domestic duck with same

coverage and sample size for each population.

(a-d) shows PCA analysis on eight wild species (one individuals each, random sample reads into 12× that is equal to Beijing individual) and a Beijing individual with different data process strategy. a) using LD pruned SNPs from all scaffolds of Beijing assemble. b) using LD pruned SNPs from the scaffolds regions which were homologous to chicken autosomes. c) using LD pruned SNPs from the scaffolds regions which were homologous to chicken Z chromosome. d) using LD pruned SNPs from all scaffolds of mallard duck genome assemble. e) 3 individuals for each of the 5 populations (mallard duck, Spot-billed, Beijig, Shaoxing and wigeon, random sample reads into 3× for each individuals).

Black cycles indicate spot-billed, mallard duck and domestic duck are clustered together.

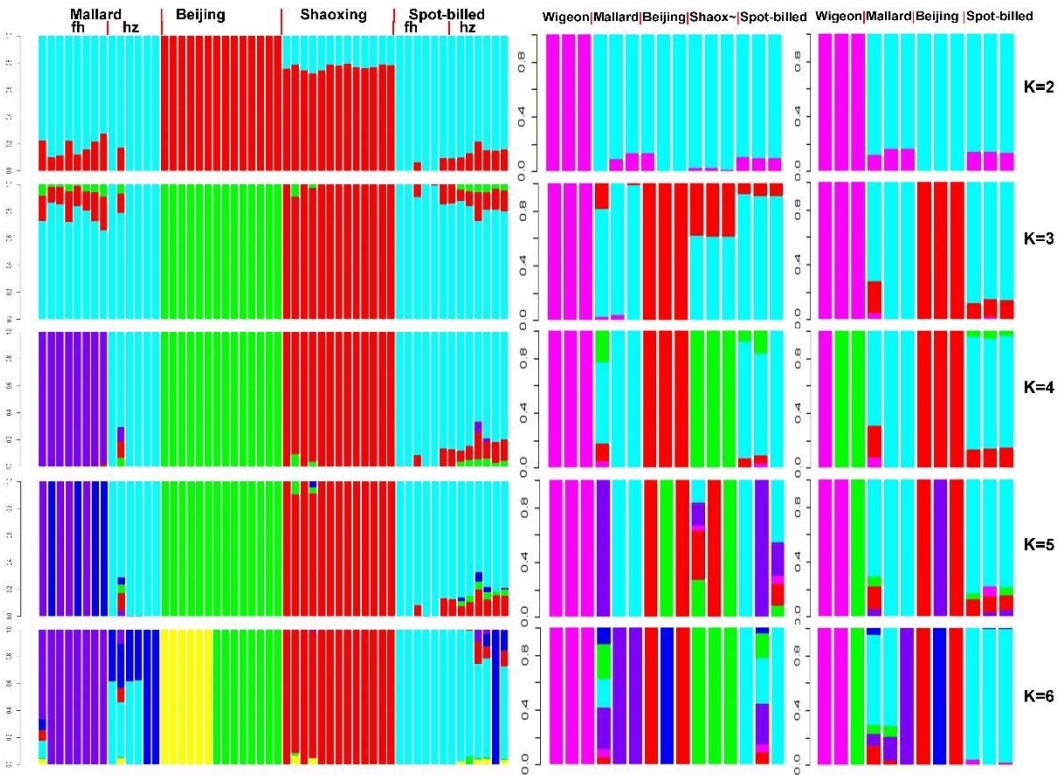


Fig. S15. Genetic structure analysis with equal sample size and same coverage of each population for different populations combinations.

Samples in left panel were random sampled to 3-fold coverage. Beijing duck samples were from a designed population, which were from a same population and formed a high growth line and low growth line after several generations selection. Samples in middle panel were random sampled to 3-fold coverage and three samples for each of the 5 populations. Samples in the right panel were random sampled to 12-fold coverage and three samples for each of the 4 populations. K range from 2 to 6.

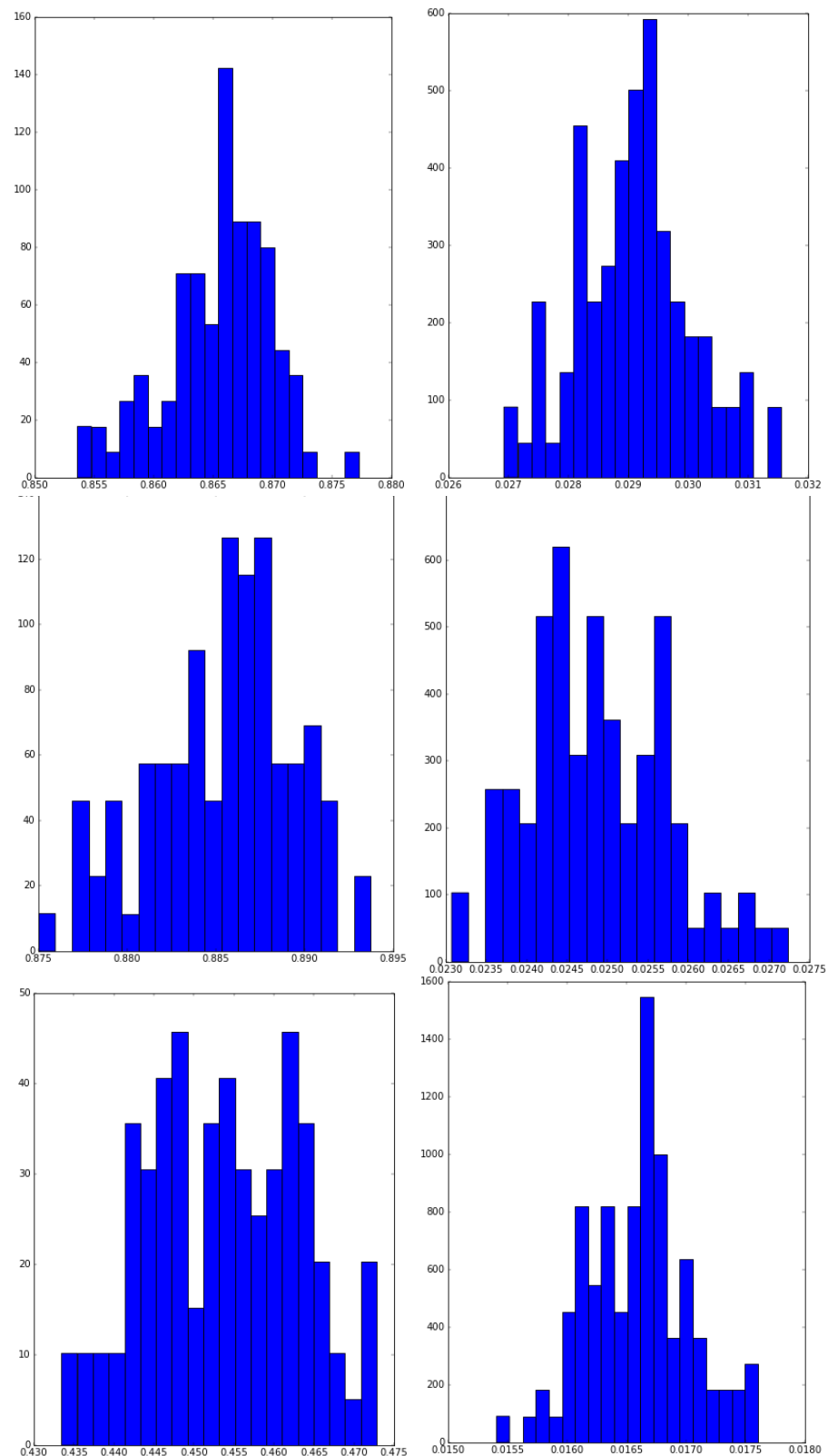


Fig. S16. Split without migration model conventional bootstrap results.

Panels from up to down corresponding to simulations of mallard and domestic duck divergence, spot-billed and domestic duck divergence, mallard duck and spot-billed duck divergence, separately. Left picture of each panels shows the distribution of s , whereas right one shows the distribution of T_s .

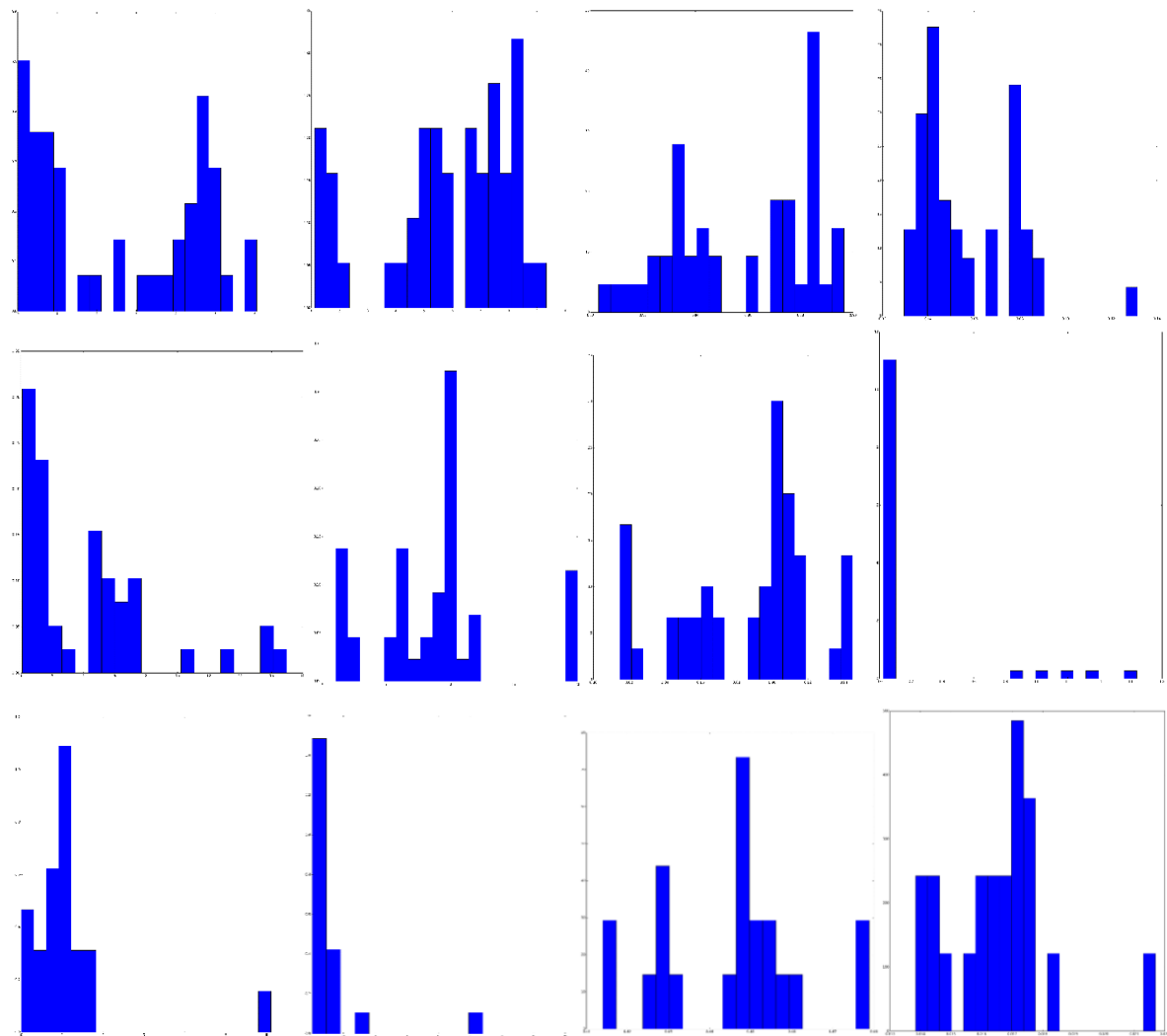


Fig. S17. Split with migration model conventional bootstrap results.

Panels from up to down corresponding to simulations of mallard and domestic duck divergence, spot-billed and domestic duck divergence, mallard duck and spot-billed duck divergence, separately. Columns from left to right corresponding the distributions of m12, m21, s, Ts parameters.

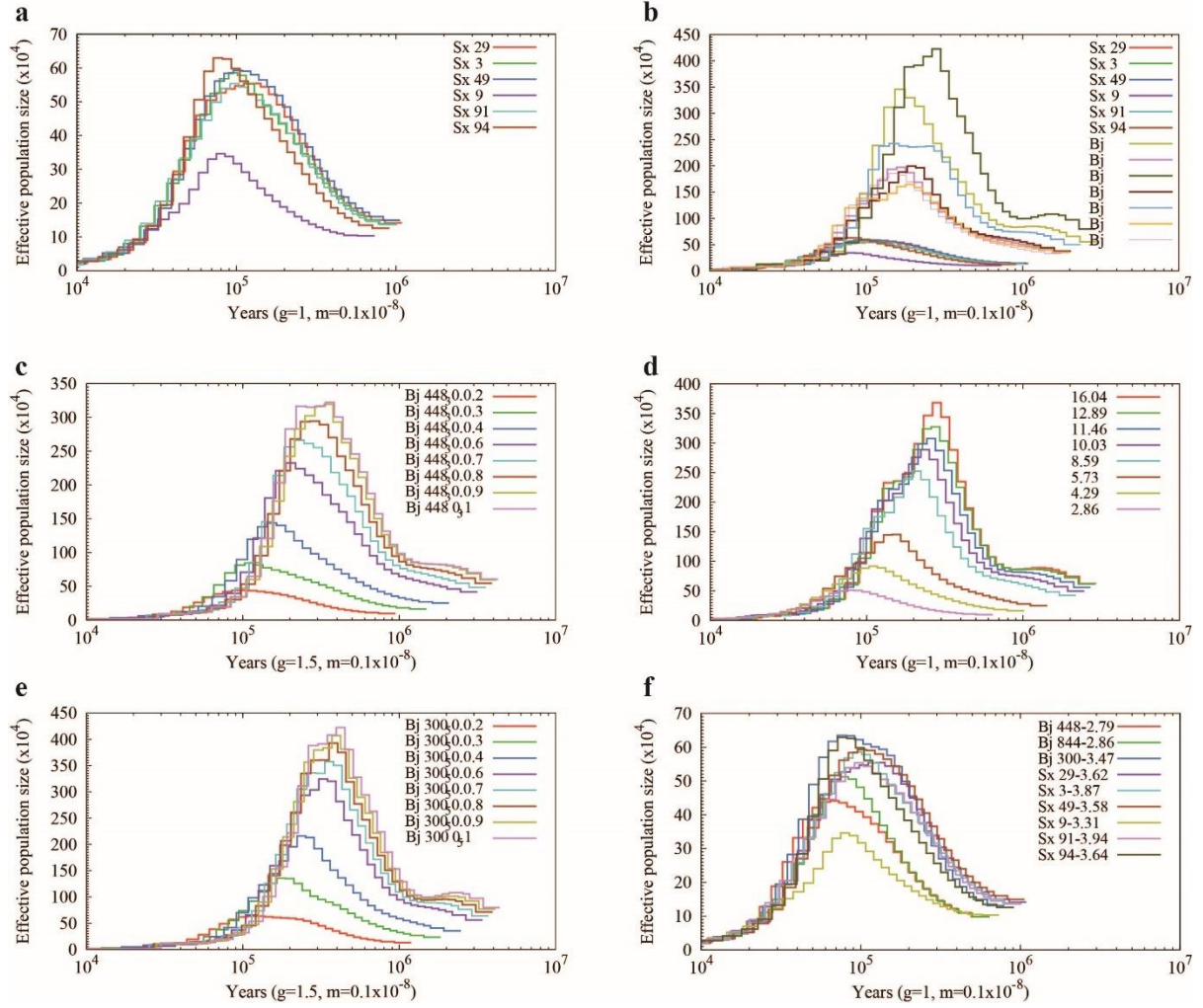
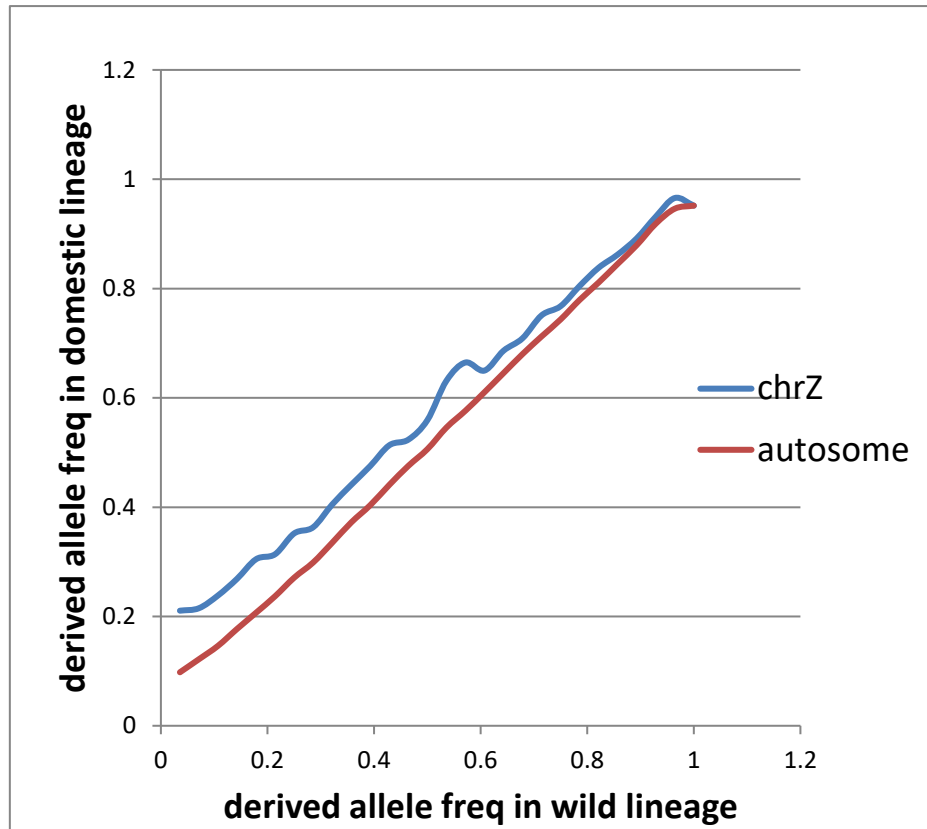


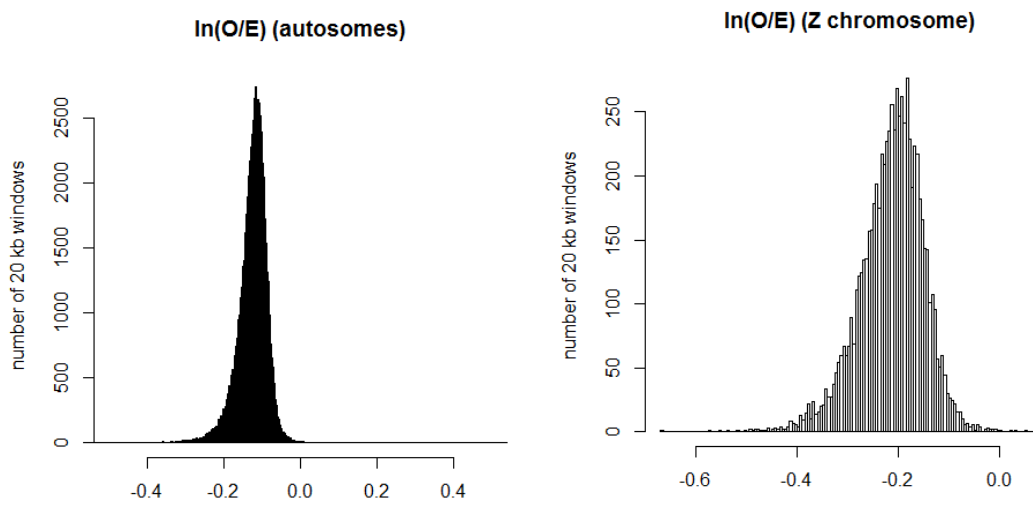
Fig. S18. Comparison of PSMC between Beijing (Bj) and Shaoxing (Sx) ducks with different coverage data.

(a) Demographic history of six Shaoxing ducks with a coverage of 2- to 3.5-fold. (b) Comparison demographic history of six Shaoxing (2- to 3.5-fold) and Beijing (13.5-fold) ducks. (c-e) PSMC analyses of Beijing or Shaoxing ducks based on different coverage of random sample reads. Sample ID in c, d and e are 44448_30, 43844_32, 41300_50, respectively. Coverage are represented in different color lines in legend. (f) PSMC curve of Beijing and Shaoxing ducks inferred with comparable coverage reads are similar.

a



b



The figure consists of two histograms side-by-side. The left histogram is titled 'S (autosomes)' and shows a very narrow, tall distribution of S values centered at 0. The y-axis is labeled 'number of 20 kb windows' and ranges from 0 to 3000. The x-axis is labeled 'S value' and ranges from -10 to 15. The right histogram is titled 'S (Z chromosome)' and shows a broader distribution of S values centered at 0. The y-axis is labeled 'number of 20 kb windows' and ranges from 0 to 150. The x-axis is labeled 'S value' and ranges from -6 to 4.

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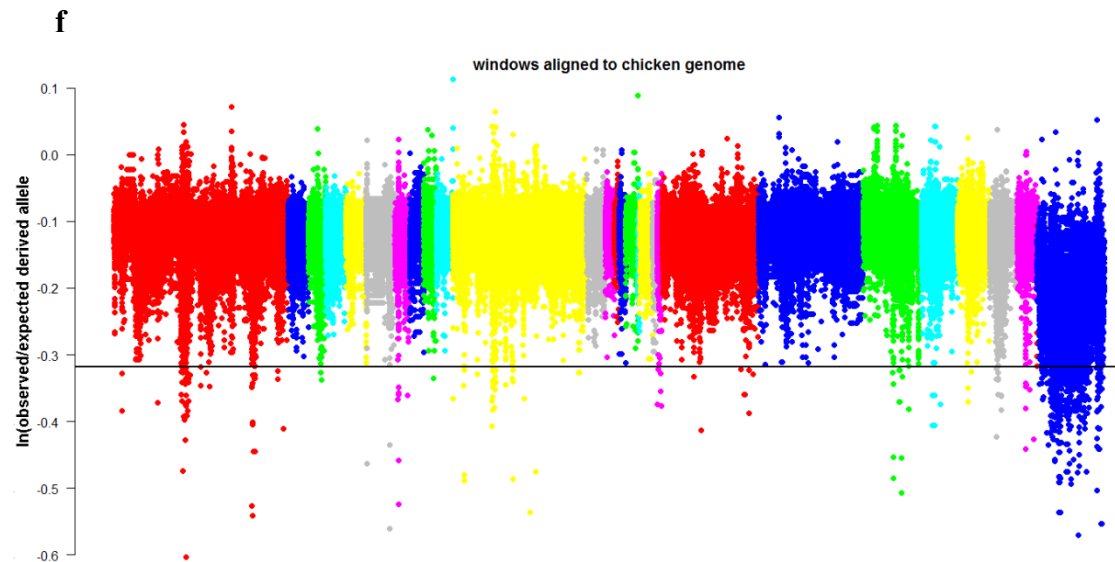
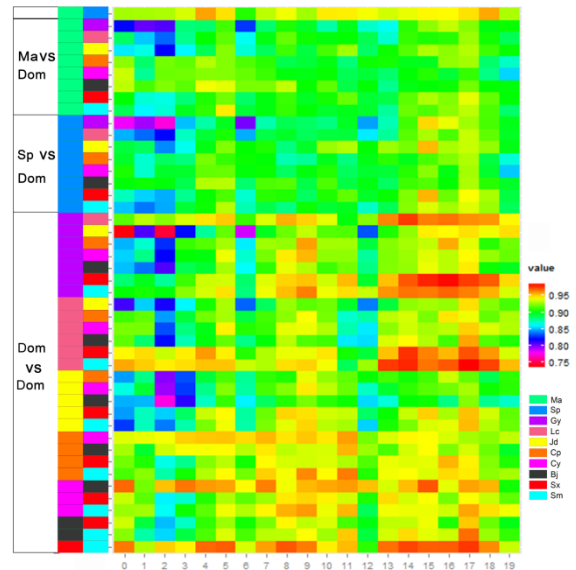
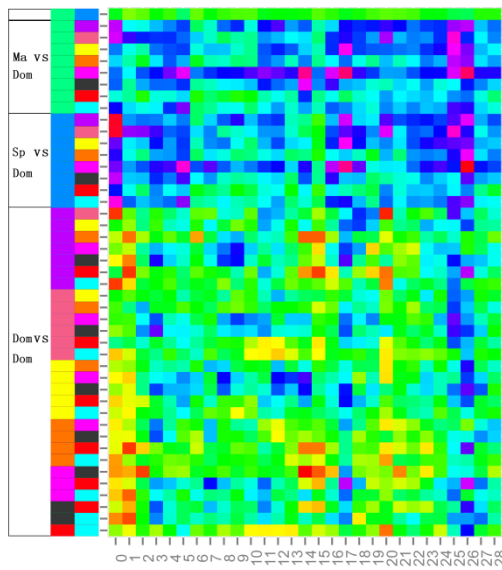
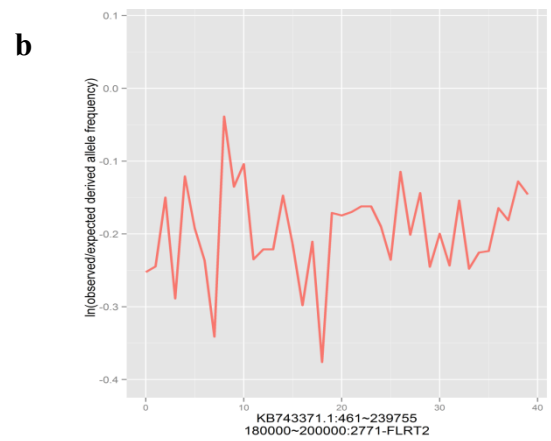
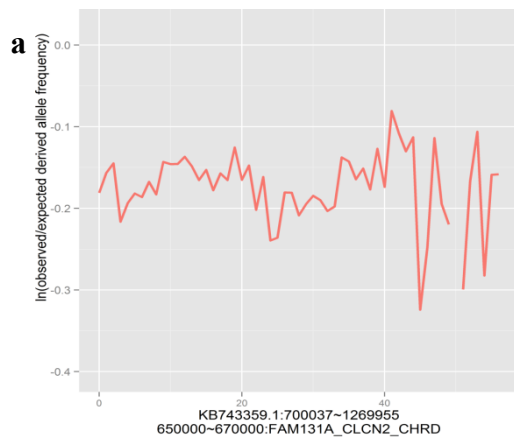


Fig. S19. Whole genome screen for S score in wild duck with 20 kb window sliding by 10 kb.

(a) Correlation of observed derived allele frequency between wild (combine mallard and spot-billed) and domestic duck (combine all 8 domestic duck breeds) lineage. (b) Distribution of the $\ln(\text{observed/expected derived alleles frequency})$ for autosomes and Z-chromosome. (c) Distribution of the S score for autosomes and Z-chromosome. (d) The negative end of the S score distribution presented along duck autosomes. Genes that located in or near selected regions were showed its symbol. (e) The negative end of the S score distribution presented along duck Z-chromosome. (f) The S score distribution presented along scaffolds that is arranged according to their alignment to the chicken genome.



1496

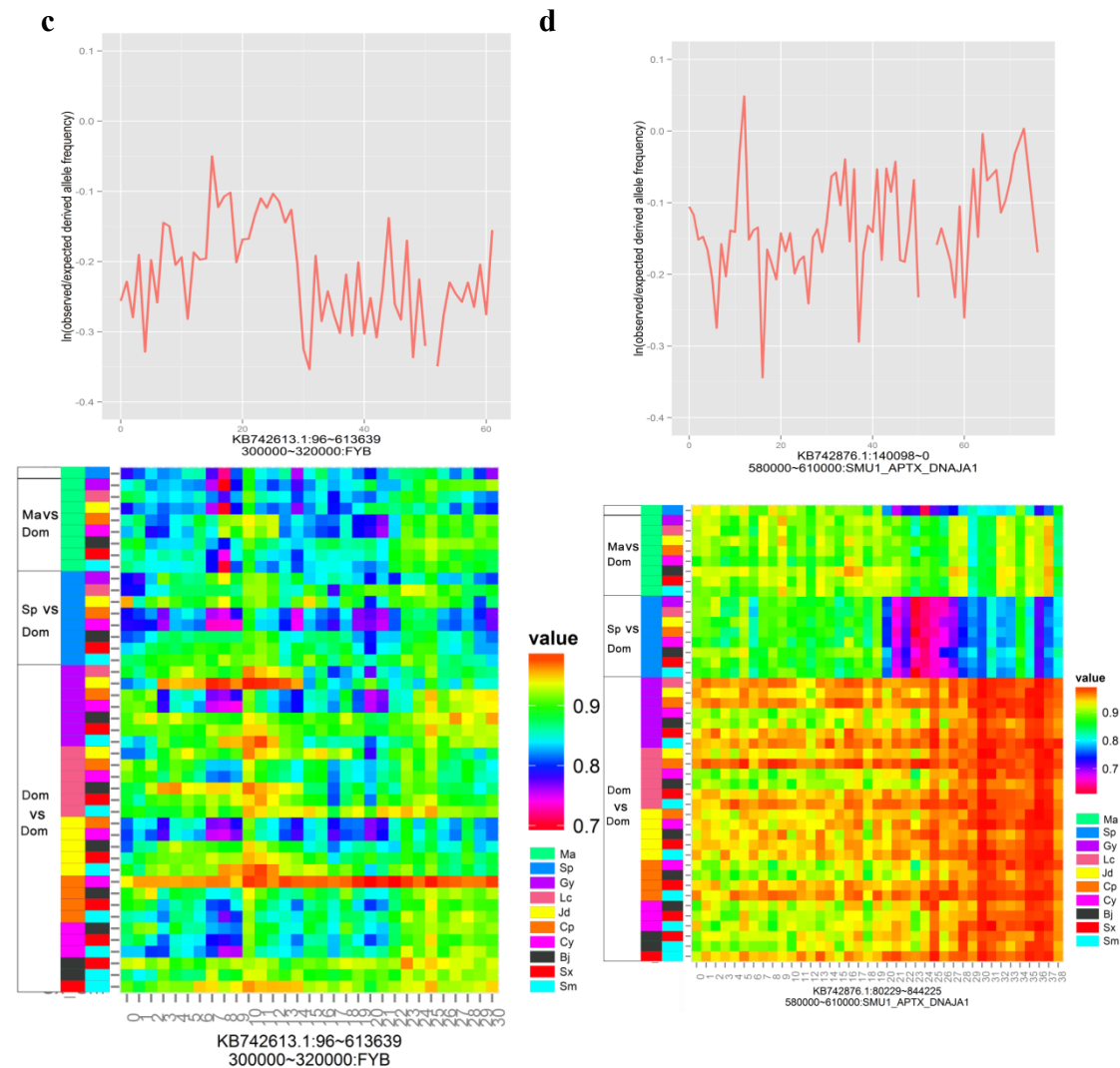


Fig. S20. Details of four putative selective sweep regions in the early stage of the wild duck lineage.

Each panel shows the distribution of the log-transformed value of the ratio for the observed/expected derive allele frequency using non-overlap 10 kb window consist (up) and scores between pairwise comparisons in 20 kb non-overlap windows (low). (a) *CHRD* locus. (b) *FLRT2* locus. (c) *FYB* locus. (d) *DNAJA1* locus.

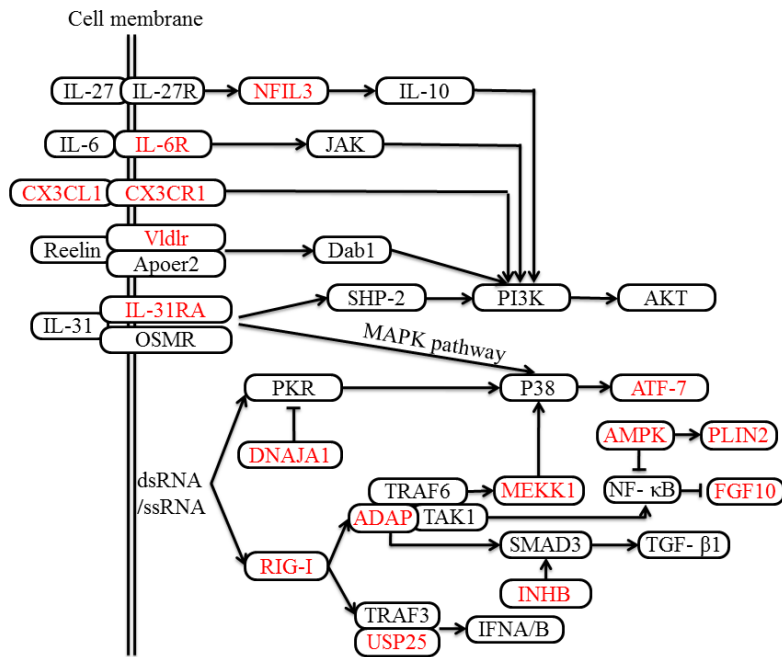


Fig. S21. The RIG-I pathway preferring to be under positive selection.

Sixteen immune genes detected to be under positive selection were in red. Three of them (RIG-I, DNAJA1 and ADAP, MAP3K1) are involved in the host immune response to influenza A viruses.

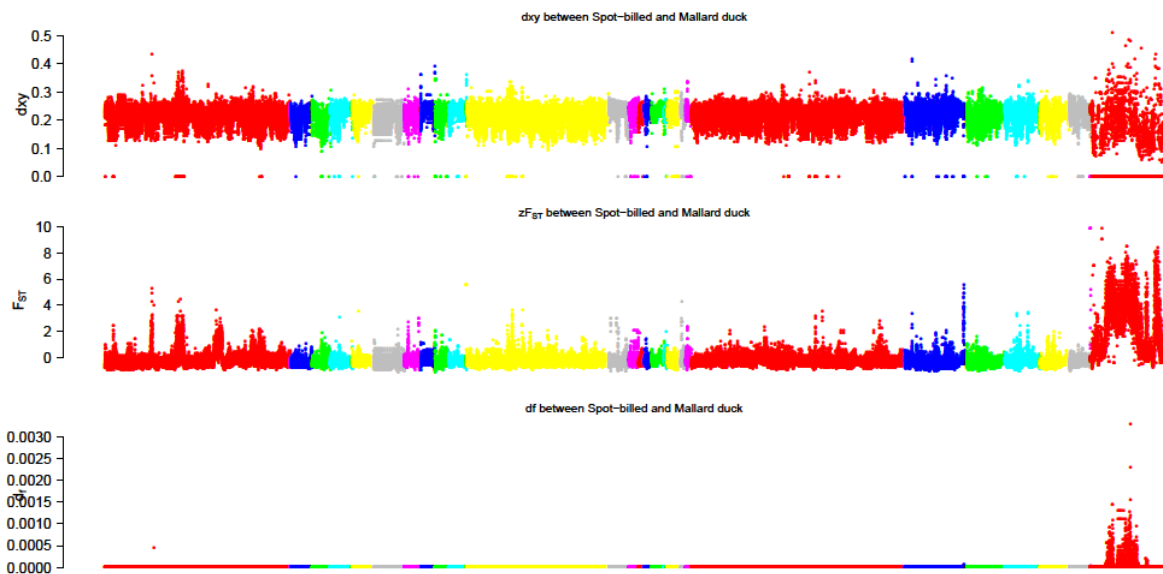
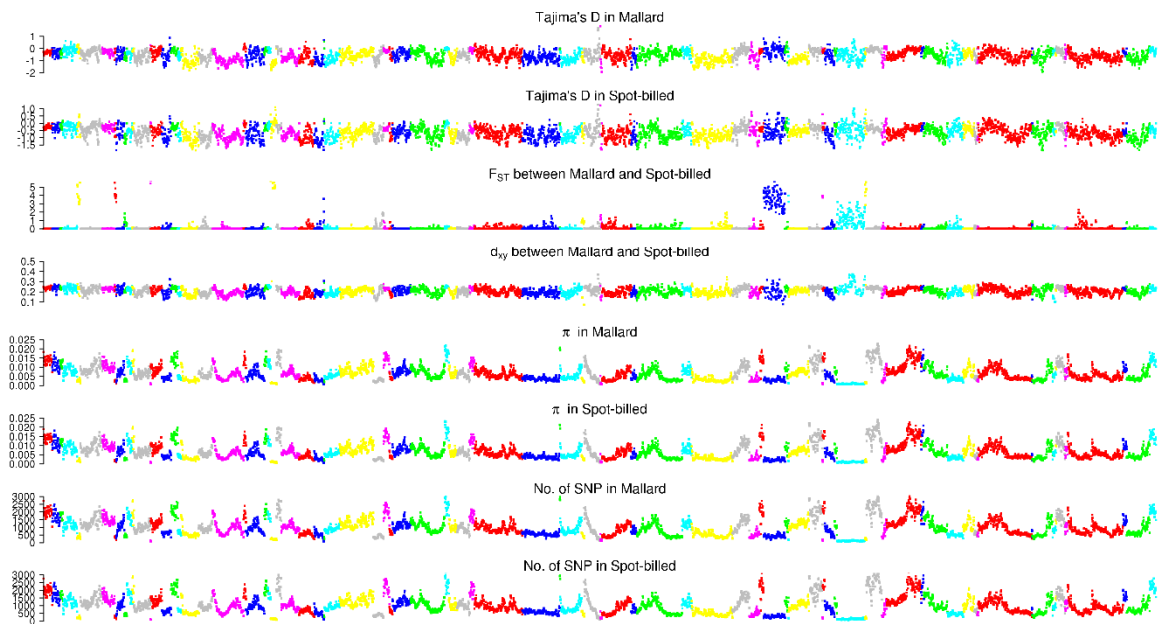


Fig. S22. Distribution of F_{ST} (fixation index), dxy and dfs (fixed difference sites) along Beijing duck scaffolds between mallard duck and spot-billed.

F_{ST} and dfs were counted with 40 kb window by step-size of 20 kb. The duck scaffolds that is shown in red and arranged at the right most are homologous to chicken Z chromosome and referred as duck Z chromosome.

1515

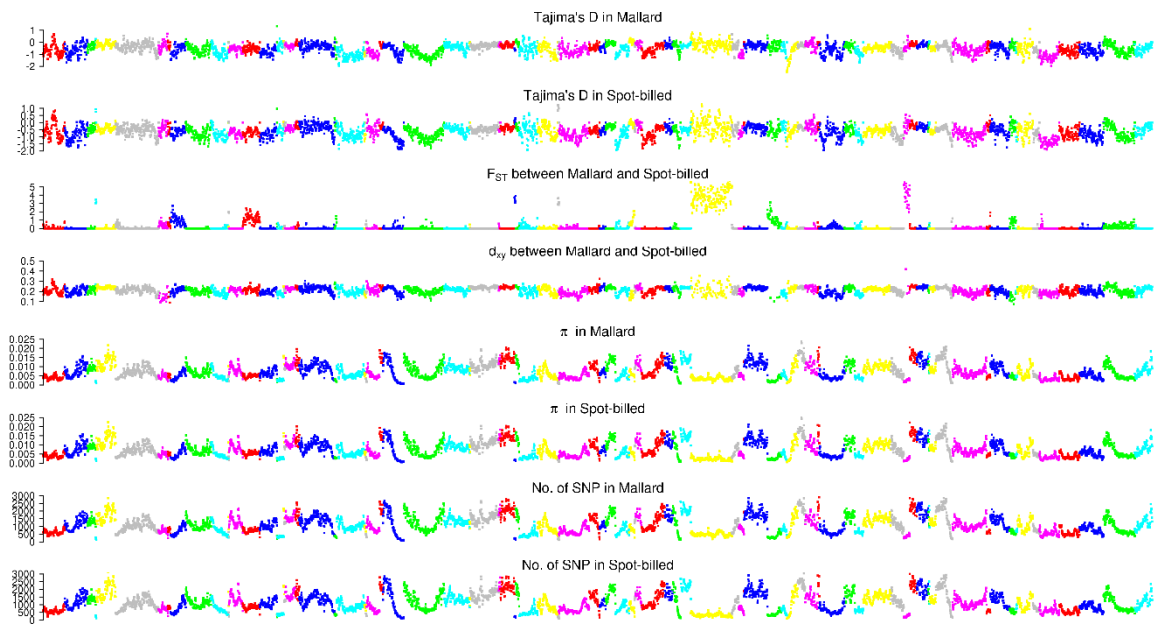
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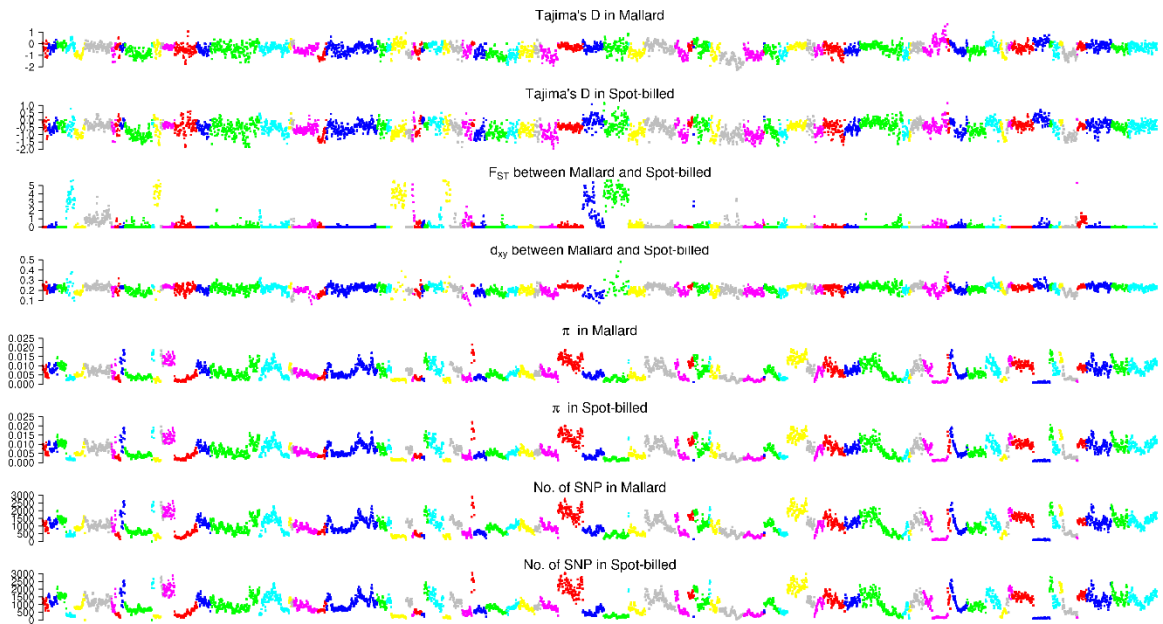
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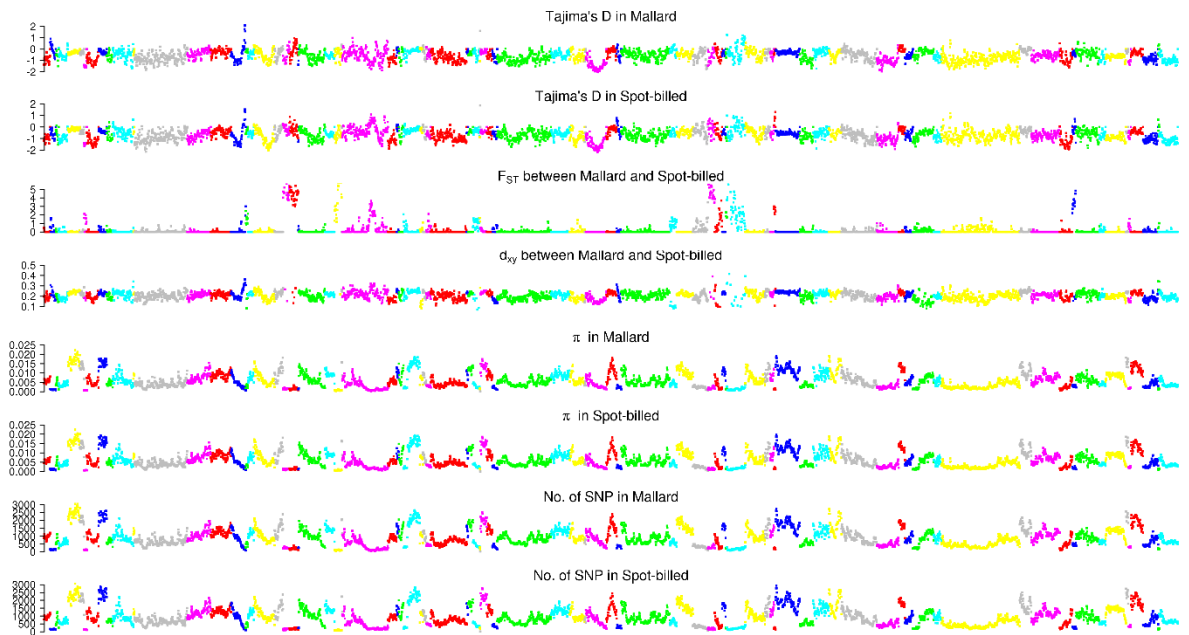
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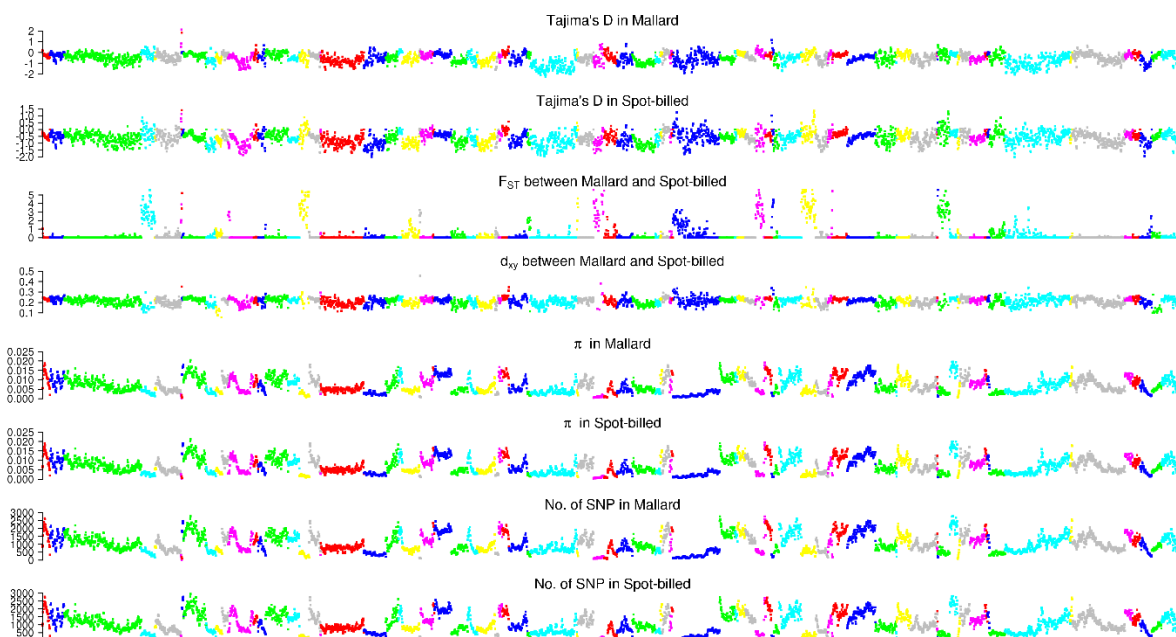
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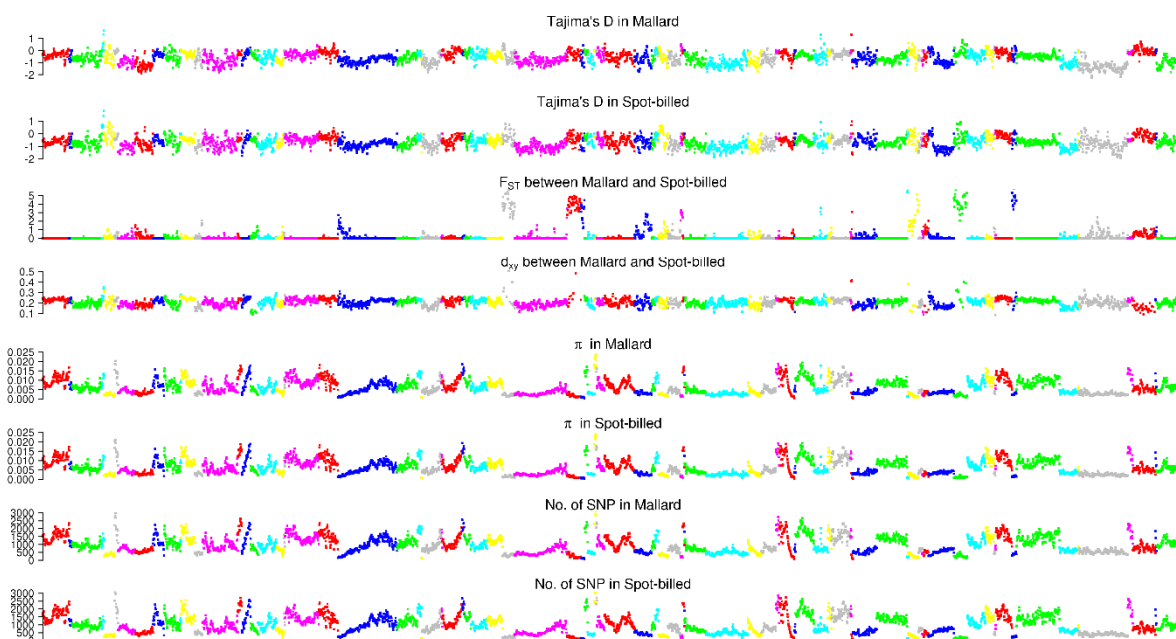
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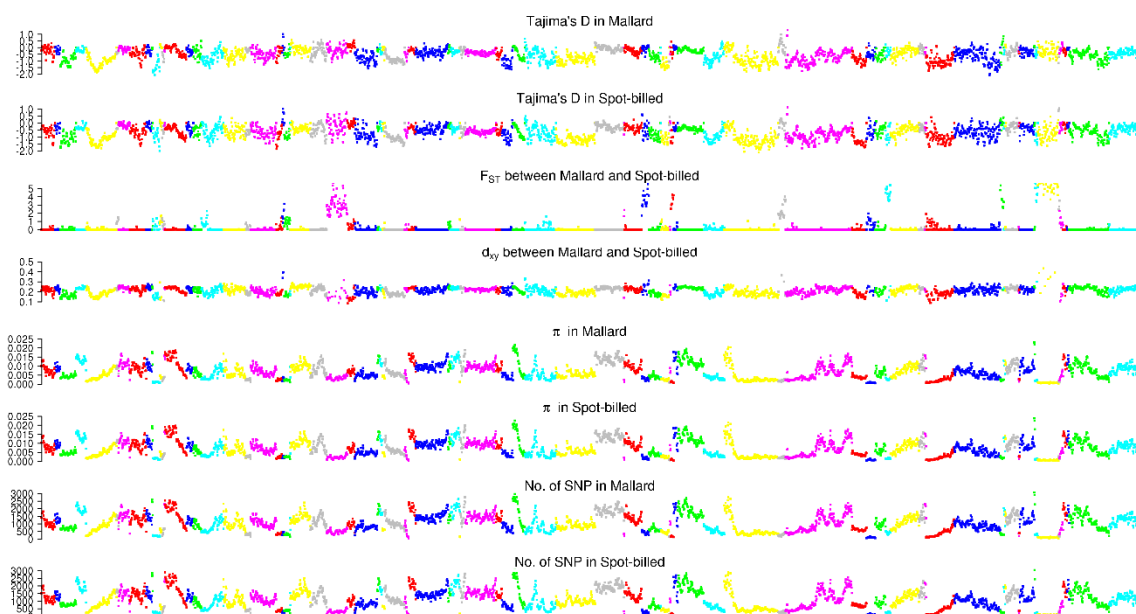
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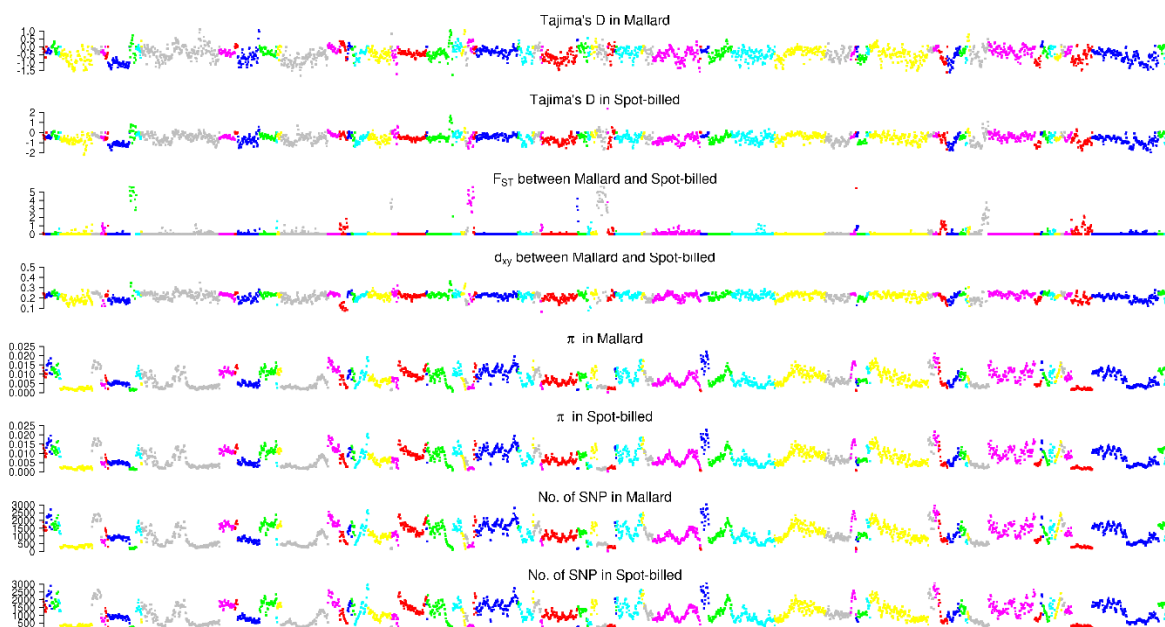
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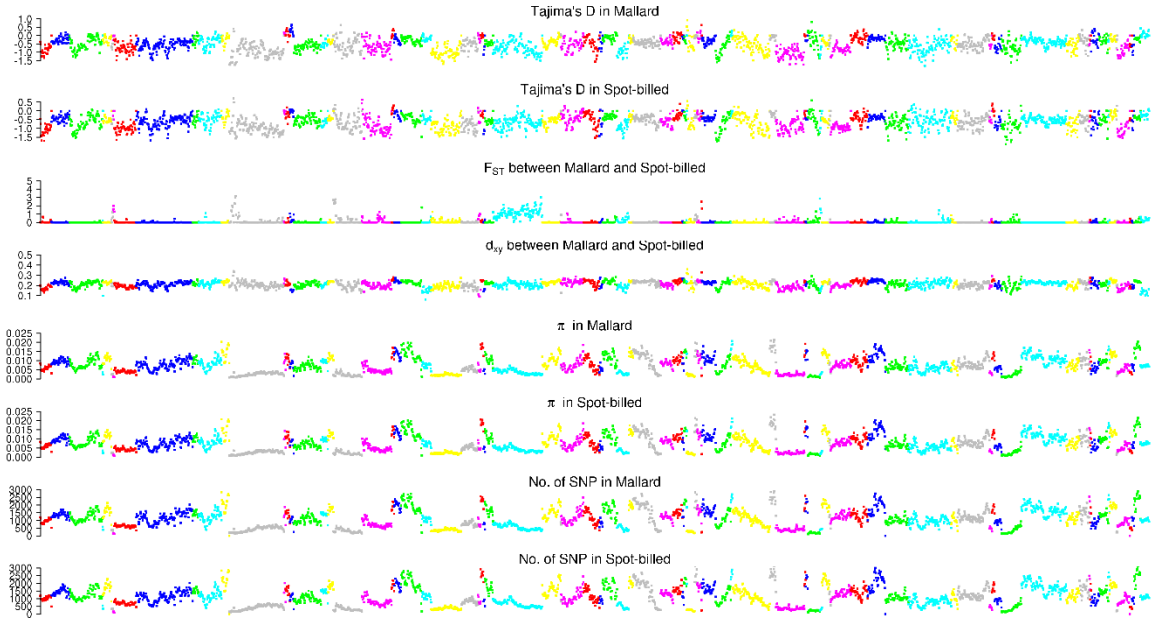
Scaffolds from KB743106.1 to KB743245.1



Scaffolds from KB743248.1 to KB743387.1



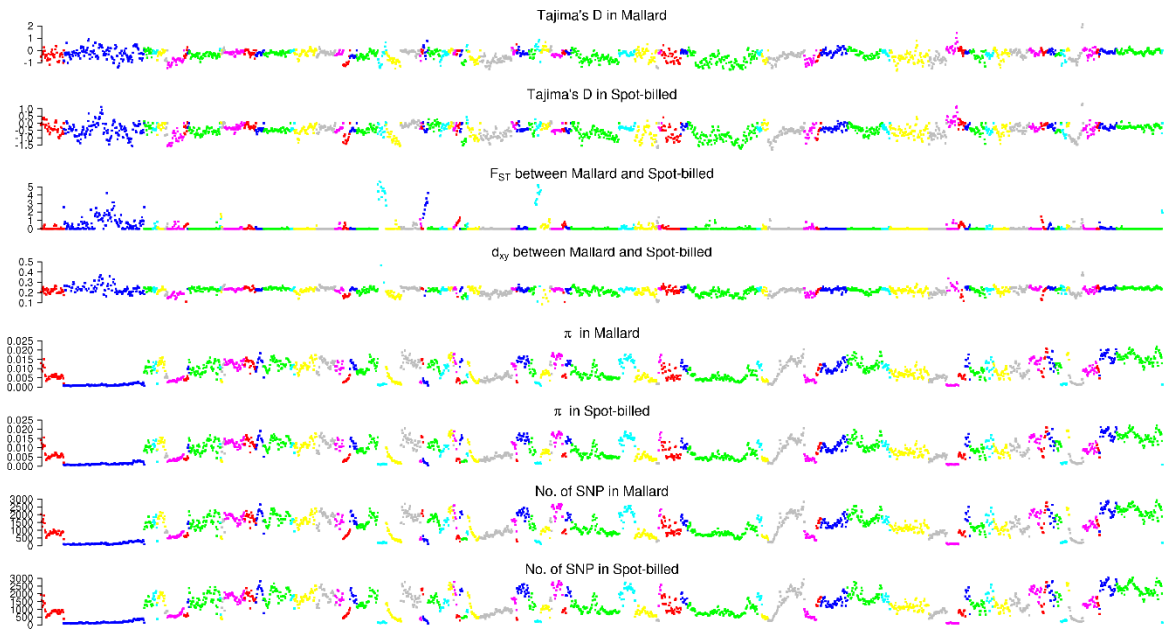
Scaffolds from KB743391.1 to KB743525.1



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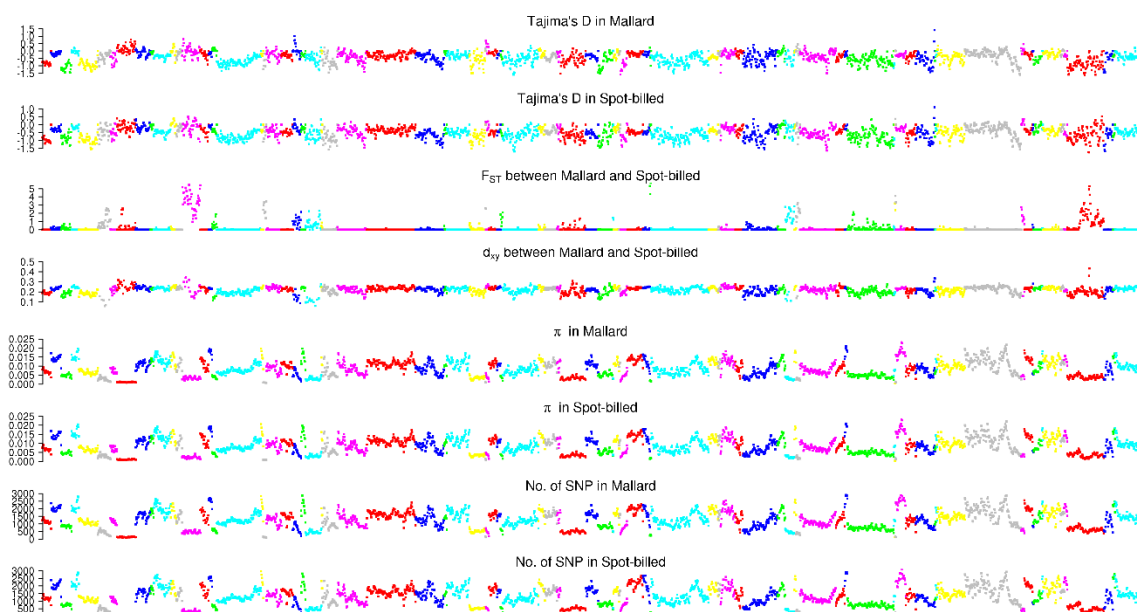
Scaffolds from KB743527.1 to KB743672.1



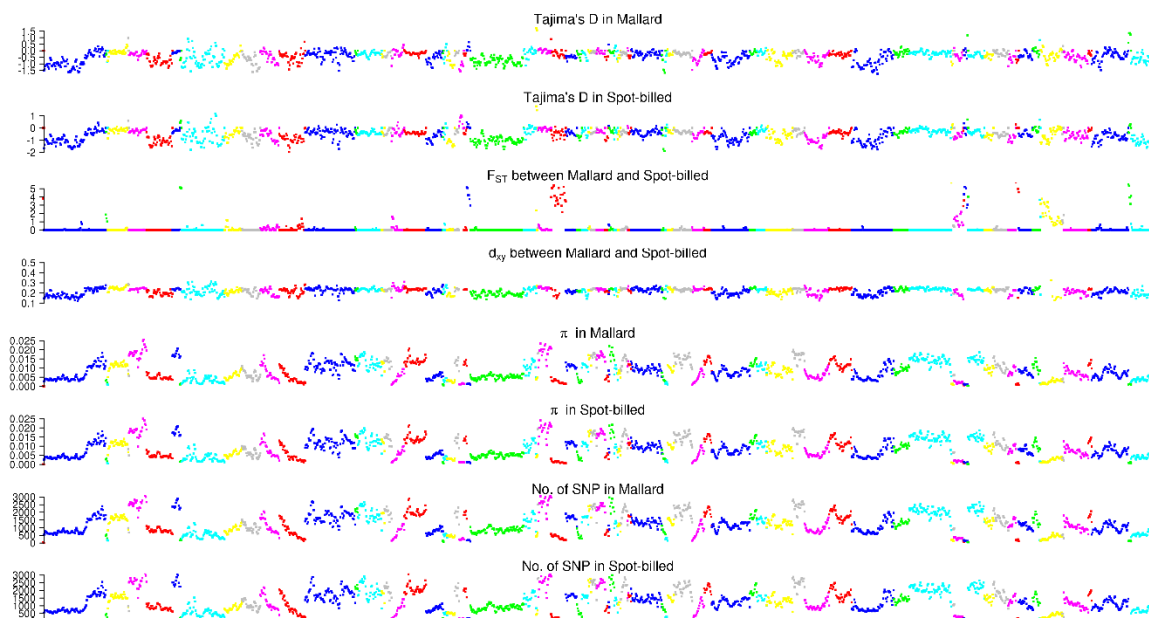
1534

1535

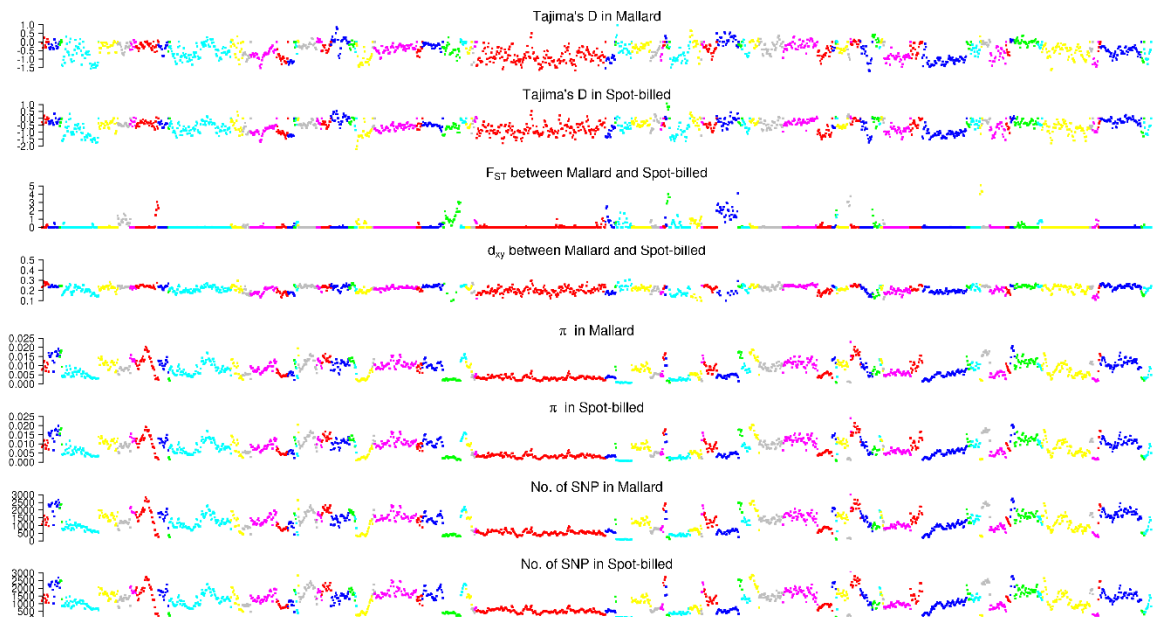
Scaffolds from KB743674.1 to KB743833.1



Scaffolds from KB743839.1 to KB744029.1



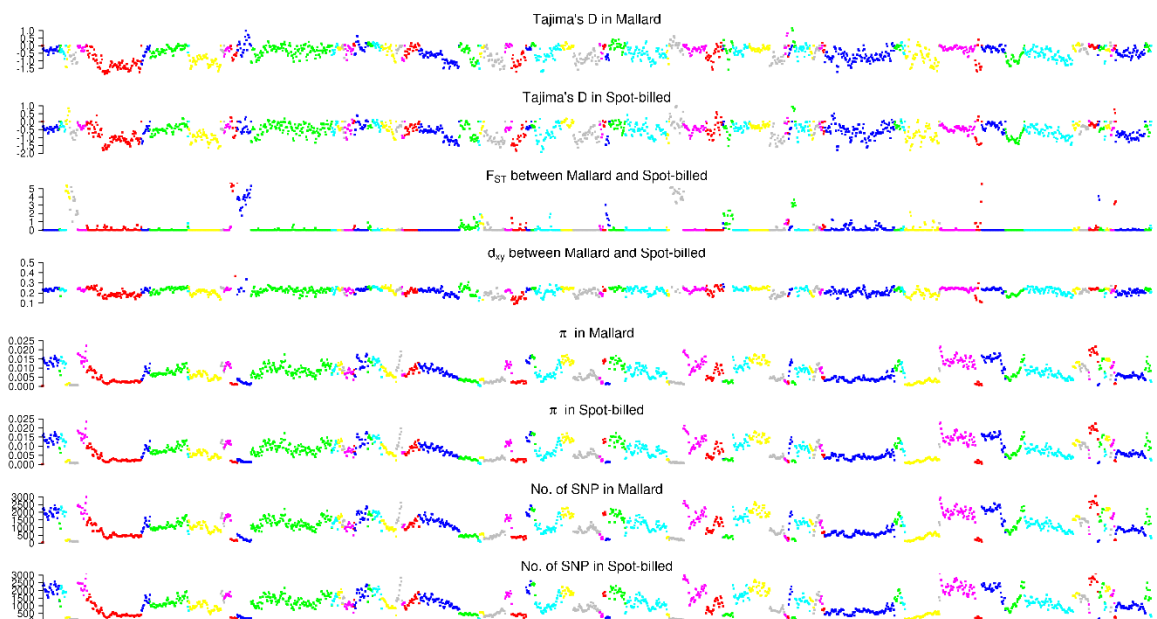
Scaffolds from KB744032.1 to KB744232.1



1540

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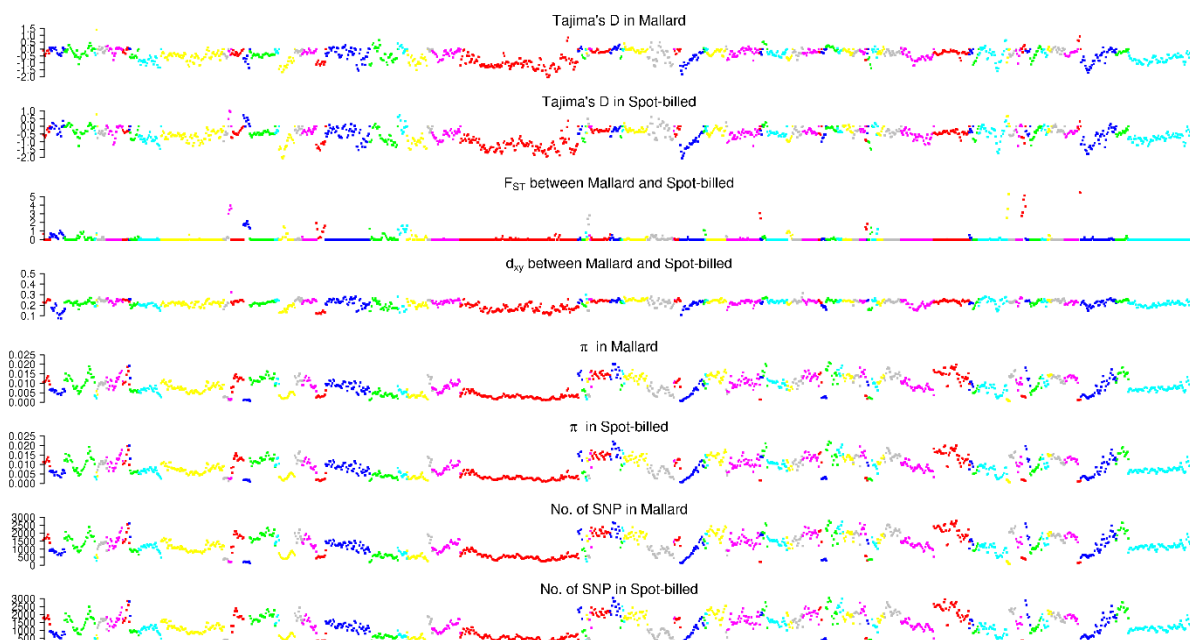
Scaffolds from KB744235.1 to KB744466.1



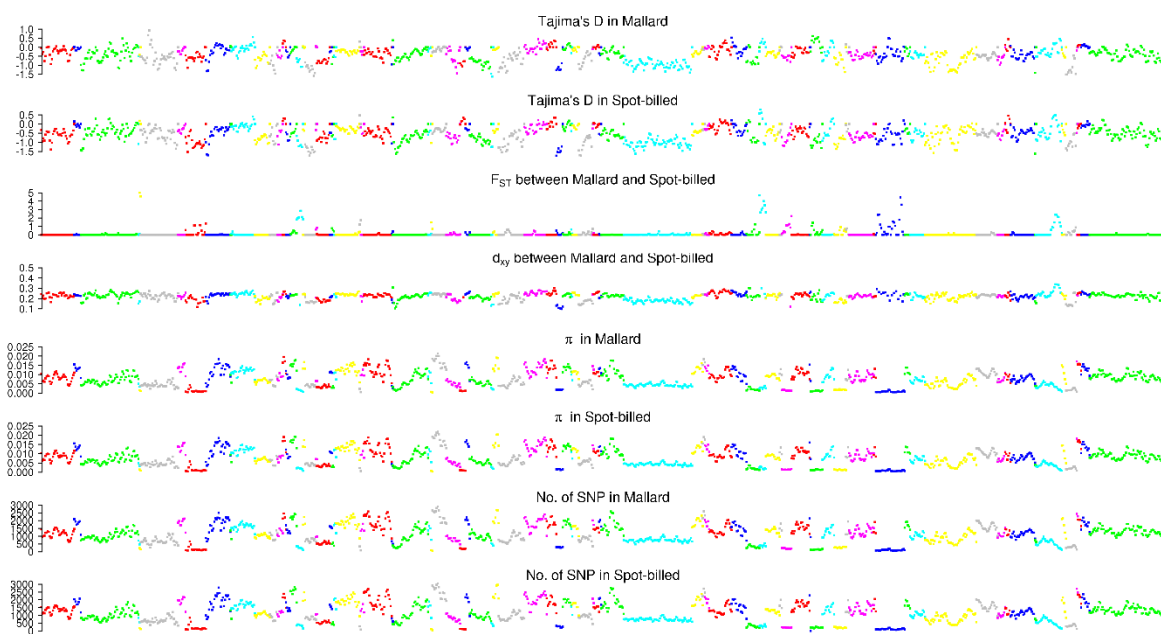
1542

1543

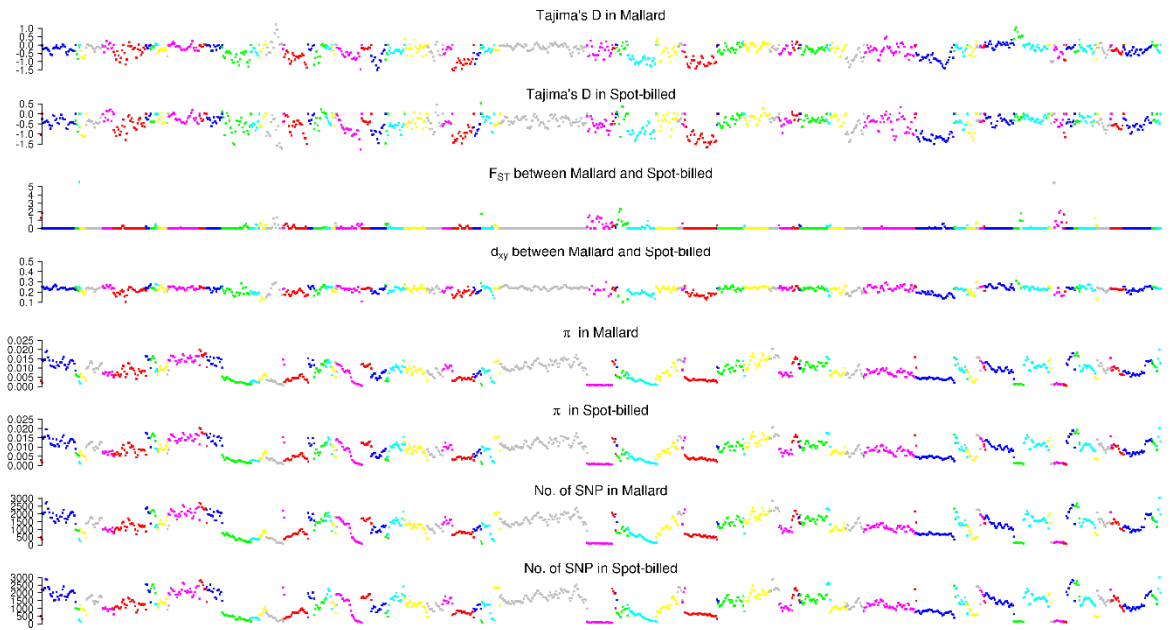
Scaffolds from KB744475.1 to KB744721.1



Scaffolds from KB744725.1 to KB745078.1



Scaffolds from KB745085.1 to KB745764.1



Scaffolds from KB745806.1 to KB747284.1

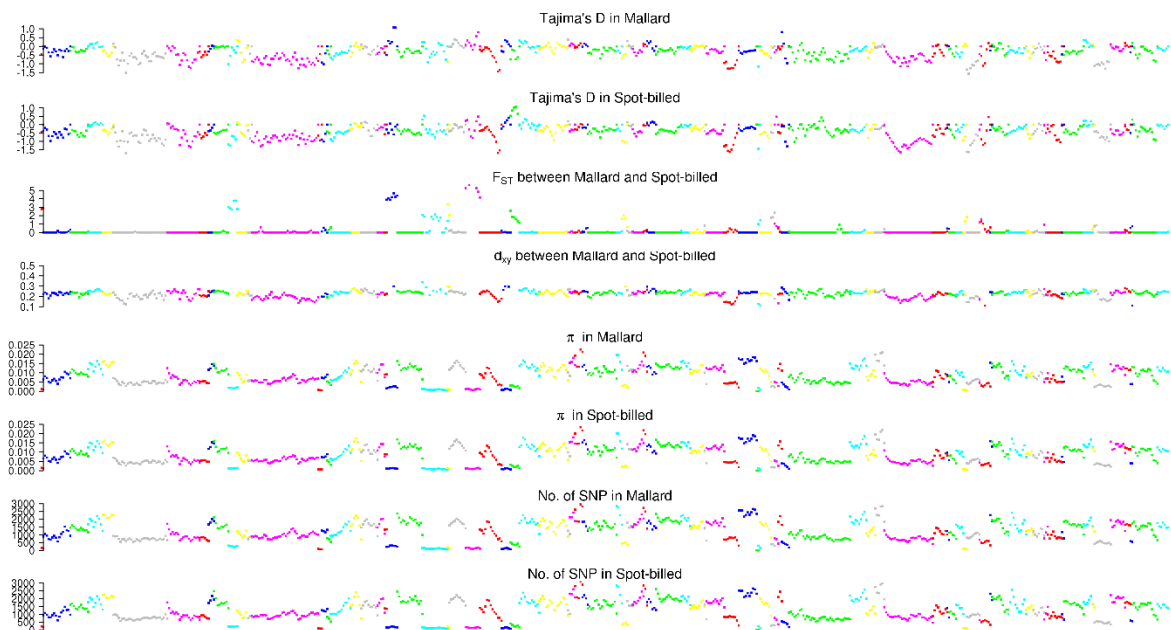


Fig. 23. Distribution of Tajima'D, positive end of zF_{ST} , d_{xy} , number of SNPs per window, nucleotide diversity (π) and zHp along scaffolds in ducks.

These numbers were counted with 40 kb window by step size of 20 kb.

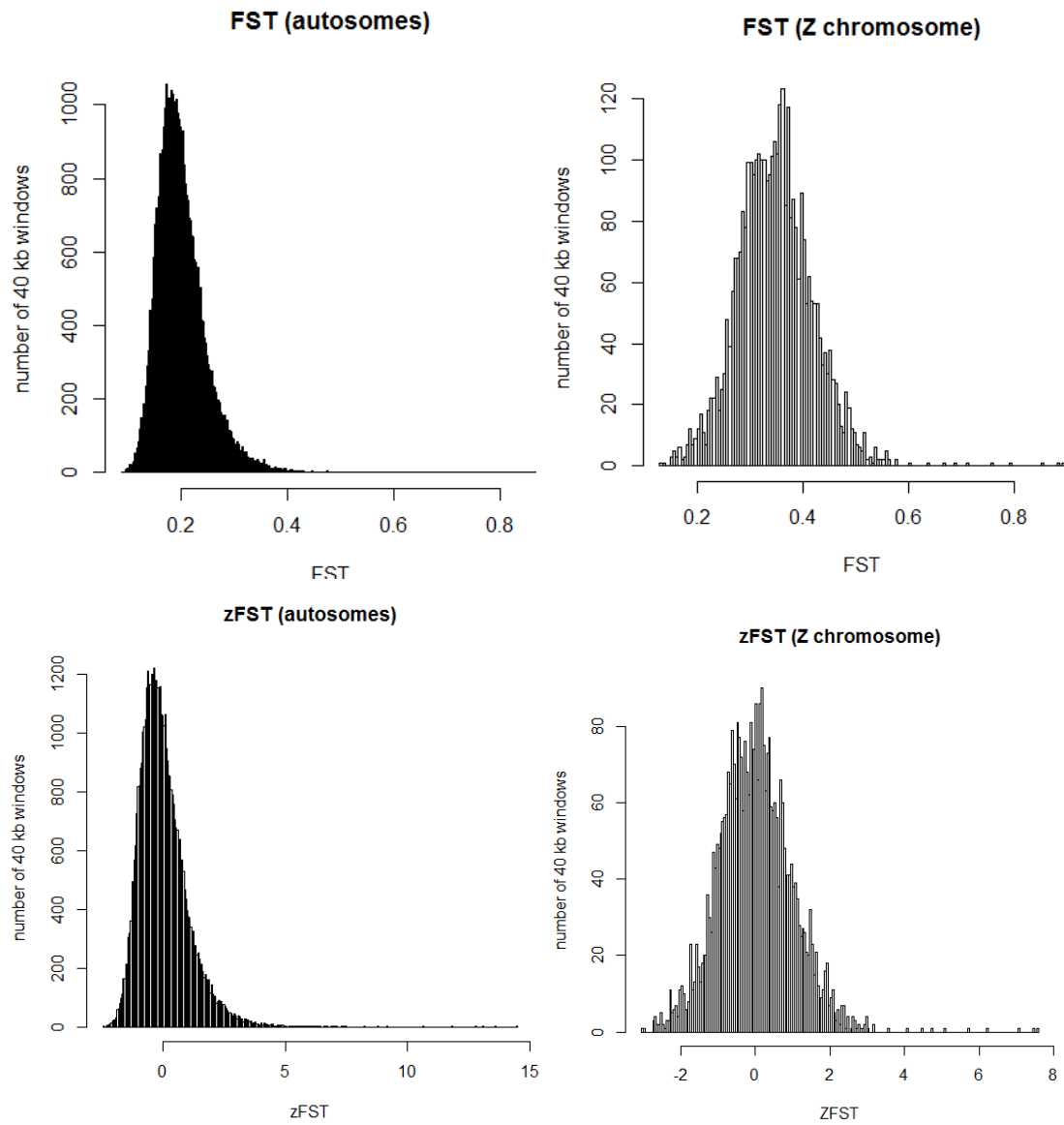


Fig. S24. Distribution of F_{ST}/zF_{ST} between the wild and the domestic duck lineage.

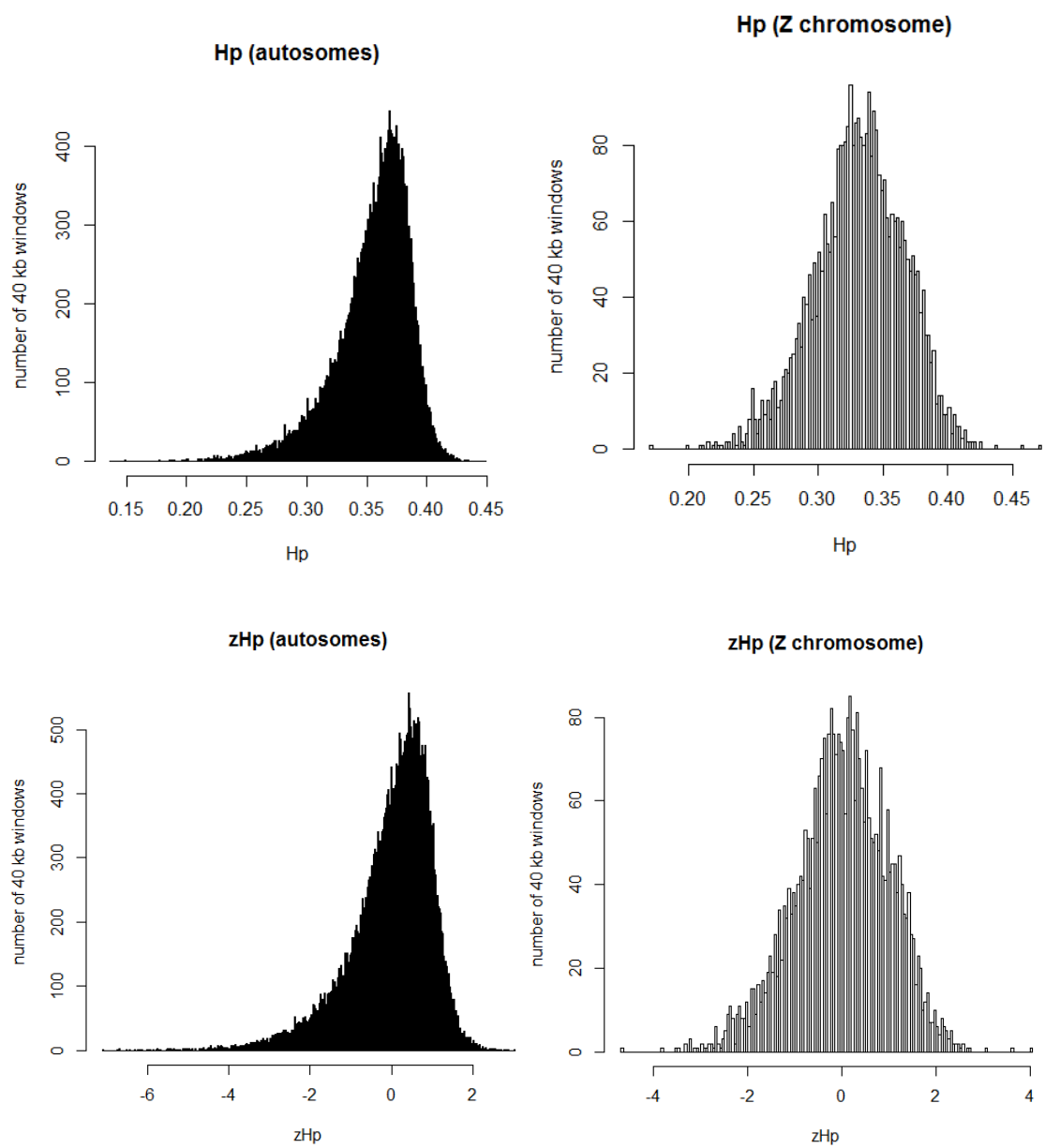


Fig. S25. Distribution of Hp/zHp in the domestic duck lineage.

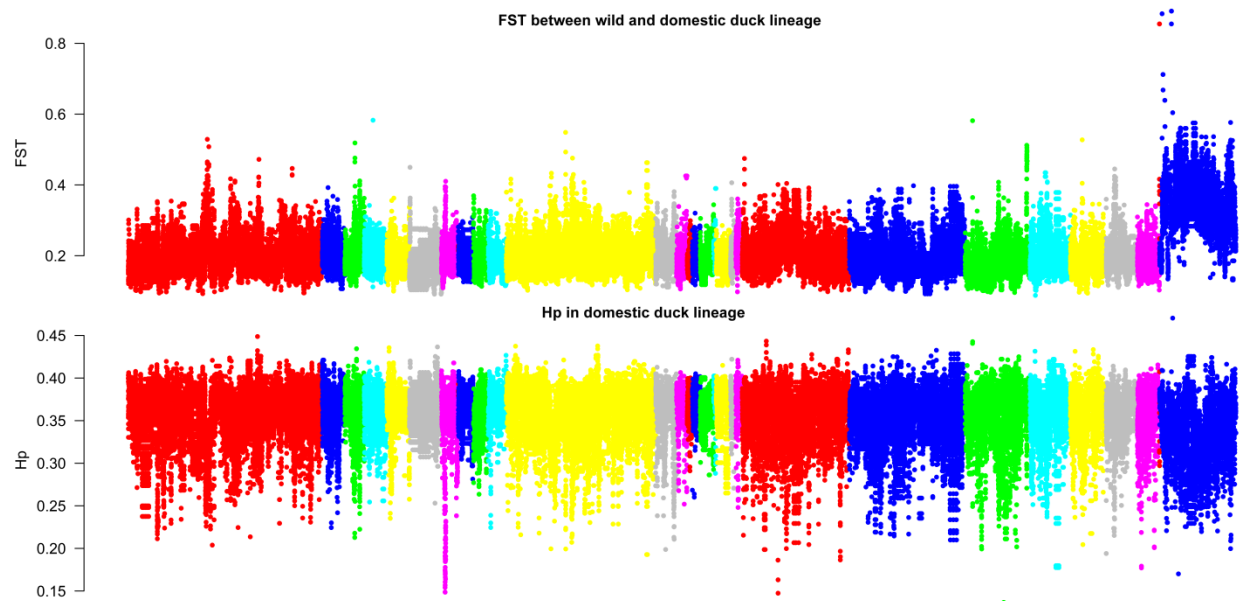


Fig. S26. Distribution of F_{ST} between the wild and the domestic duck lineage, and H_p in the domestic duck lineage.

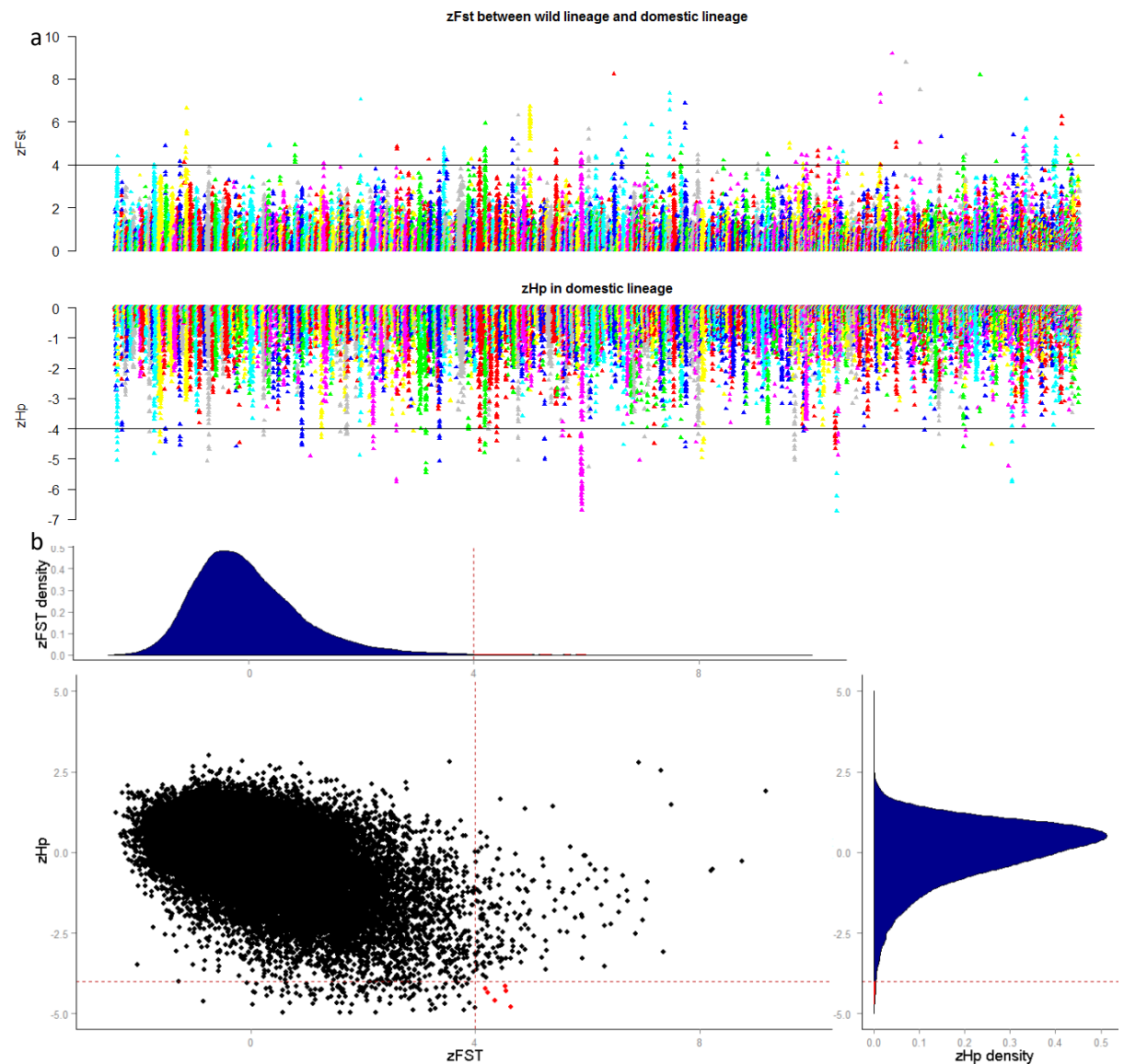


Fig. S27. Distribution of zF_{ST} and zH_p in ducks.

zF_{ST} and zH_p were counted with 40 kb window by a step size of 20 kb. (a) The positive end of the zF_{ST} between wild lineage and domestic lineage and negative end of zH_p in domestic lineage distribution plotted along scaffolds. (b) Distribution of zF_{ST} and zH_p . Red dash line indicate threshold and the windows identified as selection regions by both zH_p and zF_{ST} were show in red dots. (scaffolds homologous to chicken Z chromosome are not shown)

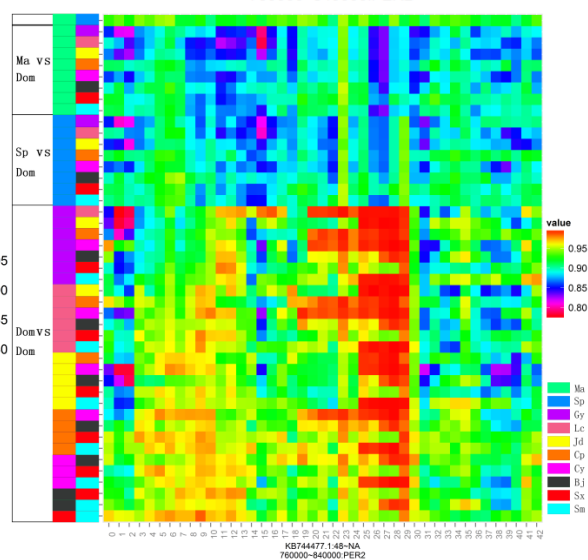
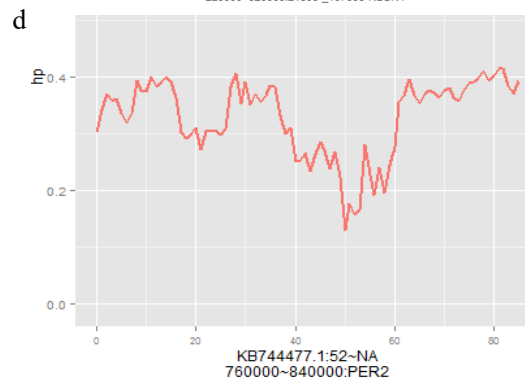
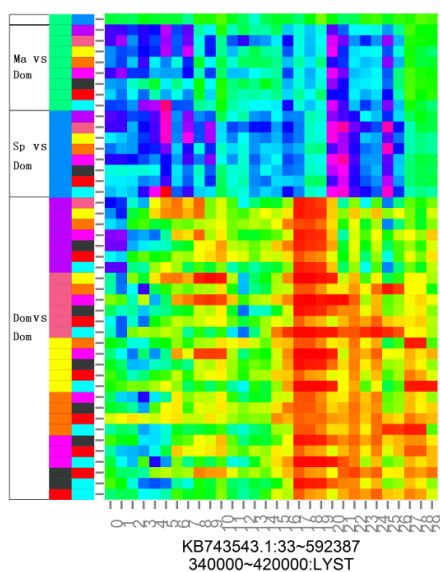
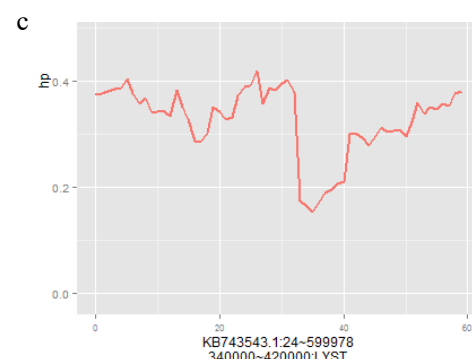
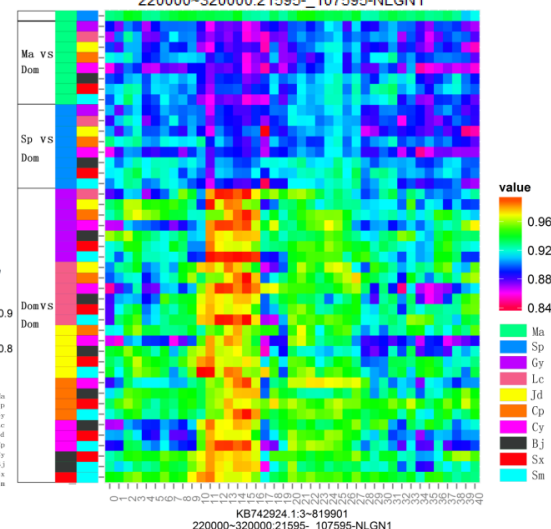
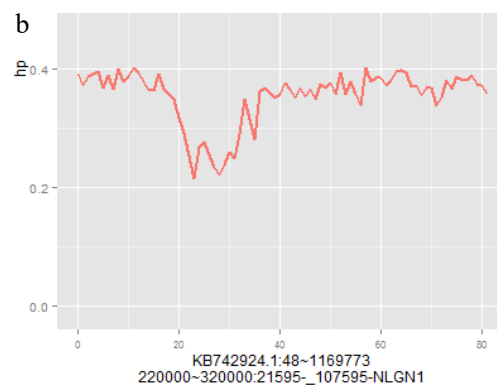
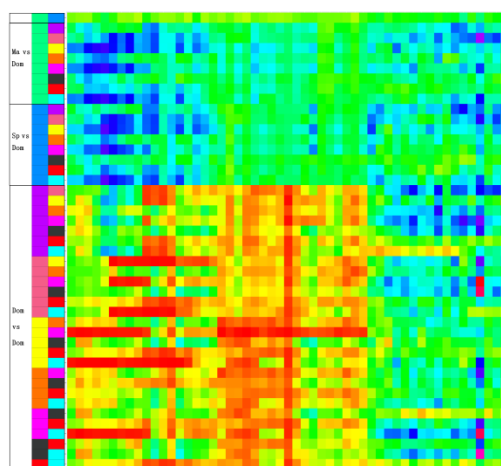
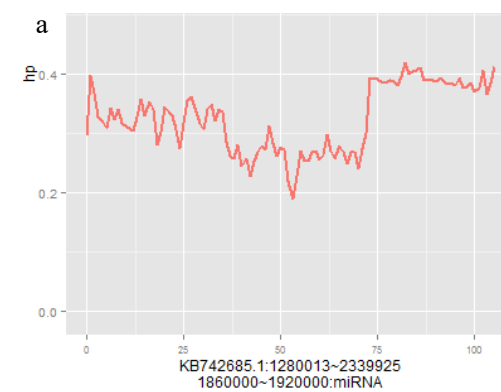


Fig. S28. Details of four putative selective sweep regions in the domestic duck lineage.

Each panel shows distribution of the hp value using non-overlap 10 kb window (upside) and scores between pairwise comparisons in 20 kb non-overlap windows (downside).
(a) *miR-3598-5p* locus. (b) *NLGN1* locus. (c) *LYST* locus. (d) *PER2* locus.

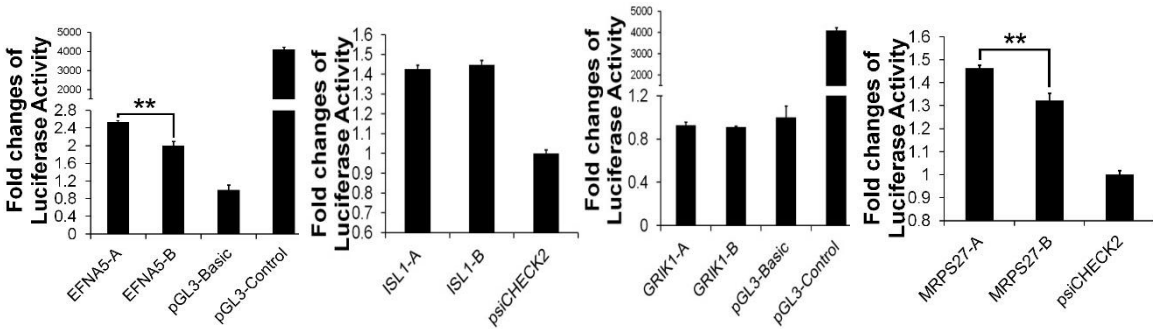
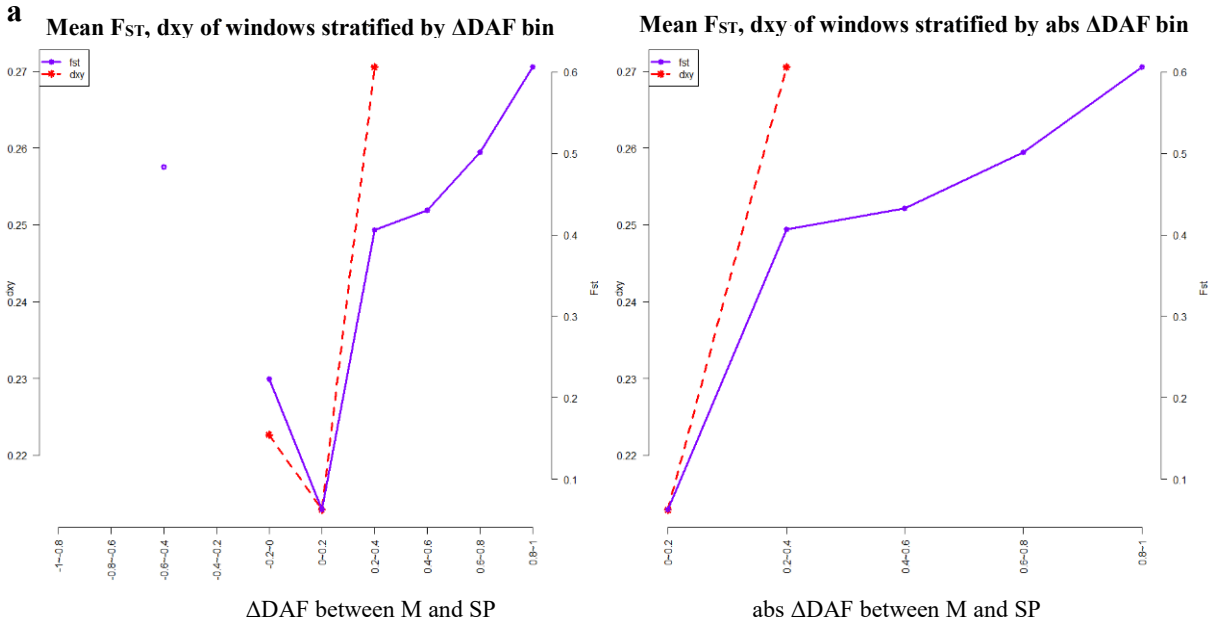


Fig. S29. Effects of casual candidate mutation on luciferase reporter activity.

Comparison in luciferase activity of alleles in wild (spot-billed and mallard) and domestic (Beijing) duck in DF1 (chicken embryonic fibroblast) cells. The *ISL1* was under positive selection in early stage of the wild lineage. The *GRIK1* and *LYST* were under positive selection in the domestic lineage. “-A”, “-B” represent wild mutation types and domestic mutation types respectively. DF1 cells were transfected with recombinant pGL3 luciferase reporter plasmids and luciferase activity was measured at 24 hours after transfection.



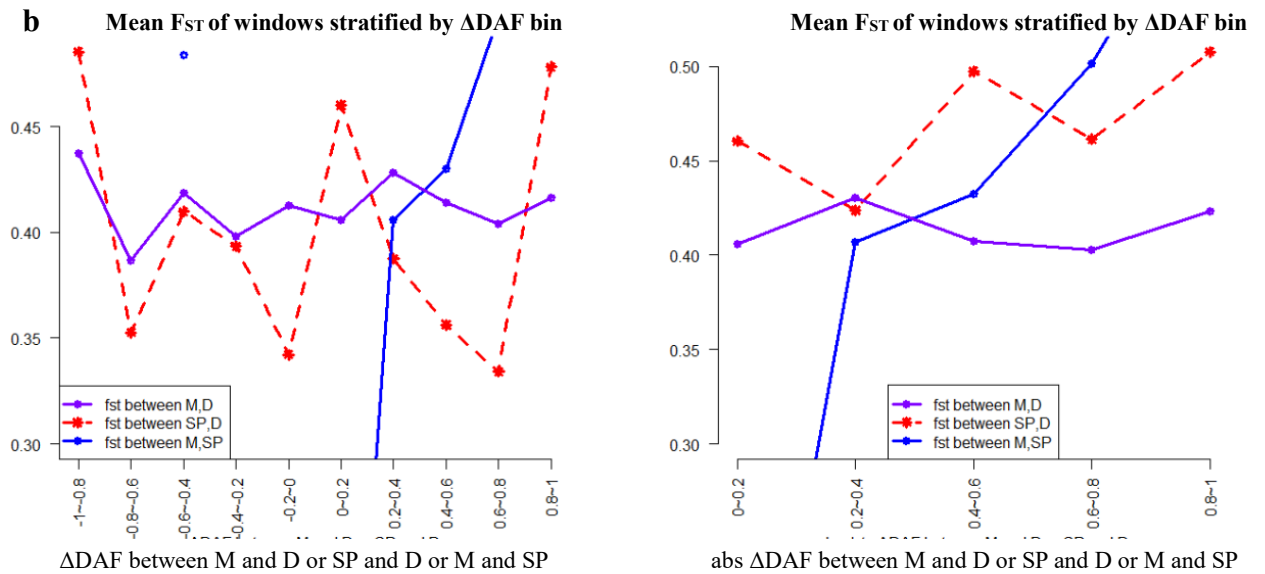
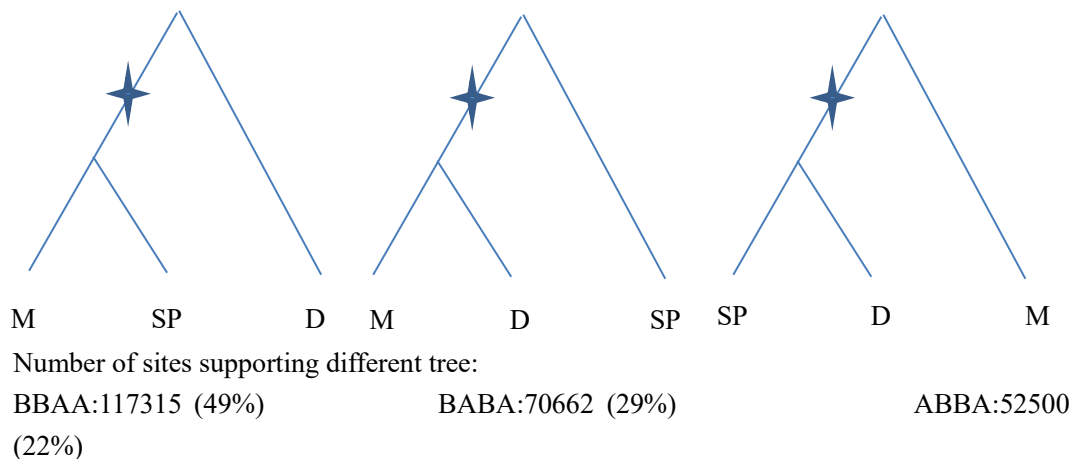


Fig. S30. Observation on genomic divergence values stratified by ΔDAF provide evidence of introgression between domestic and wild ducks.

(a) Mean F_{ST} (purple lines), dxy (red dash line) of windows stratified by ΔDAF (absolute ΔDAF in right) bins between mallard and spot-billed duck populations. dxy is not available in higher ΔDAF bins due to fixed SNPs are exclude in dxy calculation.

(b) Mean F_{ST} between domestic and wild duck populations decreased in high ΔDAF bins may due to introgression. Blue lines (corresponding to purple line in (a)) indicate no evidence of introgression between mallard and spot-billed in compare with the measurement between wild (red dash line: spot-billed, purple line: mallard) and domestic ducks. “M”, “SP” and “D” indicate mallard, spot-billed and domestic duck, respectively.



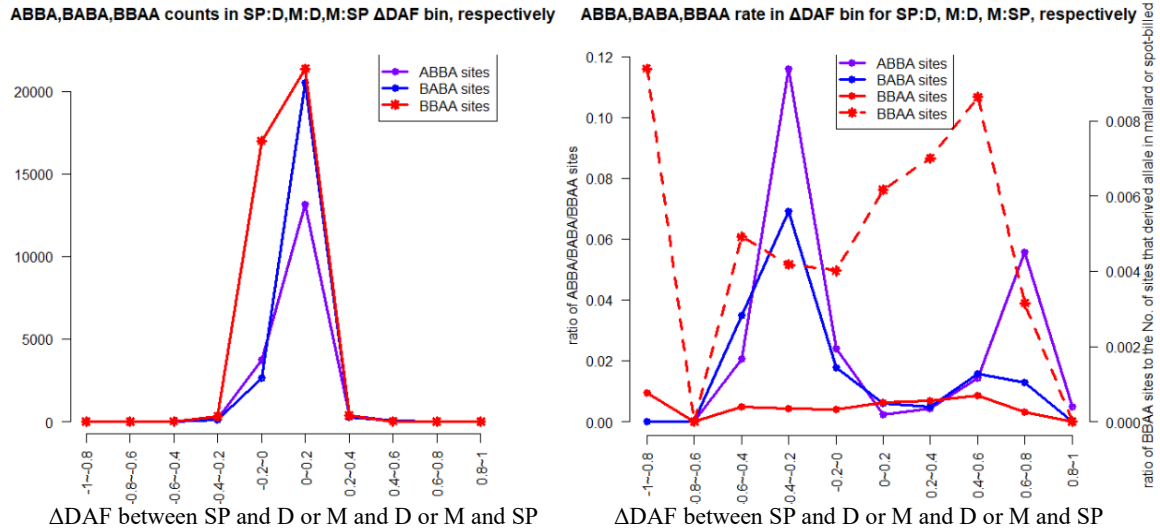


Fig. S31. Absent evidence of introgression between mallard duck and spot-billed duck population based on ABBA-BABA analysis of mallard, spot-billed and domestic duck. (a) Number of sites supporting different trees is indicated both as a percentage and as actual numbers. **(b)** Left panel, number of ABBA, BABA and BBAA sites were stratified by ΔDAF between spot-billed: domestic (purple line), mallard: domestic (blue line) and mallard: spot-billed (red line), respectively. Right panel, BBAA rate (red line) were measured as the BBAA counts divided by number of sites that derived allele present in either mallard or spot-billed population in each bins (ABBA (purple line), BABA (blue line) similarly). The red dash line is corresponding to right y-axis. “M”, “SP” and “D” indicate mallard, spot-billed and domestic duck, respectively.

1627 **Supplementary Tables**

1628 **Table S1: Statistics of the reads in mallard and spot-billed genome assemblies**

Paired-end libraries (bp)	Paired-end insert size	Number of libraries	Average read length (bp)	Total data (Gb)	Sequence coverage (X)	Physical coverage (X)
mallard	250bp	3	150_150	89.76	70.81	59.01
	500bp	2	100_100	46.18	36.94	92.35
	800bp	2	100_100	57.4	45.92	183.68
	2K	3	49_49	27.37	21.9	446.94
	5K	2	49_49	16.32	13.06	666.33
	10K	3	49_49	11.67	9.34	953.06
	20K	2	49_49	13.75	11	2,244.90
Total	/	17	/	262.4 5	208.97	4,646.27
spot-billed	250bp	3	150_150	95.64	76.51	63.76
	500bp	2	100_100	63.39	50.71	126.78
	800bp	2	100_100	45.31	36.25	145
	2K	3	49_49	26.2	20.96	427.76
	5K	2	49_49	17.11	13.69	698.47
	10K	3	49_49	15.37	12.3	1,255.10
	20K	2	49_49	12.17	9.74	1,987.76
Total	/	17	/	275.1 9	220.16	4,704.63

1629 **Table S2: Statistics of the three *de novo* ducks assemblies**

Category	Contig			Scaffold		
	Beijing Duck	Mallard	Spot-billed	Beijing Duck	Mallard	Spot-billed
Total length(kb)	1,069,961	1,176,969	1,187,333	1,105,049	1,270,808	1,317,385
Total number	227,597	172,189	194,934	78,487	61,591	67,685
Longest size(bp)	263,737	484,574	673,122	5,998,093	17,658,973	25,320,044
N50 size(bp)	26,138	38,271	33,061	1,233,631	2,489,142	2,054,876
N50 number	11,193	7,876	9,194	268	125	154
N90 size(bp)	3,085	5,381	4,386	195,458	145,390	102,224

N90 number	53,850	38,114	45,269	1,097	870	1,035
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1631 **Table S3: Comparison in length and coverage of the mallard and the spot-billed**
1632 **genome using seven Beijing duck BAC sequences**

BACs name	Length of BACs (bp)	mallard coverage	spot-billed coverage
ndae_TLR1	67,954	85.91%	70.03%
ndbaa_TLR15	79,704	98.50%	98.25%
ndbab_TLR5	72,128	98.41%	98.32%
ndbac_1_TLR3	76,489	97.53%	96.49%
ndbad_2_TLR7	97,799	95.64%	87.51%
ndbae_TLR2	75,080	98.82%	93.59%
ndbag_TLR4	77,042	97.79%	97.98%

1633 **Table S4: Evaluation of the mallard genome coverage using duck EST sequences**

Dataset	Number	Total length (bp)	Covered by assembly (%)	with >90% sequence in one scaffold		with >50% sequence in one scaffold	
				Number	Percent (%)	Number	Percent (%)
>0bp	319,996	98,257,350	99.02	294,845	92.14	315,689	98.65
>200bp	144,337	73,939,509	99.53	135,799	94.08	143,128	99.16
>500bp	43,896	43,155,260	99.84	41,595	94.76	43,687	99.52
>1000bp	13,652	22,939,026	99.99	12,831	93.99	13,613	99.71

1634 **Table S5: Evaluation of the spot-billed genome coverage using duck EST**
1635 **sequences**

Dataset	Number	Total length (bp)	Covered by assembly (%)	with >90% sequence in one scaffold		with >50% sequence in one scaffold	
				Number	Percent (%)	Number	Percent (%)
>0bp	319,996	98,257,350	99.01	294,324	91.98	315,566	98.62
>200bp	144,337	73,939,509	99.53	135,565	93.92	143,053	99.11
>500bp	43,896	43,155,260	99.85	41,531	94.61	43,670	99.49
>1000bp	13,652	22,939,026	99.99	12,802	93.77	13,606	99.66

1636 **Table S6: Type and proportion of transposable elements(TEs) in the Mallard,**
1637 **Spot-billed, Beijing duck and chicken genome**

TE type	mallard length (bp)	% genome	Spot-billed length (bp)	% genome	Beijing duck length (bp)	% genome	Chicken length (bp)	% genome
DNA	9,479,699	0.75	9,438,606	0.71	5,590,903	0.51	13,872,990	1.33

LINE	98,060,422	7.71	101,177,399	7.68	63,146,176	5.71	116,778,031	11.15
SINE	1,389,649	0.1	1,274,941	0.09	892,886	0.08	573,610	0.05
LTR	52,195,974	4.1	48,381,075	3.67	21,255,312	1.92	47,374,647	4.53
Other	21,136	0	12,303	0	3,201	0	3,338	0
Unknown	1,100,812	0.08	3,402,377	0.25	4,600,535	0.41	3,351,756	0.32
Total	136,134,153	10.71	139,051,537	10.55	80,740,993	7.31	149,208,388	14.25

1638 **Table S7: Gene annotation of the mallard and spot-billed reference gene sets using**
1639 **five protein databases**

Database	Mallard		Spot-billed	
	Number	Percent (%)	Number	Percent (%)
Total	21,056	100	21,123	100
InterPro	15,335	72.83	15,308	72.47
GO	12,817	60.87	12,815	60.67
KEGG	15,878	75.41	15,803	74.81
Swissprot	17,551	83.35	17,531	82.99
TrEMBL	18,371	87.25	18,327	86.76
All annotated	18,467	87.70	18,426	87.23
Unannotated	2,589	12.29	2,697	12.77

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1641 **Table S8: Gene family of two mammals and seven birds clustering by TreeFam**

Species	Total genes	Unclustered genes	Families	Unique families	Ave. genes per family
mallard	21,056	2,478	14,541	30	1.28
Spot-billed	21,123	2,604	14,530	42	1.27
Beijing duck	20,629	5,381	13,498	18	1.13
<i>G. gallus</i>	15,508	493	13,351	6	1.12
<i>S. camelus</i>	16,178	326	11,579	105	1.37
<i>M. gallopavo</i>	14,123	455	12,414	7	1.10
<i>F. albico</i>	15,303	716	13,284	33	1.10
<i>H. sapiens</i>	20,238	1,825	15,036	117	1.22
<i>M. musculus</i>	22,627	1,549	15,407	139	1.37

1642 **Table S9: Over represented GO terms (p-value <0.001) of wild duck expanded**
1643 **gene families**

1644 See file "Table_S9.xlsx"

1645 **Table S10: Summary for whole-genome resequencing**

Group	Species/breeds	Sample number	Sequence depth	Total data (Gb)	Read length	Sequence strategy	SNP number
Wild duck	Mallard	14	14.1×	218.2	125	Individual	32,982,331
	Spot-billed	13	14.5×	209.5	125	Individual	33,581,513

	Wigeon	3	20×	75.6	125	Individual	21,393,824
	Gadwall	1	20×	17.0	125	Individual	10,756,767
	Pintail	1	20×	25.7	125	Individual	8,551,170
	Falcata teal	1	20×	22.3	125	Individual	7,658,519
	Common teal	1	20×	22.1	125	Individual	9,146,292
	Baikal teal	1	20×	23.2	125	Individual	8,500,280
Indigenous Chinese duck breeds	Beijing	27	13.5×	400.7	125	Individual	14,748,500
	Shaoxing	27	2~3.5×	95.1	100	Individual	15,414,890
	Gaoyou	10	4.7×	56.9	100	Pooling	9,190,980
	Shanma	10	4.0×	50.4	100	Pooling	10,185,166
	Lianchengbai	10	4.7×	57.6	100	Pooling	8,978,662
	Jinding	10	4.4×	55.5	100	Pooling	8,617,344
Commercial breeds	Cherry valley	10	7.5×	94.5	100	Pooling	9,615,871
	Campbell	10	4.5×	57.3	100	Pooling	9,821,965
Outgroup	Muscovy	10	5.24×	57.9	100	Pooling	--
	Taihu goose	10	3.5×	39.7	100	Pooling	--

Table S11: Tests for population mixture of mallard, spot-billed, Beijing and Shaoxing ducks

Source 1	Source 2	Target	f3	Standard error	Z	SNPs
Mallard	Spot-billed	Beijing	0.007886	0.000748	10.546	9,997
Mallard	Spot-billed	Shaoxing	0.008662	0.000814	10.643	9,943
Mallard	Spot-billed	Mix	0.007279	0.000305	23.834	2,930,914
		Beijing, Shaoxing				
Mallard_hz	Spot-billed_hz	Beijing	0.008656	0.001076	8.045	9,975
Mallard_hz	Spot-billed_hz	Shaoxing	0.010200	0.000683	14.930	9,907
Mallard_fh	Spot-billed_fh	Beijing	0.007320	0.000603	12.147	9,986
Mallard_fh	Spot-billed_fh	Shaoxing	0.007380	0.000988	7.470	9,921

Considering the extensive substructure exist in wild ducks which reflect those individuals may trace their recent ancestry to different populations even species, we assume different clusters combinations to source 1 and source 2, and combine different domestic breeds to one target population, to minimize the population effect. All the Z were significantly greater than zero.

Table S12: Divergence without migration inferred parameters.

combinations	Parameter ^a	Conventional bootstrap 95% confidence interval	Conventional bootstrap 95% confidence interval (convert to real)	Maximum likelihood
Domestic, mallard duck	s (Nmallard)	0.85658-0.874169	1567694-1637111	1595689
	1-s(Ndomestic)	0.125831-0.143419	233297-265236	-
	Ts	0.0272-0.031008	100818-114733	102427
Domestic, spot-billed duck	s(Nspot-bill)	0.8776821-0.893130	1660302-1731619	1659603
	1-s(Ndomestic)	0.10687-0.122315	205404-233549	-

	Ts	0.02321-0.02653	89220-101346	91700
Spot-billed, mallard duck	s(Nmallard)	0.4352336-0.472079	949555-1031148	969384
	1-s(Nspot~)	0.52792-0.564766	1146439- 1239046	-
	Ts	0.015755-0.0174258	68583-76294	69432

^a see model code above.

Table S13: Summary of selective sweep regions in the wild duck lineage

These selective sweep regions occurred during their population in conjunction with or shortly after, their population divergence from domestic lineage ($S < -4$ for autosomes, $S < -1.5$ for Z-chromosomes). See file “Table_S13.xlsx”

Table S14: Enriched gene ontology of positively selected genes in the wild duck lineage.

See file “Table_S14.xlsx”

Table S15: Enriched gene ontology in biological process terms of positively selected genes in wild duck lineage.

See file “Table_S15.xlsx”

Table S16: Mean values of population genomic parameters for Z-chromosome and autosomes in mallard and spot-billed

Index		Autosomes		Z-chromosome	
		mean	SD	Mean	SD
Pi	Mallard	0.007520	0.004199	0.003397	0.002495
	Spot-billed	0.007607	0.004321	0.003020	0.002625
SNPs	Mallard	1,123	572	494	348
	Spot-billed	1,150	600	440	376
Tajima's D	Mallard	-0.525593	0.435471	-0.257150	0.445641
	Spot-billed	-0.625759	0.420070	-0.348016	0.492465
F_{ST}		0.049303	0.034864	0.259623	0.150430

Table S17: Mean values of genomic parameters inside/outside high F_{ST} (between mallard and spot-billed ducks) regions in autosomes

Species		pi (π)		No. of SNPs		Tajima's D	
		mean	sd	mean	sd	mean	sd
Mallard	Inside	0.002421	0.00217898	360	305	-0.61100	0.579419
	Outside	0.007595	0.00418171	1,134	569	-0.52064	0.433172
Spot-billed	Inside	0.002331	0.00196792	334	280	-0.47211	0.656325
	Outside	0.007683	0.00430549	1,161	597	-0.6228	0.415501

Table S18: Summary of domestic candidate regions ($zHp < -4$ for autosomes, $zHp < -2$ for Z-chromosomes)

See file “Table_S18.xlsx”

Table S19: Summary of high F_{ST} regions between the wild and domestic duck lineage.

See file “Table_S19.xlsx”

Table S20: Enriched gene ontology of positively selected genes in/close to domestic candidate regions.

See file “Table_S20.xlsx”

Table S21: Enriched gene ontology in biological process terms of positively selected genes in/close to domestic candidate regions.

See file “Table_S21.xlsx”

Table S22: Information of variant sites being performed luciferase assay to evaluate the activity.

Gene symbol	Variant position	Catalog of variant position	ΔAF	AF in wild	AF in domestic
<i>EFNA5</i>	KB744645.1: 114292,114293,114301	10 kb*	0.949,0.916,0.740	0,0.916,0	0.949,0,0.741
<i>EFNA5</i>	KB744645.1:58363	1 th intron	0.809	0.835	0.026
<i>ISL1</i>	KB743873.1:25803 (indel)	3' UTR	0.875	0.875	0
<i>MRPS27</i>	KB742554.1: 2151394,2151434,2151445	3' UTR	0.960,0.419,0.417	0,0.419,0.4	0.960,0.419,0.4
<i>GRIK1</i>	KB742951.1:543269	2 th intron	0.730	0.151	0.881
<i>LYST</i>	KB743543.1:403502	5 th intron	0.611	0.685	0.074

Note: 10 kb* means 10 kb downstream of the gene.

Table S23. Averaged all ΔDAF (absolute ΔDAF) of windows inside/outside high differentiation regions among the genome comparison of the two wild duck populations.

Mean value across SNPs	high differentiation regions	Genomic background
absolute ΔDAF	0.0385672	0.03672404
ΔDAF	0.0370516	0.03497989

Reference

1. Huang Y, *et al.* (2013) The duck genome and transcriptome provide insight into an avian influenza virus reservoir species. *Nature genetics* 45(7):776-783.
2. Huang Y, *et al.* (2011) Molecular evolution of the vertebrate TLR1 gene family--a complex history of gene duplication, gene conversion, positive selection and co-evolution. *BMC Evol Biol* 11:149.
3. Green RE, *et al.* (2010) A draft sequence of the Neandertal genome. *Science* 328(5979):710-722.
4. Kent WJ (2002) BLAT--the BLAST-like alignment tool. *Genome Res* 12(4):656-664.
5. Xu Z & Wang H (2007) LTR_FINDER: an efficient tool for the prediction of full-length LTR retrotransposons. *Nucleic acids research* 35(Web Server issue):W265-268.
6. Chen N (2004) Using RepeatMasker to identify repetitive elements in genomic sequences. *Current protocols in bioinformatics* Chapter 4:Unit 4 10.
7. Jurka J, *et al.* (2005) Repbase Update, a database of eukaryotic repetitive elements. *Cytogenetic and genome research* 110(1-4):462-467.
8. Benson G (1999) Tandem repeats finder: a program to analyze DNA sequences. *Nucleic acids research* 27(2):573-580.
9. Edgar RC (2004) MUSCLE: multiple sequence alignment with high accuracy and high throughput. *Nucleic acids research* 32(5):1792-1797.
10. Birney E, Clamp M, & Durbin R (2004) GeneWise and Genomewise. *Genome research* 14(5):988-995.
11. Trapnell C, Pachter L, & Salzberg SL (2009) TopHat: discovering splice junctions with RNA-Seq. *Bioinformatics* 25(9):1105-1111.
12. Huang Y, *et al.* (2013) The duck genome and transcriptome provide insight into an avian influenza virus reservoir species. *Nature genetics* 45(7):776-783.
13. Elsik CG, *et al.* (2007) Creating a honey bee consensus gene set. *Genome biology* 8(1):R13.
14. Hunter S, *et al.* (2009) InterPro: the integrative protein signature database. *Nucleic acids research* 37(Database issue):D211-215.
15. Ashburner M, *et al.* (2000) Gene ontology: tool for the unification of biology. The Gene Ontology Consortium. *Nat Genet* 25(1):25-29.
16. Schmitt P, Gueguen Y, Desmarais E, Bachere E, & de Lorgeril J (2010) Molecular diversity of antimicrobial effectors in the oyster *Crassostrea gigas*. *BMC evolutionary biology* 10:23.
17. Kanehisa M & Goto S (2000) KEGG: kyoto encyclopedia of genes and genomes. *Nucleic acids research* 28(1):27-30.
18. Yu XJ, Zheng HK, Wang J, Wang W, & Su B (2006) Detecting lineage-specific adaptive evolution of brain-expressed genes in human using rhesus macaque as outgroup. *Genomics* 88(6):745-751.
19. Ruan J, *et al.* (2008) TreeFam: 2008 Update. *Nucleic acids research* 36(Database issue):D735-740.
20. Huelsenbeck JP & Ronquist F (2001) MRBAYES: Bayesian inference of phylogenetic trees. *Bioinformatics* 17(8):754-755.
21. Yang Z (1997) PAML: a program package for phylogenetic analysis by maximum likelihood. *Comput Appl Biosci* 13(5):555-556.

- 1732 22. McPherson JD, *et al.* (2001) A physical map of the human genome. *Nature* 409(6822):934-941.
- 1733 23. DePristo MA, *et al.* (2011) A framework for variation discovery and genotyping using next-
1734 generation DNA sequencing data. *Nat Genet* 43(5):491-498.
- 1735 24. Karlsson EK, *et al.* (2007) Efficient mapping of mendelian traits in dogs through genome-wide
1736 association. *Nature Genetics* 39(11):1321-1328.
- 1737 25. Kofler R, *et al.* (2011) PoPoolation: a toolbox for population genetic analysis of next generation
1738 sequencing data from pooled individuals. *PLoS One* 6(1):e15925.
- 1739 26. Purcell S, *et al.* (2007) PLINK: a tool set for whole-genome association and population-based
1740 linkage analyses. *Am J Hum Genet* 81(3):559-575.
- 1741 27. Tamura K, *et al.* (2011) MEGA5: molecular evolutionary genetics analysis using maximum
1742 likelihood, evolutionary distance, and maximum parsimony methods. *Mol Biol Evol*
1743 28(10):2731-2739.
- 1744 28. Yang J, Lee SH, Goddard ME, & Visscher PM (2011) GCTA: a tool for genome-wide complex trait
1745 analysis. *Am J Hum Genet* 88(1):76-82.
- 1746 29. Li H, *et al.* (2009) The Sequence Alignment/Map format and SAMtools. *Bioinformatics*
1747 25(16):2078-2079.
- 1748 30. Zhao S, *et al.* (2013) Whole-genome sequencing of giant pandas provides insights into
1749 demographic history and local adaptation. *Nat Genet* 45(1):67-71.
- 1750 31. Harris RS (2007) *Improved pairwise alignment of genomic DNA* (The Pennsylvania State
1751 University).
- 1752 32. Groenen MA, *et al.* (2012) Analyses of pig genomes provide insight into porcine demography
1753 and evolution. *Nature* 491(7424):393-398.
- 1754 33. Li C, *et al.* (2015) The Immune Adaptor ADAP Regulates Reciprocal TGF-beta1-Integrin Crosstalk
1755 to Protect from Influenza Virus Infection. *PLoS Pathog* 11(4):e1004824.
- 1756 34. Barber MR, Aldridge JR, Jr., Webster RG, & Magor KE (2010) Association of RIG-I with innate
1757 immunity of ducks to influenza. *Proc Natl Acad Sci U S A* 107(13):5913-5918.
- 1758 35. Wu LQ & Dickman JD (2012) Neural correlates of a magnetic sense. *Science* 336(6084):1054-
1759 1057.
- 1760 36. Liedvogel M, *et al.* (2007) Chemical Magnetoreception: Bird Cryptochrome 1a Is Excited by Blue
1761 Light and Forms Long-Lived Radical-Pairs. *Plos One* 2(10).
- 1762 37. Anderson RM, Lawrence AR, Stottmann RW, Bachiller D, & Klingensmith J (2002) Chordin and
1763 noggin promote organizing centers of forebrain development in the mouse. *Development*
1764 129(21):4975-4987.
- 1765 38. Wu F, *et al.* (2015) Two transcription factors, Pou4f2 and Isl1, are sufficient to specify the retinal
1766 ganglion cell fate. *Proc Natl Acad Sci U S A* 112(13):E1559-1568.
- 1767 39. Ellegren H, *et al.* (2012) The genomic landscape of species divergence in Ficedula flycatchers.
1768 *Nature* 491(7426):756-760.
- 1769 40. Lavretsky P, *et al.* (2015) Speciation genomics and a role for the Z chromosome in the early
1770 stages of divergence between Mexican ducks and mallards. *Mol Ecol* 24(21):5364-5378.
- 1771 41. Han F, *et al.* (2017) Gene flow, ancient polymorphism, and ecological adaptation shape the
1772 genomic landscape of divergence among Darwin's finches. *Genome Res* 27(6):1004-1015.
- 1773 42. Axelsson E, *et al.* (2013) The genomic signature of dog domestication reveals adaptation to a
1774 starch-rich diet. *Nature* 495(7441):360-364.
- 1775 43. Harr B (2006) Genomic islands of differentiation between house mouse subspecies. *Genome*

1776 *Research* 16(6):730-737.

1777 44. de Manuel M, *et al.* (2016) Chimpanzee genomic diversity reveals ancient admixture with
1778 bonobos. *Science* 354(6311):477-481.

1779 45. Li M, *et al.* (2013) Genomic analyses identify distinct patterns of selection in domesticated pigs
1780 and Tibetan wild boars. *Nat Genet* 45(12):1431-1438.

1781 46. Prado-Martinez J, *et al.* (2013) Great ape genetic diversity and population history. *Nature*
1782 499(7459):471-475.

1783 47. Danecek P, *et al.* (2011) The variant call format and VCFtools. *Bioinformatics* 27(15):2156-2158.

1784 48. Pfeifer B, Wittelsburger U, Ramos-Onsins SE, & Lercher MJ (2014) PopGenome: an efficient
1785 Swiss army knife for population genomic analyses in R. *Mol Biol Evol* 31(7):1929-1936.

1786 49. Palm EC, *et al.* (2015) Mapping migratory flyways in Asia using dynamic Brownian bridge
1787 movement models. *Movement ecology* 3(1):3.

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